THE HORIZONTAL GEOMETRY OF PRODUCTION NETWORKS

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Abstract

Modern economies are organised as dense production networks, yet the transmission of shocks is conventionally understood through observed Input-Output linkages and vertical cascade effects. This paper argues that propagation additionally operates through horizontal complementarities: the extent to which sectors share upstream suppliers or downstream buyers. These interdependencies shape a latent network of connections linking also otherwise unconnected sectors, where a shared buyer makes its suppliers rivals: a gain in one moves the others' sales the opposite way. Because it moves Domar weights, horizontal propagation is hidden in Hulten's (1978) theorem while governing the effect of sectoral volatility on aggregate output, since horizontal reallocation among sectors shifts the average level of GDP. This channel complements vertical propagation, vanishes under Cobb-Douglas technologies, and gains relevance as networks expand. Using Input-Output data, I show that sectoral responses vary systematically with horizontal relations, forging novel implications for the transmission of sectoral and aggregate shocks.

JEL: C67, D57, E32, J21

KEYWORDS: production networks, Input-Output economy, horizontal transmission, network distance, Hulten theorem, sectoral comovement, aggregate fluctuations

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1. INTRODUCTION

A hallmark of contemporary production systems is the comovement of economic activity across sectors. Through Input-Output linkages production networks connect these sectors, allowing idiosyncratic shocks to propagate across the economy and to generate synchronized adjustments in production that contribute to aggregate fluctuations. In this framework, a shock travels *vertically*, cascading from suppliers to customers along parallel supply chains, through the tangle of sectoral ties.[1] Yet a shock often travels through hidden network corridors. Comovement may nonetheless emerge between productions that never trade with each other, because they share a downstream buyer (*e.g.*, February 2, 2026: the HBM4 memory breakthrough reshuffles demand across competing chip providers) or an upstream supplier (*e.g.*, the 2025 AI-driven surge in cloud demand for DRAM and HBM chips strained capacity for other users).[2]

This paper argues that production networks connect sectors not only through existing trade linkages, but also by demand and supply similarities embedded in their common structure of inter-sectoral trade. I conceptualize these interdependencies as *network economic distances* (*e.g.*, Conley and Dupor 2003). Specifically, sectors are close in input demand when they source intermediate inputs from the same upstream suppliers, and close in input supply when they sell to the same downstream buyers. Viewed through this lens, an Input-Output structure carries two geometries. It reflects *vertical* interdependencies, describing who supplies whom and how intensively (*e.g.*, Long and Plosser 1983; Shea 2002), as much as *horizontal* complementarities, where network participants are additionally linked through structural demand and supply relations from a resembling Input-Output geometry.

Network-based comovement is therefore endogenous. Since sectors mutually adjust in response due to shared economic relationships, horizontal geometry becomes a source of systematic comovement across economically proximate sectors. The transmission of shocks depends not only on the sequentiality proper of observed connections, but also on the exposure to the same Input-Output neighbourhoods: sectors that share a buyer compete for the same demand; sectors that share a supplier draw on the same production capacity. The question is not only whether shocks propagate through production networks, but also whether the horizontal geometry of these networks matters for determining who is exposed to whom, and by how much.

The two geometries propagate shocks in different ways. Vertically, a shock cascades from suppliers to customers through input requirements, with recursive feedback and amplification along supply chains (*e.g.*, Acemoglu, Carvalho, et al. 2012). Horizontally, it does not cascade: it reallocates. A buyer facing a cheaper input shifts its budget across suppliers, and what one supplier gains its rivals lose. This is why horizontal geometry is easy to overlook: since buyers reallocate optimally, these shifts leave aggregate output unchanged at first order, and the

[1] A positive shock to a sector improves its productive performance, triggering: *(i)* a *downstream effect*, as its buyers expand their demand for the cheaper commodity; *(ii)* an *upstream effect*, as its higher production raises its demand for intermediate inputs, benefiting its suppliers; and *(iii)* vertical *network spillovers*, as these effects ripple beyond immediate trading partners. When elasticities of substitution are not unitary, the transmission of shocks along these chains also depends on how easily inputs can be combined (*e.g.*, Baqaee and Farhi 2019). On the verticality of Input-Output economies refer to Pasinetti (1973).

[2] Refer to Supplemental Appendix D for a discussion about some documented real-world examples.

economy seems to respond to shocks through the size of sectors alone. Yet reallocation is not neutral: it moves workers across sectors and it changes how aggregate output responds.

Indeed, building a theory of horizontal transmission that separates it from vertical propagation and traces its implications for sectoral employment and aggregate output, and validating its predictions with U.S. sectoral data, constitute the central contributions of this paper.

Section 2 lays down the horizontal geometry. The same Input-Output matrix can be read in two ways: along supply chains, by composing relationships from supplier to customer, or across nodes, by comparing their profiles of suppliers and of buyers. The first reading builds the familiar Leontief inverse; the second builds two distance matrices, one for common suppliers and one for common buyers. Distance has an intuitive reading as common exposure: sectors that buy from the same suppliers inherit correlated cost shocks, and sectors that sell to the same buyers face correlated shifts in demand. As Theorem 1 establishes, a common neighbour makes two sectors economically connected even if they never trade, and the same structure generally places a pair of sectors at different distances on the side of suppliers and on that of customers.

Section 3 turns horizontal geometry into a transmission object. It first isolates a single shared partner, and finds that the two horizontal channels differ in nature. A common supplier transmits only through its own price: when one of its customers expands, the supplier becomes more expensive if its capacity is congested and cheaper under increasing returns, so that comovement between co-customers is a vertical round trip whose sign depends on the supplier's costs (Proposition 1). A common buyer, instead, generates genuine horizontal competition: when one upstream supplier becomes cheaper, the downstream buyer reallocates a given budget, and the shocked sector and its rivals move in opposite directions (Proposition 2). The elasticity of substitution among intermediate inputs does not alter this opposition; it only decides who wins: the shocked sector if inputs are substitutes, its rivals if they are complements. Demand distances therefore make comovement ambiguous, whereas supply distances make it transparent.

Idiosyncratic shocks are then allowed to propagate across the whole production network. By Theorem 2, general equilibrium reallocation unfolds in three steps: supply chains carry a shock to the cost of every commodity, each buyer reallocates its budget among intermediate inputs whose costs have moved unevenly, and parallel supply chains carry the resulting change in demand back to suppliers. Vertical structure is the carrier, horizontal geometry is where reallocation happens. In models with Cobb-Douglas technologies buyers never reallocate (*e.g.*, Acemoglu, Carvalho, et al. 2012), which explains why horizontal transmission is absent (Corollary 1). Horizontal geometry is also an independent dimension of the network: economies with identical sectoral sizes, and hence identical aggregate responses at first order, can make the same pair of sectors rivals in one case and apart in the other (Corollary 2). Finally, horizontal linkages proliferate as the production network expands: every new demand-supply connection makes the new supplier a rival of all the suppliers its buyer already has (Corollary 3).

Because horizontal transmission moves sales shares, it also moves workers. Sectors spend a constant share of revenue on labour, so that employment moves relative to the aggregate only if intermediate input shares move (Proposition 3). Labour reallocation is thus zero-sum: a shock moves the employment of a sector and of its close common-buyer neighbours in opposite directions, whereas horizontally distant sectors comove only through the common, aggregate

component. For aggregate output, instead, Hulten's (1978) theorem seems to leave no room for horizontal geometry: at first order, aggregate output responds to sectoral shocks through the Domar size of sectors alone. Theorem 3 reveals, however, that horizontal geometry is not absent, but hidden and plays a fundamental allocative role. Shared Input-Output structures matter through the endogenous response of Domar weights to shocks: the horizontal reallocation of intermediate input shapes how sectoral output shares adjust, thereby moving aggregate GDP beyond its benchmark where idiosyncratic shocks average out (Lucas 1977) whatever the distribution of Domar weights. This horizontal dimension gives rise to a non-vanishing effect of micro volatility on mean aggregate output. Positive and negative shocks may cancel out in the aggregate, but their dispersion within downstream buyers' input bundles does not.

Section 4 brings the two geometries to U.S. Input-Output data. Vertical propagation is concentrated in a few hubs supplying a wide range of activities, whereas horizontal relations are diffuse, as most sectors share partners with many others. Removing transactions below a progressively higher threshold steadily weakens vertical transmission, while horizontal propagation remains stable and becomes the dominant channel: horizontal relations rest on many shared partners, and survive sparsification as Corollary 3 predicts. A Bartik-style design then studies whether horizontal geometry shapes the transmission of aggregate disturbances beyond vertical exposure. The relevant channel depends on the nature of the disturbance and its sectoral exposures: aggregate productivity shocks, which move relative prices, travel mainly through common downstream buyers, whereas government spending shocks, which are demand impulses, travel mainly through common upstream suppliers. Horizontal exposure thus contains information on shock transmission that conventional Leontief-based linkages do not capture.

Section 5 studies sectoral comovement directly, relying on the two-stage approach of Barattieri and Cacciatore (2023) through which I estimate the way identified employment shocks propagate across sectors by type and degree of network "economic" distance. The empirical results are threefold. First, an increase in the intermediate inputs circulating through a sector raises its employment, with minimal and alternating effects from similar changes in distant sectors. Second, sectors that are close in supply distance exhibit opposite employment comovement, as the common-buyer channel predicts: a rise in employment in a nearby sector tends to reduce employment in its rivals. For sectors close in demand distance, responses are ambiguous, as the common-supplier channel predicts, although they mostly point to a dampening of vertical transmission. Third, positive comovement emerges among sectors that are distant in both dimensions, where vertical and aggregate forces prevail.

Overall, theory and evidence point in the same direction: horizontal linkages complement the canonical Leontief inverse, vertical transmission of shocks. Production networks shape the propagation of economic activity not only through who trades with whom, but also through who competes for the same buyers and draws on the same suppliers.

Literature review.– The objective of this paper is to advance both theoretically and empirically the understanding of how Input-Output economies work. On the theoretical side, it builds on the modern and rapidly expanding literature on production networks. In the wake of Gabaix (2011)'s "granular hypothesis", the seminal observation of Long and Plosser (1983) – that comovements across sectors are not dictated by a shared disturbance but rather by sec-

toral interdependencies –, has been modernized through a series of papers of the early 2010s (*e.g.*, Acemoglu, Carvalho, et al. 2012; Carvalho and Gabaix 2013; Carvalho 2014; Barrot and Sauvagnat 2016). This renewed perspective gave rise to studies on production networks related to efficient economies (*e.g.*, Baqaee and Farhi 2019; vom Lehn and Winberry 2022; Liu and Tsyvinski 2024), monetary policy and nominal rigidities (*e.g.*, La'O and Tahbaz-Salehi 2022; Rubbo 2023; Ghassibe and Nakov 2025), inefficiencies (*e.g.*, Jones 2011; Grassi 2017; Baqaee 2018; Baqaee and Farhi 2020), policy-oriented issues (*e.g.*, Liu 2019; Grassi and Sauvagnat 2019; Lane 2025), endogenous network formation (*e.g.*, Oberfield 2018; Acemoglu and Azar 2020; Ghassibe 2024; Taschereau-Dumouchel 2026), economic growth (*e.g.*, Hausmann and Hidalgo 2011; Gualdi and Mandel 2019; McNerney et al. 2022), and international contexts (*e.g.*, Caliendo et al. 2022; Qiu et al. 2026; Huo et al. 2025). A common mechanism explored in these works emphasizes the vertical propagation.[3] My contribution broadens this perspective by introducing horizontal linkages measured by common upstream or downstream Input-Output relationships. These novel interdependencies forge horizontal complementarities, allowing asymmetric shocks to propagate across the network in ways beyond vertical chains.

On the empirical side, relatively few studies examine the role of sectoral production networks on macroeconomic outcomes (*e.g.*, Acemoglu, Akcigit, et al. 2015; Ghassibe 2021; Barattieri and Cacciatore 2023; Barattieri, Cacciatore, and Traum 2023), with some illustrative spillovers underlying horizontal transmission (*e.g.*, Boehm and Oberfield 2018; Dragomirescu-Gaina and Elia 2021; Huremović, Nutarelli, et al. 2025). From this standpoint, the paper contributes novel empirical evidence on the role of Input-Output linkages in shaping interlinked dynamics, particularly in explaining the comovement of employment across sectors. Synchronized employment patterns are an important characteristic of business cycles, as most sectors tend to move together over time (*e.g.*, Christiano and Fitzgerald 1998). Several studies explore the nature and the sources of this comovement. Cooper and Haltiwanger (1990) find that sectoral employment levels are positively correlated, in ways not fully explained by aggregate shocks. Cassou and Vázquez (2014) attribute high employment comovement to similar shock transmission channels. In Yedid-Levi (2016) comovement is stronger along the extensive (number of workers) than the intensive (hours per worker) margin. Room for structural changes is in Kim (2020), where positive technology shocks in manufacturing raise employment in both manufacturing and services. While these studies provide valuable insights, they do not explicitly address how production networks shape employment comovement. Empirically, the paper contributes by: *(i)* advancing the comprehension of shocks' transmission exploiting network "economic" distances; and *(ii)* providing unprecedented evidence on short-run employment comovement from a production network perspective.

[3] As noted by Shea (2002), a positive sectoral supply shock has a price (decrease in its price level, and reduction in its nominal expenditure for intermediate inputs) and a quantity (its supply to other sectors increase, sector's production rises, thereby increasing its input demand) effect. These opposing forces affect upstream suppliers but, under Cobb-Douglas technology, they offset each other leaving intermediate inputs demand unchanged, thereby affecting only downstream suppliers. In general, only demand shocks can propagate upstream (*e.g.*, Acemoglu, Akcigit, et al. 2015; Barrot and Sauvagnat 2016; Ferrari 2024).

2. FOUNDATIONAL THEORY

The economy is populated by a finite set of sectors, $i, j, s \in \Phi = \{1, \ldots, m\}$, with Φ_i denoting the set of all others not including sector-i. Each produces a single commodity using labour and a set of circulating intermediate inputs from other sectors. The whole economic system is thus represented by an $m \times 1$ vector of sectoral production outputs, $\boldsymbol{y} = \left[y_i > 0\right]$, an $m \times 1$ vector of sectoral employment levels, $\boldsymbol{n} = \left[n_i > 0\right]$, each associated to its output elasticity of labour $\alpha_i \in (0,1)$ and wage w_i, and by an $m \times m$ square matrix, $\boldsymbol{X} = \left[x_{ij} \geq 0\right]$, indicative of the inter-sectoral trade of intermediate input quantities, with $x_{ij} = 0$ stating that sector-i does not make use of the commodity produced by sector-j at price p_j. Finally, an $m \times m$ square Input-Output matrix, $\boldsymbol{\mathcal{A}} = \left[\alpha_{ij} \geq 0\right]$, collects the intensity of the commodities produced by sectors as intermediate inputs in others. Its i-th row, $\boldsymbol{a}_i^{row} = \left[\alpha_{i1}, \ldots, \alpha_{ij}, \ldots, \alpha_{im}\right]$, is sector-$i$'s input profile: whom it buys from, and how intensively. Its i-th column, $\boldsymbol{a}_i^{col} = \left[\alpha_{1i}, \ldots, \alpha_{ji}, \ldots, \alpha_{mi}\right]'$, is sector-$i$'s output profile: who buys from it, and how intensively. Denote the Leontief inverse as $\boldsymbol{\mathcal{L}} = \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \left[\ell_{ij} \geq 0\right]$, with $\boldsymbol{I}$ being an $m \times m$ identity matrix.

A competitive final producer combines the production of different sectors, $Y = \prod_{i \in \Phi} y_i^{\omega_i} \propto \boldsymbol{\omega}' \boldsymbol{y}$, where $\boldsymbol{\omega}$ is an $m \times 1$ vector of $\{\omega_i\}_{\forall i \in \Phi} \in (0,1)$, weighting the relative importance of each sectoral commodity in aggregate output Y. Total consumption aggregates all final demands: let $\boldsymbol{c} = \left[c_i > 0\right]$ be an $m \times 1$ vector of sectoral consumption, it holds that $C = \prod_{i \in \Phi} c_i^{\omega_i} \propto \boldsymbol{\omega}' \boldsymbol{c}$. A representative household gets utility from consumption and leisure dis-utility from working: $\mathcal{U} = f^u\left[c_i\,,\, n_i\right]_{\forall i \in \Phi}$, with consumption of all sectoral commodities and labour supplied characterized by a price, p_i and w_i respectively. Moreover, since the household spends a share ω_i of final expenditure on commodity-i, Domar weights are $\boldsymbol{\lambda}' = \boldsymbol{\omega}' \boldsymbol{\mathcal{L}}$.[4]

A representative firm characterizes each sector-i. Given an exogenous Hicks-neutral productivity z_i, the production function is

$$y_i = z_i\, f^y\left[n_i\,,\; x_i := \left\{x_{ij}\right\}_{\forall j \in \Phi}\right]$$

ASSUMPTION 1 (Production technology requirements) *The production function: (i) has constant returns to scale in either n_i and x_i so that $\alpha_i + \sum_{j \in \Phi} \alpha_{ij} = 1$; (ii) is differentiable, continuous, strictly quasi-concave, homogeneous of degree one, and increasing in z_i, n_i, and x_i; (iii) the case $f^y\left[0\,,\,\cdot\right]$ is ruled out as labour input is essential to production; (iv) at least two elements of x_i have to be positive, thus $f^y\left[\cdot\,,\,0\right]$ cannot exist; consequentially, (v) part of the sectoral output is directly produced by the sector, $x_{ii} > 0$, $\forall i \in \Phi$, that is a production plan with a bundle of intermediate inputs as $f^y\left[\cdot\,,\,x_{ij}\right]$, given any $j \in \Phi_i$, is precluded.*

The first two sub-assumptions are common in production technology (*e.g.*, Acemoglu, Carvalho, et al. 2012). By way of *A1.iii*, workers cannot be fully substituted by any combination of intermediate inputs, and changes in the purchased quantity of intermediate inputs would deter-

[4] Domar weights are defined as $\lambda_i = \frac{p_i y_i}{\sum_{i \in \Phi} w_i n_i}$, that is sectoral sales relative to nominal GDP (labour income only, since profits are zero and labour is the only factor). Multiplying the market-clearing condition for sector-i, $y_i = c_i + \sum_{j \in \Phi} x_{ji}$, by $\frac{p_i}{\sum_{i \in \Phi} w_i n_i}$, using $p_i c_i = \omega_i \sum_{i \in \Phi} w_i n_i$ and, by definition of cost shares, $p_i x_{ji} = \alpha_{ji}\, p_j y_j$, gives $\lambda_i = \omega_i + \sum_{j \in \Phi} \lambda_j \alpha_{ji}$. In matrix form, $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}' \boldsymbol{\mathcal{A}}$. Since every row of $\boldsymbol{\mathcal{A}}$ sums to $1 - \alpha_i < 1$, $\boldsymbol{I} - \boldsymbol{\mathcal{A}}$ is invertible, and $\boldsymbol{\lambda}' = \boldsymbol{\omega}' \left(\boldsymbol{I} - \boldsymbol{\mathcal{A}}\right)^{-1} = \boldsymbol{\omega}' \boldsymbol{\mathcal{L}}$.

mine subsequent variations in the labour force size. The assumptions characterizing the bundle of intermediates, *A1.iv* and *A1.v*, ensure the sector not only to participate in the network, but also that part of the production process is roundabout: for sector-i delivering its commodity y_i requires to be at least vertically connected (buying from one upstream sector), while imposing that a fraction of intermediate inputs must be directly produced within the sector.

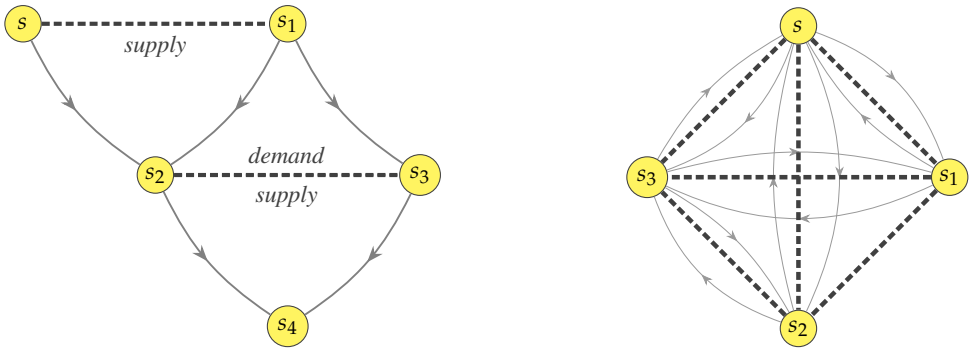
Each perfectly competitive representative firm maximizes profits given input prices, w_i and p_j, $\forall j \in \Phi$. For any functional form of y_i satisfying Assumption 1, by Shephard's lemma the share of commodity-j in sector-i's total cost $\kappa_i = f^{\kappa}(p, w)$ is $\alpha_{ij} = \frac{\partial \log \kappa_i}{\partial \log p_j} \geq 0$, and the labour share is $\alpha_i = \frac{\partial \log \kappa_i}{\partial \log w_i}$, so that the constraint $\alpha_i + \sum_{j \in \Phi} \alpha_{ij} = 1$ holds exactly for any $\{p, w\}$.

Vertical and horizontal contractions.– In a networked economy, comovement of production inputs is endogenously induced. Each sector both relies on and supplies to others, so that changes in one sector affect not only its own optimal input composition but also the demand and supply conditions faced by other sectors. Sectors that trade with similar partners therefore face shared network exposure, providing a natural basis for a rigorous comparison. Indeed, the same Input-Output matrix can be contracted over a common partner in three ways.

Consider a pair of sectors, say $\{i, j\}$, related to a common one, say s. Then, for the pair:

- $\left[\boldsymbol{\mathcal{A}}^2\right]_{ij} = \sum_{s \in \Phi} \alpha_{is}\, \alpha_{sj}$ denotes a recursive production chain;
- $\left[\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'\right]_{ij} = \sum_{s \in \Phi} \alpha_{is}\, \alpha_{js}$ denotes the common upstream neighbourhood;
- $\left[\boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}\right]_{ij} = \sum_{s \in \Phi} \alpha_{si}\, \alpha_{sj}$ denotes the common downstream neighbourhood.

The first is a recursive chain (i buys from s, which buys from j), generalisable to k-stages beyond the common sector-s, $\left[\boldsymbol{I}\right]_{ij} + \left[\boldsymbol{\mathcal{A}}\right]_{ij} + \left[\boldsymbol{\mathcal{A}}^2\right]_{ij} + \cdots + \left[\boldsymbol{\mathcal{A}}^k\right]_{ij}$; the second is a connection through a common supplier (both i and j buy from s); the third is a connection through a common buyer (s buys from both). Under this perspective, network connections extend beyond their flows of intermediate inputs. These additional interdependencies highlight the complexity of inter-sectoral dynamics and complement the *Leontief inverse*, a reminiscent result from Leontief (1936): it captures how sectors not directly trading are indirectly related – *e.g.*, a sector buying from an upstream one introjects such sector's purchases as well the other of more upstream sectors –, emphasizing the depth of vertical linkages through layers of Input-Output relationships. Here, powers of $\boldsymbol{\mathcal{A}}$ compose relationships along supply chains and build the Leontief inverse: they are *vertical* and operate across stages of production. Common connections, by contrast, consider how participants are mapped horizontally: sectors are additionally related because they are relying on the same suppliers or serving the same customers. Here, Gram products compare the profiles of two sectors at the same production stage: they are *horizontal*, and they link also sectors that need not trade with each other at all. These broader notions of linkages provide a richer understanding about systemic interdependencies and explain *horizontal complementarities* in demand and supply that classical Input-Output analysis overlook. They reveal how seemingly unrelated sectors can experience synchronized fluctuations, and how the structure of common upstream or downstream connections shapes the propagation of independent and idiosyncratic shocks. In other words, the notion of network interdependencies is not confined to "who trades with whom", but also emerges from "who depends on whom".



(A) DISTANCE DEFINITION (B) DISTANCE NETWORK

FIGURE 1: HORIZONTAL INPUT-OUTPUT GEOMETRIES

Note: the figure presents a stylized production network and its corresponding network of sectoral distances. Production network linkages are depicted with oriented thin lines, while dashed thick lines indicate Input-Output economic distance relationships.

DEFINITION 1 (Horizontal proximity and network distance) *The common-supplier and common-buyer proximity matrices are*

$$\mathcal{B}^{fd} = \mathcal{A}\mathcal{A}' \qquad \text{and} \qquad \mathcal{B}^{fs} = \mathcal{A}'\mathcal{A}$$

so that $b^{fd}[i,j] = \langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row} \rangle$ *and* $b^{fs}[i,j] = \langle \boldsymbol{a}_i^{col}, \boldsymbol{a}_j^{col} \rangle$. *The associated distances are* $d^q[i,j] = \left(b^q[i,i] + b^q[j,j] - 2\,b^q[i,j]\right)^{1/2}$, *for* $q = \{fd, fs\}$, *that is* $d^{fd}[i,j] = \left\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\right\|$ *and* $d^{fs}[i,j] = \left\|\boldsymbol{a}_i^{col} - \boldsymbol{a}_j^{col}\right\|$, *collected in* $\mathcal{D}^q$.

Paraphrasing Conley and Dupor (2003), two types of network "economic" distances between sectors, sketched in Panel 1a of Figure 1, can be defined: *(a) factor input demand*, when sectors are buying their intermediate inputs from similar upstream sectors; and *(b) factor input supply*, when sectors are selling their commodities as intermediate inputs to similar downstream sectors. Panel 1b sketches a stylized economy associated to its distance-based network: distances effectively double Input-Output linkages and bound disconnected sectors.

EXAMPLE 1 (Distance in the network) *Consider an economy populated by four sectors,* $\{s, s_1, s_2, s_3\} \in \Phi$, *where some trade with all others, while some do not. In Panel 1b, focus on the pair* $\{s_1, s_2\}$*: they do not mutually trade, so* $\alpha_{s_1 s_2} = 0 = \alpha_{s_2 s_1}$. *Nevertheless, their linkages with all the remaining sectors generate a common structure of inter-sectoral trade. Yet, while the underlying trade and distance structures are unique, the horizontal metrics vary under alternative specifications of network distance, as Examples 2-3 clarify.*

Indeed, the same primitive Input-Output structure governing the trade of commodities not only determines how connections spread through production chains, but also how sectors are positioned relative to each other in the space of interdependencies. In this sense, the network matrix induces both a dynamic vertical structure and a geometric delineation of sectoral similarities. A single production network generates two distinct but tightly connected objects. On one hand, its Leontief inverse captures a vertical system: it is the equilibrium mapping encoding how commodities are produced by the means of other commodities (Sraffa 1960) through successive layers of production from direct to increasingly indirect effects along length-k production chains. On the other hand, the same matrix defines a horizontal embedding: sectors

are related insofar as they share similar demand and supply relationships, interacting with common segments of the network; a collection of Input-Output profiles $\left\{\boldsymbol{a}_i^{row}, \boldsymbol{a}_i^{col}\right\}_{i\in\Phi}$ encoding a geometric representation of sectoral similarities. This induces a set of sectoral profiles which embeds sectors according to their commonalities in the Input-Output space. Key is that these objects are not separated but different features of the same primitive structure: one capturing recursive chains, the other describing the overlap of network linkages.

The discussion is in terms of the primitive coefficient matrix $\boldsymbol{\mathcal{A}}$, but it applies also to its Leontief inverse $\boldsymbol{\mathcal{L}}$, since it carries over the primitive structure into its cumulative form rather than creating a new one. The entries of $\boldsymbol{\mathcal{A}}$ are the direct Input-Output coefficients, while its powers aggregate their propagation across production chains of length-k.

REMARK 1 *The Leontief inverse inherits the dual representation, since* $\boldsymbol{\mathcal{L}} = \sum_{k\geq 0} \boldsymbol{\mathcal{A}}^k$.

So, the established duality is intrinsic to any Input-Output system. The primitive matrix $\boldsymbol{\mathcal{A}}$ and its Leontief inverse $\boldsymbol{\mathcal{L}}$ are two representations of the same underlying system, so that constructing similarities from either objects is logically equivalent, although the two carry different economic interpretations.

2.1. RECOVERING DEMAND AND SUPPLY MATRICES

Consider the case in which two sectors, say $\{i, j\}$, are buying intermediate inputs from the same sector, say s. Let the price of the common input to vary by $d\log p_s$, holding all other prices fixed. The two sectors' costs respond as $d\log \kappa_{i,s} = \alpha_{is} \cdot d\log p_s$ and $d\log \kappa_{j,s} = \alpha_{js} \cdot d\log p_s$. The comovement induced by the common sector-s is therefore the covariance of the two responses:

$$b^{fd}_{s\rightarrow}[i,j] \;=\; \text{Cov}\left[d\log \kappa_{i,s}\,,\, d\log \kappa_{j,s}\right] \;=\; \alpha_{is}\alpha_{js} \;\geq\; 0 \tag{1}$$

Three properties follows. First, eq. (1) selects genuinely shared linkages, since $b^{fd}_{s\rightarrow}[i,j] > 0$ if and only if both sectors purchase from sector-s, for strictly positive connections $\alpha_{is}\alpha_{js}$. Second, it weights by the network intensity of both: sharing a supplier from which each buys a large share of its inputs counts more than sharing one that is marginal for either. Third, since these α's are cost shares, $b^{fd}_{s\rightarrow}[i,j]$ is the probability that one unit of expenditure drawn at random from sector-i and one drawn independently from sector-j end up at sector-s.

Iteratively, the above condition can be rewritten accounting for all sectors from which the pair is buying: $\boldsymbol{b}^{fd}[i,j] = \left[\alpha_{ii}\alpha_{ji}\,,\, \alpha_{ij}\alpha_{jj}\,,\, \ldots\,,\, \alpha_{im}\alpha_{jm}\right]$, which is the $1\times m$ vector of the pair's shared intermediate usage when sectors $\{i, j\}$ are buying from others. The vector carries two distinct pieces of information. Its support identifies the non-negative entries through which suppliers the pair is connected, and is the extensive margin; and its magnitudes identifying how much each channel weighs, and are the intensive margin. Summing across channels and stacking such condition across all sectors delivers the $m\times m$ demand-side proximity matrix $\boldsymbol{\mathcal{B}}^{fd}$, whose generic element is the sum of eq. (1) over common nodes, $b^{fd}[i,j] = \sum_{s\in\Phi}\alpha_{is}\alpha_{js}$. The greater the value of each cell, the closer the sectors are within the production network. Co-exposure is increasing in both trade intensities: for $\alpha_{is} < \alpha_{js}$, a shock transmitted to the pair through s moves the two sectors' costs by unequal amounts and the induced comovement is weak, signaling low horizontal complementarity; conversely, when both intensities are large the

same shock moves both strongly, and $b^{fd}_{s\rightarrow}[i,j]$ is large (the same logic applies to the common-supply case, eq. 3). Using the demand composition of sector-i, $\boldsymbol{a}^{row}_i = [\alpha_{i1},\ldots,\alpha_{im}]$, each entry is an inner product between profiles, $b^{fd}[i,j] = \langle \boldsymbol{a}^{row}_i, \boldsymbol{a}^{row}_j \rangle$, so that $\boldsymbol{\mathcal{B}}^{fd} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'$.

Matrix $\boldsymbol{\mathcal{B}}^{fd}$ measures economic *proximity*: large entries denote strongly co-exposed pairs. The corresponding economic *distance* follows from the standard Gram-to-metric conversion,

$$d^{fd}[i,j]^2 = b^{fd}[i,i] + b^{fd}[j,j] - 2\,b^{fd}[i,j] = \left\|\boldsymbol{a}^{row}_i - \boldsymbol{a}^{row}_j\right\|_2^2 \tag{2}$$

where the second equality follows by expanding the squared norm through bi-linearity of the inner product. Explicitly,

$$d^{fd}[i,j] = \sqrt{[\alpha_{ii}-\alpha_{ji}]^2 + [\alpha_{ij}-\alpha_{jj}]^2 + \cdots + [\alpha_{im}-\alpha_{jm}]^2}$$

thereby representing a certain network "economic" distance between two sectors, with larger values denoting how dissimilar they are in terms of shared Input-Output structure when buying from upstream sellers. Henceforth, the corresponding $m \times m$ matrix $\boldsymbol{\mathcal{D}}^{fd}$ collects such distances, whose characterizing elements $d^{fd}[i,j] > 0$, $\forall i \neq j \in \Phi$, are symmetric outside the main diagonal (the distance from i to j is the same as from j to i), with $d^{fd}[i,i] = 0$. Proximity and distance are thus two readings of one object: $\boldsymbol{\mathcal{B}}^{fd}$ is foundational on the theory side, where larger entries mean closer sectors, while the conversion delivers the metric for $\boldsymbol{\mathcal{D}}^{fd}$ used in the empirics of Sections 4-5, where larger entries mean the opposite.

EXAMPLE 2 (Factor input demand metrics) *Consider the pair of sectors $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors from which both are purchasing are $\{s, s_3\}$, and assume fictional trade intensities for sector-s_1 to be $\alpha_{s_1 s} = 0.2$ and $\alpha_{s_1 s_3} = 0.1$, while that of sector-s_2 are $\alpha_{s_2 s} = 0.1$ and $\alpha_{s_2 s_3} = 0.2$. Accordingly, the co-exposure vector expressing their factor input demand relation is $\boldsymbol{b}^{fd}[s_1,s_2] = [.02,.02]$ when buying from sector-s and sector-s_3, respectively. Then $d^{fd}[s_1,s_2] = \sqrt{.02}$. Demand-based proximity is positive even in the absence of a direct linkage between the two sectors in the pair.*

Consider now the case in which two sectors, say $\{i,j\}$, are selling their own commodity as an intermediate input to the same sector, say s. By definition of cost share, the value of sector-i's sales to s is $p_i x_{si} = \alpha_{si} \cdot \kappa_s$. A variation in the buyer's scale at constant shares therefore moves both suppliers' sales in the same direction, and the covariance of the two responses per unit of shock variance delivers

$$b^{fs}_{\rightarrow s}[i,j] \;=\; \mathrm{Cov}\left[d\log[p_i x_{si}],\, d\log[p_j x_{sj}]\right] \;=\; \alpha_{si}\alpha_{sj} \;\geq\; 0 \tag{3}$$

Collecting across all common buyers, $\boldsymbol{b}^{fs}[i,j] = [\alpha_{ii}\alpha_{ij},\ldots,\alpha_{mi}\alpha_{mj}]$, and stacking delivers the supply-side proximity matrix $\boldsymbol{\mathcal{B}}^{fs}$, with generic element $b^{fs}[i,j] = \sum_{s\in\Phi}\alpha_{si}\alpha_{sj} = \langle \boldsymbol{a}^{col}_i, \boldsymbol{a}^{col}_j \rangle$, where $\boldsymbol{a}^{col}_i = [\alpha_{1i},\ldots,\alpha_{mi}]'$ is the supply profile of sector-i. In matrix form $\boldsymbol{\mathcal{B}}^{fs} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$; its $m \times m$ distance matrix $\boldsymbol{\mathcal{D}}^{fs}$ follows from eq. (2) applied to supply profiles.

EXAMPLE 3 (Factor input supply metrics) *Consider the pair of sectors $\{s_1, s_2\}$ in an Input-Output matrix mirroring Example 1, where the common sectors to which both are selling are $\{s, s_3\}$, and assume fictional trade intensities for sector-s_1 to be $\alpha_{ss_1} = 0.4$ and $\alpha_{s_3 s_1} = 0.3$,*

while that of sector-s_2 are $\alpha_{ss_2} = 0.6$ and $\alpha_{s_3s_2} = 0.6$. Accordingly, the co-exposure vector expressing their factor input supply relation is $\boldsymbol{b}^{fs}[s_1, s_2] = [.24, .18]$ when selling to sector-s and sector-s_3, respectively. Then $d^{fs}[s_1, s_2] = \sqrt{.13}$. Supply-based proximity is positive even in the absence of a direct trade linkage between the two sectors in the pair.

An important aspect to consider when introducing economic distances in the production network is that, under the same configuration of the inter-sectoral trade structure, sectors can be in a different "distance relation" depending on whether they are buying from or selling to other sectors. Asking when the two sides coincide amounts to asking when co-exposure fails to distinguish $\boldsymbol{\mathcal{A}}$ from $\boldsymbol{\mathcal{A}}'$, which is the content of the following result.

THEOREM 1 (On characteristic properties) *Demand- and supply-based matrices satisfy:*

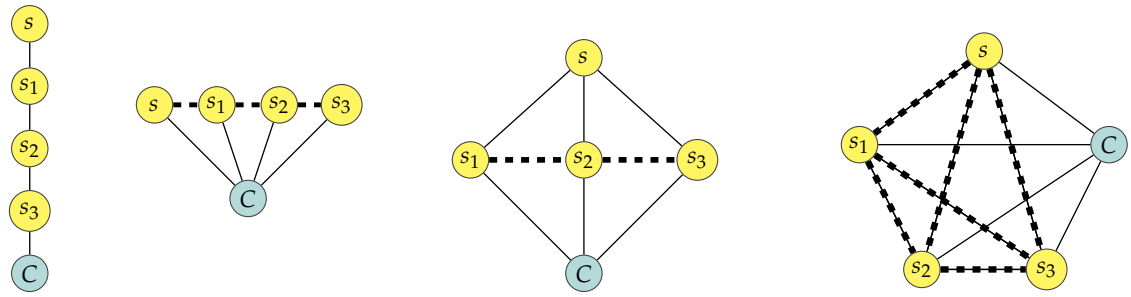
- *$\boldsymbol{\mathcal{B}}^{fd}$ and $\boldsymbol{\mathcal{B}}^{fs}$ are non-negative, symmetric and positive semi-definite;*
- *proximity is coupled with $d^q[\cdot]$, a Euclidean metric on Input-Output profiles: $d^q[i,j] = 0$ if and only if the two profiles coincide; $\boldsymbol{\mathcal{D}}^{fd}$ and $\boldsymbol{\mathcal{D}}^{fs}$ have a zero diagonal and, unless null, are not positive semi-definite; for profiles normalized to sum to one at the common node, $d^q[i,j] \in [0, \sqrt{2}]$ for $q = \{fd, fs\}$;*
- *proximity and distance matrices make all nodes simultaneously connected, with each cell strictly positive if the pair shares at least one neighbour, so that both matrices are full;*
- *they differ entry-wise unless the corresponding bilateral trade with the common neighbour coincides: $\boldsymbol{\mathcal{B}}^{fd} = \boldsymbol{\mathcal{B}}^{fs}$ if and only if $\boldsymbol{\mathcal{A}}$ is a normal matrix, as when $\boldsymbol{\mathcal{A}}$ is symmetric.*

Proof: for the first claim, note that for any $\boldsymbol{\xi} \in \mathbb{R}^m$, then $\boldsymbol{\xi}'\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}'\boldsymbol{\xi} = \|\boldsymbol{\mathcal{A}}'\boldsymbol{\xi}\|^2 \geq 0$ and $\boldsymbol{\xi}'\boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}\boldsymbol{\xi} = \|\boldsymbol{\mathcal{A}}\boldsymbol{\xi}\|^2 \geq 0$; symmetry is immediate. Since $b^{fd}[i,j] = \sum_{s\in\Phi} \alpha_{is}\alpha_{js}$ is a sum of non-negative terms, it is positive if and only if $\alpha_{is}\alpha_{js} > 0$ for some s, *i.e.*, if i and j share a supplier; likewise for b^{fs} and buyers. For the second claim, by bilinearity, $b^q[i,i] + b^q[j,j] - 2\,b^q[i,j]$ equals the squared Euclidean norm of the difference between the two profiles, which defines a metric on profiles. $\boldsymbol{\mathcal{D}}^q$ is symmetric with zero trace; if it has a non-zero entry, its eigenvalues sum to zero and are not all zero, so at least one is negative. For non-negative profiles summing to one, $\|\boldsymbol{a}_i^{row} - \boldsymbol{a}_j^{row}\|^2 = \|\boldsymbol{a}_i^{row}\|^2 + \|\boldsymbol{a}_j^{row}\|^2 - 2\langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row}\rangle \leq 2$, since $\|\boldsymbol{a}_i^{row}\|^2 = \sum_{s\in\Phi} \alpha_{is}^2 \leq \left(\sum_{s\in\Phi} \alpha_{is}\right)^2 = 1$ and $\langle \boldsymbol{a}_i^{row}, \boldsymbol{a}_j^{row}\rangle \geq 0$, and analogously for columns. The third claim follows from eqs. (1)-(3). The fourth claim is immediate from the definition of a normal matrix: if $\boldsymbol{\mathcal{A}}$ is normal, then $\boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{A}}' = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$. ■

The final property formalizes that the same Input-Output structure places a pair of sectors at different horizontal positions depending on whether one looks at their suppliers or buyers. Their relation is uniquely determined if the two buy and sell the same relative quantity of intermediate inputs with a common third sector, and their input demand and input supply co-exposures coincide. Absolute quantities may differ; what matters is that, for each sector individually, its purchases or sales with the common sector match in own-share relative terms. The first two properties, instead, emerge in Examples 2-3 and Figure 1. Proximity is the object of the theory (larger means closer); distance is a metric used in Sections 4-5 to sort sector pairs (larger means farther). The two are inversely related for given concentration of the two profiles, as in eq. (2).

2.2. STYLIZED ECONOMIES AND HORIZONTAL GEOMETRY

Figure 2 illustrates how the vertical and horizontal dimensions combine, by showing the economic relevance of the network's horizontal geometry through highly stylized production structures. I consider three benchmark configurations: *(i) sequential economies*, where connections



(A) SEQUENTIAL (B) RHOMBOID (C) STAR

FIGURE 2: BENCHMARK ECONOMIES

Note: the figure illustrates some benchmark economic systems and their associated horizontal geometry (dashed) highlighting common Input-Output structures (solid). Sectors are in yellow, while the teal circles with C represent the final consumer.

occurs purely along a vertical chain (sector-by-sector towards the final consumer) or, alternatively, purely horizontally (distinct sectors supplying only the final consumer); *(ii) rhomboid economies*, where sectors depend on a common upstream supplier while simultaneously selling to the final consumer; and *(iii) star economies*, with all sectors mutually connected while serving the final consumer.

Sequential economies.– Consider the "snake" in the left graph of Figure 2a. Connections are purely vertical, the upstream commodities flow sequentially along the supply chain, with no scope for horizontal interdependencies. By contrast, in the "half-spider" on the right graph, sectors sell directly to the final consumer without trading among themselves. This structure is a characteristic of standard multi-sector models with no inter-sectoral trade, and supply-based horizontal complementarities are central: even without explicit trade linkages, paired sectors are tied together by their shared exposure to final demand. Adjustments in one sector reshapes consumers' expenditure across all others, moving horizontally through final demand reallocation and generating simultaneous responses absent any network structure.

Rhomboid economies.– A simple network in Figure 2b combines both sequential economies into a unique structure. This configuration blends vertical and horizontal propagation, as commodities travel vertically (via sector-specific supply chain) and horizontally across sectors simultaneously exposed to common supplier and final demand. Rhomboid economies thus epitomize both types of horizontal complementarities, as even sectors not directly trading with each other become tightly synchronized through their shared dependence on upstream and downstream markets.

Star economies.– Figure 2c classifies three archetypes. In fully symmetric and balanced star networks, where all sectors buy and sell with identical intensity, network distances collapse to zero: propagation is only vertical, but the network structure becomes irrelevant as sectoral shocks average out (*e.g.*, Lucas 1977; Acemoglu, Carvalho, et al. 2012). In partially symmetric star networks, trade intensities in buying and selling are equal but sector-specific; vertical and horizontal dimensions coexist, though the two distance measures coincide. Finally, asymmetric star networks (*i.e.*, different trade intensities across sectors) generate parallel vertical and horizontal channels, with comovement emerging endogenously from network distances: horizontal complementarities in production and shared exposure to final demand weave a dense web of

horizontal interdependencies, and sectors with differing upstream and downstream connections experience unrelated effects.

These illustrative examples represent extreme cases. In more realistic, highly interconnected systems, the boundaries between these benchmark economies blur: vertical supply chains coexist with horizontal interdependencies through shared buyers and suppliers, and sectors participate simultaneously in multiple propagation layers.

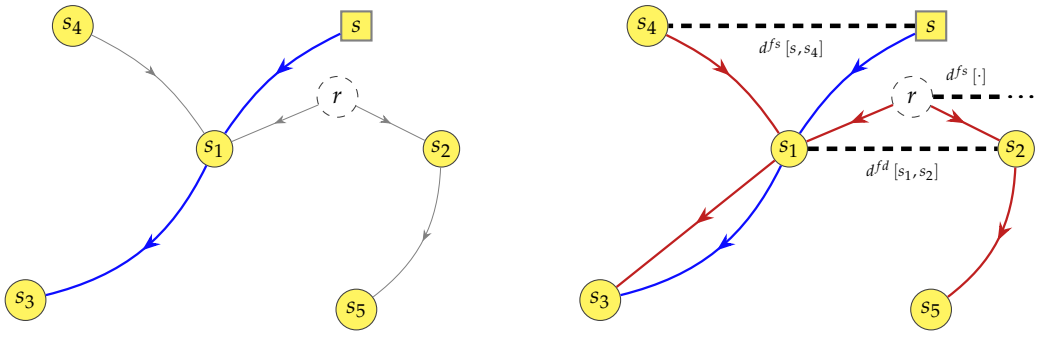
3. An Inspection into the Horizontal Transmission

Outlined horizontal geometry introduces novel implications for the propagation of asymmetric, independent and idiosyncratic shocks across the network. Crucially, the transmission depends not only on the size of existing connections among sectors, but also on their network "economic" distance: in Figure 3, a shock originating in a given sector propagates vertically through downstream buyers along the supply chain, while concurrently transmitting horizontally.

Example 4 (Propagating behaviour under horizontal distances) *Consider an economy populated by* $\{s, s_1, s_2, s_3, s_4, s_5, r\} \in \Phi$ *sectors, with* r *representing the rest of the Input-Output network. Some trade with others, while some others do not:* s_1 *is buying from* $\{s, s_4, r\}$ *while selling to* s_3*;* s_2 *trades from* r *and to* s_5*. Suppose a supply shock to* s*. In Panel 3a, the shock propagates downstream: it directly affects* s_1 *and indirectly reaches* s_3*, yielding a purely vertical transmission with others unaffected.*[5] *Unknown remain the potential effects of induced adjustments in* s_1*, even though it shares linkages with others. Panel 3b illustrates both vertical and horizontal propagation: a shock to* s *not only affects its immediate trading connections but also alters its interaction with* s_4*: changes in* s_1 *triggered by the initial shock generate an horizontal effect that links* $\{s, s_4\}$ *through their common supply-based connection with* s_1*. By analogy, this reasoning falls under supply distances of* $\{s, s_4\}$ *with the rest of other sectors-*r *selling to* s_1*,* $d^{fs}[\cdot]$*. With due demand-based scheme, analogous is the logic underlying* $d^{fd}[s_1, s_2]$*, simultaneously buying from* r*. Subsequent adjustments in* $\{s_4, r\}$ *feed back into* s_1*, compounding the indirect propagation towards* s_3*, and into* s_2*: reverberating the initial shock on* s*, via* $\{s_1, r\}$ *to* s_2*, an additional vertical effect further downstream, from* s_2 *to* s_5*, is generated.*

Figure 3 summarizes the guiding question of the paper. A shock to sector-s propagates downstream along supply chains (Panel 3a). Along the way it changes the prices and quantities faced by the buyers and suppliers of every sector it reaches, and thereby the conditions of sectors that share those buyers and suppliers (Panel 3b). This section characterizes the sign and the magnitude of these horizontal effects.

[5] Upstream responses vanish because price and quantity effects offset each other (*e.g.*, Acemoglu, Carvalho, et al. 2012). Yet, the horizontal mechanism uncovers these latent transmission channels via common network structures instead of tracking non-linearities in production (*e.g.*, Baqaee and Farhi 2019). Note that the example is presented as the more intuitive case of a sectoral supply shock propagating downstream but the specular, reversed logic applies in the case of a demand shock.



(A) VERTICALITY (B) VERTICALITY AND DISTANCES

FIGURE 3: HORIZONTALITY OF A TRANSMISSION

Note: stylized example of how a sectoral shock propagates across the production network. Coloured and oriented bold-curves trace the propagation mechanism: vertical transmission is in blue, while additional transmission paths from the horizontal geometry are in red. Oriented thin-grey curves indicate further existing Input-Output connections, and network economic distances through common nodes are represented by dashed lines. The underlying network structure is identical in both panels. Panel 3a shows a purely vertical propagation of a shock to sector-i, following standard sectoral supply-chain logics. Panel 3b depicts a richer mechanism, where the shock also spreads horizontally through sectors sharing common upstream suppliers or downstream buyers.

3.1. DISTANCE-LEAD COMOVEMENT

A shadowed implication of the framework developed so far is that intermediate inputs generally function as substitutes, allowing sectors to freely adjust their input mix in response to shocks. However, the horizontal geometry of a network can embed two aspects: *(i)* vertical propagation through the supply chain may remain the dominant channel, and what may appear as a horizontal transmission can in fact be the by-product of vertical propagation; and *(ii)* intermediate inputs complementarity in production may constrain the transmission of shocks. This tension between horizontal complementarities at the network level and substitutability at the input level sets the stage for a more precise characterization of how sectoral comovement occurs. In the context of factor input supply, alleviating the tension is doable since competition for common downstream buyers yields a more direct and unambiguous form of horizontal interdependence. Conversely, the structure of upstream connections renders demand-based linkages ambiguous to interpret, since the coexistence of a dual transmission makes the direction and the strength of induced comovement subtle. Below, comovement is characterised at a single shared node – *i.e.*, at "first round": the shock is traced to the shared partner and back, holding fixed the prices of all others – and shows that demand and supply distances differ along these two dimensions.

Two local mechanisms.– The production function is indeed characterised by a Constant Elasticity of Substitution (CES) aggregator over the intermediate bundle, with substitution $\sigma_i > 0$ between intermediate inputs. Since the household has Cobb-Douglas preferences over commodities, with constant expenditure shares $\{\omega_i\}_{i\in\Phi}$, all horizontal transmission runs through the production network. Throughout, refer to representative household with index-0.

Let $\widehat{\alpha}_{ij} = \frac{\alpha_{ij}}{1-\alpha_i}$ denote the share of sector-j in sector-i's intermediate spending, $\widehat{\boldsymbol{a}}_i^{row} = \left[\widehat{\alpha}_{i1}, \ldots, \widehat{\alpha}_{im}\right] = \frac{\boldsymbol{a}_i^{row}}{1-\alpha_i}$ the corresponding row vector – buyer-i's intermediate portfolio, whose entries sum to one –, and $\boldsymbol{C}_i = \text{Diag}\left[\widehat{\boldsymbol{a}}_i^{row}\right] - \left(\widehat{\boldsymbol{a}}_i^{row}\right)'\widehat{\boldsymbol{a}}_i^{row}$ its covariance matrix, where $\text{Diag}\left[\cdot\right]$ is the $m \times m$ diagonal matrix with the entries of its argument on its diagonal: for any $\boldsymbol{u}$ and $\boldsymbol{v}$, then $\text{Var}_{\widehat{\boldsymbol{a}}_i^{row}}\left[\boldsymbol{u}\right] = \boldsymbol{u}'\boldsymbol{C}_i\boldsymbol{u}$ and $\text{Cov}_{\widehat{\boldsymbol{a}}_i^{row}}\left[\boldsymbol{u}, \boldsymbol{v}\right] = \boldsymbol{u}'\boldsymbol{C}_i\boldsymbol{v}$ are the variance and covariance of their

entries across buyer-i's inputs, weighted by the shares. Let also η_i to be the elasticity of demand faced by buyer-i for its own output, so that $d\log y_i = -\eta_i\, d\log p_i$, and let the marginal cost of a shared supplier-s have elasticity $\varphi_s = \frac{\partial \log p_s}{\partial \log y_s}$. Moreover, $e_{si} = \frac{\lambda_i\, \alpha_{is}}{\lambda_s}$ is the share of s's sales to buyer-i, and consider an exogenous expansion in the scale of buyer-j.

PROPOSITION 1 (Common-supplier transmission) *At first round, the purchases of downstream sector-i from the shared upstream supplier-s respond as*

$$\delta_{s\to}\,[i,j] = \frac{\partial \log x_{is}}{\partial\, \log x_{js}} = -\,\chi_{is}\;\widetilde{\varphi}_s\; e_{sj}$$

with $\chi_{is} \;=\; \sigma_i\big(1-\widehat{\alpha}_{is}\big) + \alpha_i\,\widehat{\alpha}_{is} + \eta_i\,\alpha_{is} \;>\; 0$ *and* $\widetilde{\varphi}_s \;=\; \varphi_s \big/ \big[1 + \varphi_s\big(e_{s0} + \sum_{i\in\Phi} e_{si}\chi_{is}\big)\big]$. *Sector-$i$'s output responds as* $\frac{\partial \log y_i}{\partial\, \log x_{js}} = -\,\eta_i\,\lambda_j \sum_{s\in\Phi} \alpha_{is}\,\frac{\widetilde{\varphi}_s}{\lambda_s}\,\alpha_{js}$.

Proof: let me derive each step separately. Start by fixing the prices of all commodities other than s, and let $\mathbf{1}_{ij}$ equal one if $i=j$ and zero otherwise. *Step 1.* Each buyer-i of s prices at unit cost, so that $d\log p_i = \alpha_{is}\, d\log p_s$. *Step 2.* CES cost minimization implies $d\log \widehat{\alpha}_{is} = (1-\sigma_i)\,(1-\widehat{\alpha}_{is})\, d\log p_s$, and $d\log\big[p_i y_i\big] = (1-\eta_i)\,\alpha_{is}\, d\log p_s + \mathbf{1}_{ij}\, d\log x_{js}$. *Step 3.* Subtracting $d\log p_s$ from the sum of the two terms, and using $\alpha_{is} = (1-\alpha_i)\,\widehat{\alpha}_{is}$, gives $d\log x_{is} = \mathbf{1}_{ij}\, d\log x_{js} - \chi_{is}\, d\log p_s$. *Step 4.* Market clearing, $y_s = c_s + \sum_{i\in\Phi} x_{is}$, with final demand $c_s = \frac{\omega_s}{p_s} wN$, gives $d\log y_s = -\mu_{s0}\, d\log p_s + \sum_{i\in\Phi} e_{si}\, d\log x_{is}$, and $d\log p_s = \varphi_s\, d\log y_s$. Combining, $d\log p_s = \varphi_s\big[e_{sj}\, d\log x_{js} - d\log p_s\big(e_{s0} + \sum_{i\in\Phi}\mu_{si}\chi_{is}\big)\big]$, so that, provided a stability condition $1 + \varphi_s\big(e_{s0} + \sum_{i\in\Phi}\mu_{si}\chi_{is}\big) > 0$, then $d\log p_s = \widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$. *Step 5.* By Step 3, $d\log x_{is} = -\chi_{is}\,\widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$ for $i \neq j$. For the second part, the expansion of j raises its demand for every supplier-s by $d\log x_{js}$, so that, treating each supplier's market as above, $d\log p_s = \widetilde{\varphi}_s\, e_{sj}\, d\log x_{js}$ for every s; summing over all shared suppliers, the unit cost of i changes by $\sum_{s\in\Phi}\alpha_{is}\, d\log p_s$, its output by $-\eta_i$ times this amount, and $e_{sj} = \frac{\lambda_j \alpha_{js}}{\lambda_s}$. ■

Opposite comovement between downstream sectors arises through upstream adjustments in supply. Intuitively, when the demand for j's output expands, it purchases more from s. If s faces rising marginal costs, its price increases (with pass-through $\widetilde{\varphi}_s e_{sj}$), raising the i's unit cost by $\alpha_{is}\, d\log p_s$ and lowering its output by η_i. Every other buyer of s contracts, both by substituting away from s (at rate σ_i) and by shrinking its own scale (at rate η_i). From i's viewpoint, the shock is a vertical cost shock originated at s; it is horizontal only in that it was caused by j, whose sign is $-\,\text{Sign}\,[\varphi_s]$. Hence, the common-supplier channel is horizontal in origin but vertical in mechanism: from i's viewpoint, the shock is a vertical cost shock originated at s; it is horizontal only in that it was caused by j. Demand-driven comovement reflects not automatic crowding-out, but the interplay of upstream price pass-through (from vertical propagation), and downstream demand adjustments (indirectly generated by horizontal complementarities). Both features make comovement between sectors that share suppliers hard to read.

The common-buyer channel behaves differently. Let s be a common buyer of i and j, and consider a shock that changes the price of j, holding other prices fixed. Let $p_i y_i$ denoting the total sales of i, the following holds.

PROPOSITION 2 (Common-buyer transmission) *At first round, the sales of upstream sector-i to the shared downstream buyer-s respond as*

$$\delta_{\to s}\,[i,j] = \frac{\partial \log\big[p_i\, x_{si}\big]}{\partial \log p_j} = \big[\,(\sigma_s - 1) + (1-\alpha_s)\,(1-\eta_s)\,\big]\,\widehat{\alpha}_{sj}$$

and, summing over the common buyers $\forall s \neq i,j$, the corresponding component of i's total sales responds as $\frac{\partial \log p_i y_i}{\partial \log p_j} = \frac{1}{\lambda_i}\sum_{s\in\Phi_{i,j}} \lambda_s\big[\,(\sigma_s-1) + (1-\alpha_s)\,(1-\eta_s)\,\big]\,\alpha_{si}\,\widehat{\alpha}_{sj}$.

Proof: fix all prices other than p_j. The price of s changes by $d\log p_s = \alpha_{sj}\, d\log p_j = (1-\alpha_s)\,\widehat{\alpha}_{sj}\, d\log p_j$. Hence, the scale term is $d\log[p_s y_s] = (1-\eta_s)(1-\alpha_s)\,\widehat{\alpha}_{sj}\, d\log p_j$. Under CES and a common buyer, with $d\log p_i = 0$, the substitution term is $d\log\widehat{\alpha}_{si} = (\sigma_s - 1)\,\widehat{\alpha}_{sj}\, d\log p_j$. Adding the two terms gives the first part. Total sales are $p_i y_i = p_i c_i + \sum_{s\in\Phi} p_i x_{si}$, where final demand spending on i is a constant share of income and does not respond to p_j, and the share of i's sales going to s is $\frac{p_i x_{si}}{p_i y_i} = \lambda_s \frac{\alpha_{si}}{\lambda_i}$; summing with these weights over the common buyers $\forall s \neq i, j$ gives the second part. ■

Opposite comovement between upstream suppliers mechanically arises from competition for downstream buyers. The common-buyer channel separates cleanly into a horizontal and a vertical component. The first term in square brackets is competition between upstream suppliers for the intermediate budget of their common buyer: when the price of j falls, s reallocates a given budget across its suppliers. This downstream reallocation necessarily comes at the expense of the competing supplier-i and has no vertical counterpart. The second term is vertical: the lower price of j lowers the price of s and changes its scale, and hence its demand for all of its inputs in the same proportion, whether or not they compete with j. With the buyer's budget given ($\eta_s = 1$), the vertical term vanishes and transmission is purely horizontal: the rivals' sales to s move by $(\sigma_s - 1)\,\widehat{\alpha}_{sj}\, d\log p_j$ and j's own by $(1-\sigma_s)\left(1-\widehat{\alpha}_{sj}\right) d\log p_j$. Upstream reallocation of downstream expenditure is then zero-sum: the shocked sector and its rivals move in opposite directions whatever $\sigma_s \neq 1$. Hence the response of rivals to a shock in an horizontally-related sector reveals whether shared buyers treat their intermediate inputs as substitutes or complements.

Discussion.– A networked economy inherently embeds complementarities in intermediate inputs, as sectors depend on overlapping sets of buyers and suppliers. In the demand-distance case, comovement is ambiguous: it reflects a combination of vertical propagation along the supply chain and horizontal complementarities from shared suppliers. In the supply-distance case, comovement is transparent: the horizontal transmission directly arises from the reallocation of inputs to adjustments in relative prices, and the interaction between the two dimensions is mechanically determined. Importantly, the discrepancy between the ambiguous effects in demand and the clear responses in supply is in accordance with non-linear network theories (*e.g.*, Atalay 2017; Baqaee and Farhi 2019), where downstream complementarities – and hence complementarities between *purchased* intermediate inputs at a common node – matter the most. This dimension is embodied into the supply-based distances through the downstream pass-through effect, which create horizontal interdependencies among sectors selling to the same buyers: when one supplier experiences a positive shock, downstream complementarities of common buyers cause other suppliers to adjust their production in response, and the propagation operates directly through the interdependence of upstream inputs used together in downstream production. Such amplification is largely absent under demand-based distances, since the upstream pass-through reflects only substitution patterns across common upstream suppliers.

The propositions deliberately hold fixed everything beyond the common node. Two questions therefore remain: whether the common-buyer mechanism survive when shocks travel along the entire length of supply chains and downstream buyers' budgets adjust, and what does it imply for sectoral employment (the other input in production) and aggregate output.

3.2. GENERAL EQUILIBRIUM: WIDESPREAD HORIZONTAL GEOMETRY

In general equilibrium, across the complete Input-Output system, three movements happen at once: two refer to the "(anti-)cascade effects" operating vertically; the other is fully horizontal. A shock initially travels along supply chains, lowering the cost of every commodity that embodies the commodity produced by the shocked sector. Each downstream buyer then reallocates its intermediate expenditure across upstream inputs whose relative costs have moved unevenly (Proposition 2). Finally, due to these adjustments, the size of each buyer's budget is itself endogenous: travelling back upstream, a supplier's sales are what its own buyers decide to spend on it. The next result shows that, in the benchmark economy (*i.e*, exploiting the full production network structure), these three forces organize into a product of three matrices.

THEOREM 2 (On the vertical-horizontal factorization) *To first order, Domar weights respond to sectoral shocks as*

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}\, d\log \mathbf{z} \qquad \textit{with} \quad \boldsymbol{\mathcal{H}} = \sum_{s\in\Phi} (\sigma_s - 1)\, \lambda_s\, (1-\alpha_s)\, \boldsymbol{\mathcal{C}}_s$$

Equivalently, $\frac{\partial \lambda_j}{\partial \log z_i} = \sum_{s\in\Phi} (\sigma_s - 1)\, \lambda_s\, (1-\alpha_s)\, \mathrm{Cov}_{\widehat{\boldsymbol{a}}_s^{row}} \left[\boldsymbol{\ell}_i^{col}, \boldsymbol{\ell}_j^{col}\right]$*, where* $\boldsymbol{\ell}_i^{col}$ *is the i-th column of* $\boldsymbol{\mathcal{L}}$ *and the covariance is across buyer-s's intermediate inputs with probabilities* $\widehat{\boldsymbol{a}}_s^{row}$*.*

Proof: Domar weights $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$ solve $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$ at every equilibrium. Differentiating, with $\boldsymbol{\omega}$ constant, $d\boldsymbol{\lambda}' = \boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}\, \boldsymbol{\mathcal{L}}$: the vector $\boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}$ is the change in spending on each commodity due to buyers changing their shares at given size, and $\boldsymbol{\mathcal{L}}$ passes it upstream. Let $\boldsymbol{\pi} = -d\log \boldsymbol{p} = \boldsymbol{\mathcal{L}}\, d\log \mathbf{z}$ be the cost reduction of each commodity. Since $\alpha_{sj} = (1-\alpha_s)\widehat{\alpha}_{sj}$ with α_s constant, CES minimization gives $\lambda_s\, d\alpha_{sj} = (\sigma_s - 1)\lambda_s(1-\alpha_s)\left(\widehat{\alpha}_{sj}\,\pi_j - \widehat{\alpha}_{sj}\,\widehat{\boldsymbol{a}}_s^{row}\boldsymbol{\pi}\right)$. Summing over buyers, $\left[\boldsymbol{\lambda}' d\boldsymbol{\mathcal{A}}\right]_j = \left[\boldsymbol{\pi}'\boldsymbol{\mathcal{H}}\right]_j$, because $\left[\boldsymbol{\pi}'\,\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right]\right]_j = \pi_j\widehat{\alpha}_{sj}$ and $\left[\boldsymbol{\pi}'\left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}\right]_j = \left(\widehat{\boldsymbol{a}}_s^{row}\boldsymbol{\pi}\right)\widehat{\alpha}_{sj}$. Hence $d\boldsymbol{\lambda}' = d\log\mathbf{z}'\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ and, $\boldsymbol{\mathcal{H}}$ being symmetric, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\, d\log\mathbf{z}$. Covariance follows from $\left[\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\right]_{ji} = \sum_{s\in\Phi}(\sigma_s-1)\lambda_s(1-\alpha_s)\left(\boldsymbol{\ell}_j^{col}\right)'\boldsymbol{\mathcal{C}}_s\boldsymbol{\ell}_i^{col}$. Rows of $\boldsymbol{\mathcal{C}}_s$ sum to zero, since $\boldsymbol{\mathcal{C}}_s\mathbf{1} = \left(\widehat{\boldsymbol{a}}_s^{row}\right)' - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\left(\widehat{\boldsymbol{a}}_s^{row}\mathbf{1}\right) = \mathbf{0}$; hence so do rows of $\boldsymbol{\mathcal{H}}$. Finally, since $(1-\alpha_s)\,\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] = \mathrm{Diag}\left[\boldsymbol{a}_s^{row}\right]$ and $(1-\alpha_s)\left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row} = \frac{\left(\boldsymbol{a}_s^{row}\right)'\boldsymbol{a}_s^{row}}{1-\alpha_s}$, then $\boldsymbol{\mathcal{H}} = \mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{w}\right] - \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{A}}$ with $w_s = (\sigma_s-1)\lambda_s$ and $\boldsymbol{\mathcal{W}} = \mathrm{Diag}\left[(\sigma_s-1)\frac{\lambda_s}{1-\alpha_s}\right]_{s\in\Phi}$, so that $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\,\mathrm{Diag}\left[\boldsymbol{\mathcal{A}}'\boldsymbol{w}\right]\boldsymbol{\mathcal{L}} - \boldsymbol{\mathcal{T}}'\boldsymbol{\mathcal{W}}\boldsymbol{\mathcal{T}}$, with $\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$. ■

Reading from right to left, the vertical-horizontal factorization retraces the three movements. The Leontief inverse representation $\boldsymbol{\mathcal{L}}$ transports a shock vertically, cascading from sector-i to the cost of every downstream commodity embedding it, $\boldsymbol{\ell}_i^{col}$. The horizontal operator $\boldsymbol{\mathcal{H}}$ then reallocates each buyer's intermediate budget, $\lambda_s(1-\alpha_s)$, according to how unevenly the costs of its intermediate inputs have moved: it is the substitution term $(\sigma_s - 1)$ of Proposition 2, applied at every downstream and further downstream buyer at once. Finally, $\boldsymbol{\mathcal{L}}'$ transports the change in the demand for each commodity back up to the suppliers of that commodity: this anti-cascade adjustment is how the scale effects of Proposition 2 are resolved in general equilibrium, where a buyer's budget is not given but equals what its own buyers spend on it. The vertical structure enters only through $\boldsymbol{\mathcal{L}}$; everything that happens at shared buyers is in $\boldsymbol{\mathcal{H}}$. Its off-diagonal entries display the horizontal reallocation in sector-j after a shock in i,

$$\mathcal{H}_{j,i} = -\sum_{s\in\Phi} (\sigma_s - 1)\, \lambda_s\, (1-\alpha_s)\, \widehat{\alpha}_{sj}\, \widehat{\alpha}_{si} \qquad \text{for} \quad j \neq i$$

as common-buyer economic proximities weighted by the size of each buyer's intermediate budget and by its substitutability: $\boldsymbol{\mathcal{B}}^{fs} = \boldsymbol{\mathcal{A}}'\boldsymbol{\mathcal{A}}$ is their unweighted counterpart, and the first-tier term coincides with Proposition 2 with the buyer's budget given. Moreover, each row of

$\mathcal{H}$ sums to zero: within every buyer's bundle, what one supplier gains its rivals lose. Writing $\mathcal{L} = I + (\mathcal{L} - I)$, where $\mathcal{L} - I = \mathcal{A} + \mathcal{A}^2 + \cdots + \mathcal{A}^k$ collects the requirements embodied in intermediate inputs at each stage-k, and the reallocation operator $\mathcal{R} = \mathcal{L}'\mathcal{H}\mathcal{L}$ decomposes as $\mathcal{R} = \mathcal{H} + \left[(\mathcal{L} - I)' \mathcal{H} + \mathcal{H} (\mathcal{L} - I) \right] + (\mathcal{L} - I)' \mathcal{H} (\mathcal{L} - I)$: the response of sector-j to a shock in sector-i is "direct rivalry" between the two sectors at their common buyers (between sectors i and j both supplying at s), plus "indirect rivalry" between each of these sectors and the other sectors' commodities embodied in their respective intermediate inputs bundle (between j's output and the intermediate inputs required by i, and vice versa), plus "embodied rivalry" between the intermediate-input bundles embodying the two sectors (between the intermediate inputs being the production requirements of j and i). From the proof, the $\mathcal{R}$'s "rivalry" component is $-\mathcal{T}'\mathcal{W}\mathcal{T}$, with $\mathcal{T} = \mathcal{A}\mathcal{L} = \mathcal{L} - I$ being the $m \times m$ matrix of buyers' total requirements and the $m \times m$ matrix $\mathcal{W} = \text{Diag}\left[(\sigma_s - 1) \frac{\lambda_s}{1-\alpha_s} \right]_{s \in \Phi}$: two sectors are rivals when the same buyers require both, directly or indirectly. This is the logical step linking theory and measurement of Sections 4-5: the operator that governs the horizontal transmission is a weighted common-buyer Gram matrix computed on total requirements. The covariance form is similar to the second-order network centrality covariance operator of Baqaee and Farhi's (2019) Proposition 7 which, due to the network's horizontal geometry, now applies at first-order to each downstream buyer's intermediate nest: the factorization in Theorem 2 makes it directly readable along the co-existing network's vertical and horizontal dimensions in $\mathcal{L}'\mathcal{H}\mathcal{L}$, which turns the second-order into a measurable common-buyer Gram matrix.

The theorem therefore separates the network response to a shock into two conceptually distinct dimensions. The vertical structure determines which commodities are affected through the propagation of costs along supply chains, while the horizontal structure determines how these changes in relative costs reallocate downstream expenditure across common upstream suppliers. This decomposition yields two implications that are useful for interpreting the role of horizontal geometry. The first concerns the direction of these reallocations: whether common-buyer relationships generate rivalry or complementarity. The second concerns their informational content: horizontal geometry is not encoded in the Domar weights and therefore cannot be recovered from the vertical structure alone.

COROLLARY 1 (Signs) *Let sector-s to be the shared node of sectors $\{i, j\}$. Then:*

- *under Cobb-Douglas intermediates ($\sigma_s = 1$) it holds $\mathcal{H} = \mathbf{0}$;*
- *if $\sigma_s \geq 1$ ($\sigma_s \leq 1$), $\forall s \in \Phi$, then $\mathcal{H}$ and $\mathcal{R}$ are positive (negative) semi-definite, and all off-diagonal entries of $\mathcal{H}$ are non-positive (non-negative), and first-tier co-suppliers are rivals (complements) in sales.*

Proof: if $\sigma_s = 1$ for all $s \in \Phi$, then downstream expenditure shares do not adjust, and every weight in $\mathcal{H}$ is zero. Moreover, each $\mathcal{C}_s$ is positive semi-definite, since $\boldsymbol{\xi}'\mathcal{C}_s\boldsymbol{\xi} = \text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}[\boldsymbol{\xi}] \geq 0$; its diagonal entries, $\widehat{\alpha}_{si}(1 - \widehat{\alpha}_{si})$, are non-negative and its off-diagonal entries, $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$, non-positive. A combination with non-negative (non-positive) weights inherits these properties (with reversed signs), and $\mathcal{R} = \mathcal{L}'\mathcal{H}\mathcal{L}$ inherits semi-definiteness by congruence. ■

Corollary 1 turns Proposition 2 into a general-equilibrium statement and makes the first implication precise. The horizontal operator inherits a transparent sign structure from the CES

substitution mechanism.[6] With Cobb-Douglas intermediate bundle, expenditure shares on inputs are fixed and invariant to the shocks (*e.g.*, Carvalho and Tahbaz-Salehi 2019) and horizontal transmission disappears altogether. With elasticities above (below) one, co-suppliers are rivals (complements): a shock raises the sales weight of some suppliers at the expense of their horizontal common-buyer neighbours, or vice versa. In both cases sectoral reallocations have opposite signs, and the shock moves the first-tier Domar weights of a sector and of its common-buyer neighbours in opposite directions, whatever the size of the elasticities: a shock reallocates sales shares between a sector and its common-buyer neighbours, and the size of the elasticity only decide who gains across horizontally-related sectors.

COROLLARY 2 (Domar weights do not span horizontal geometry) *There exist networked economies with identical elasticities and identical Domar weights – hence identical first-order responses of aggregate output to any vector of shocks – in which the same pair of sectors are rivals in one economy and unrelated in the other.*

Proof: by market clearing, Domar weights $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$ depend on the flows $\lambda_s \alpha_{si}$ only through buyers' budgets, $\lambda_s (1 - \alpha_s)$, and suppliers' intermediate sales, $\lambda_i - \omega_i$. Let s, with $\lambda_s > 0$ and $\sigma_s \neq 1$, be the only common buyer of $\{i, j\}$, and m a buyer that spends at least $\lambda_s \alpha_{sj}$ on some $(m - 2) \notin \{i, j\}$ but nothing on i. Shifting s's purchases from j to $m - 2$, and the same value of m's purchases from $m - 2$ to j, leaves these margins, hence Domar weights and $d \log Y = \boldsymbol{\lambda}' d \log \mathbf{z}$, unchanged, but i and j no longer share a buyer: $\mathcal{H}_{i,j}$ goes from $-(\sigma_s - 1) \lambda_s (1 - \alpha_s) \widehat{\alpha}_{si} \widehat{\alpha}_{sj} \neq 0$ to zero, and so does $\mathcal{R}_{i,j} = \mathcal{H}_{i,j}$ if $\{i, j, m - 2\}$ use only labour and sell only to s and m, which sell only to final demand. ■

Corollary 2 establishes the second implication by providing a precise sense in which horizontal geometry is a dimension of the network distinct from the vertical one: the sufficient statistics for first-order aggregate effects, the Domar weights, carry no information on it. Domar weights summarize the vertical importance of sectors for aggregate output, but they contain no information about how sectors are arranged across common buyers. Consequently, two economies can have the same elasticities and exactly the same Domar weights – and hence the same first-order aggregate response to every shock – while exhibiting different horizontal relationships between sectors. Hence, the corollary postulates a dimensional separation: that, within an Input-Output economy, the horizontal geometry of the network – captured by demand- and supply-driven shared relationships – remains outside the domain of its trade intensities, encoded in the Input-Output matrix. This distinction is crucial for understanding the economics of production networks: while the vertical structure reflects the strength of sectoral relationships, the horizontal structure identifies demand/supply complementarities across sectors related through common upstream suppliers or downstream buyers. In other words, by separating the horizontal from the vertical geometry, it clarifies that whatever distance matrix is not merely a transformation of the Input-Output matrix, but a complementary object that reveals otherwise hidden propagation channels.

[6] When also households have CES preferences over final goods of elasticity $\sigma_0 \neq 1$, final demand would become a further common buyer, of size one and with no labour share, and each result below would gain a household term with weight $\sigma_0 - 1$: $\boldsymbol{\mathcal{H}}$ would add $(\sigma_0 - 1) \left\{ \text{Diag}\left[\boldsymbol{\omega}\right] - \boldsymbol{\omega}\boldsymbol{\omega}' \right\}$, the rivals' sales to final demand in Proposition 2 would move by $(\sigma_0 - 1)\, \omega_j\, d \log p_j$, the unweighted counterpart of $\boldsymbol{\mathcal{H}}$ would become $\boldsymbol{\mathcal{B}}^{fs} + \boldsymbol{\omega}\boldsymbol{\omega}'$, and the sign conditions of Corollary 1 would extend to σ_0. The household channel is active even without intermediate inputs – it is the horizontal channel of standard multi-sector models – and adds to the second-order term of Theorem 3 a component with the sign of $\sigma_0 - 1$ and absolute value at least $\frac{1}{2}\nu^2 |\sigma_0 - 1| \left(1 - \|\boldsymbol{\lambda}\|^2\right)$. It reinforces, and does not alter, the conclusions on the supply side. See more in Supplemental Appendix A.

Theorem 2 also explains why horizontal relations become increasingly important as the network expands. At the first tier, a shock to sector-i travels vertically to each buyer of i, through the positive entries of $\boldsymbol{\mathcal{A}}$, and horizontally to each sector with which i shares a buyer, through the non-zero off-diagonal entries of $\boldsymbol{\mathcal{H}}$. The two sets of relations grow at different rates, because vertical relations are shared: when a buyer adds a supplier, the new supplier becomes a rival of every supplier the buyer already has.

COROLLARY 3 (Horizontal proliferation) *It holds $\mathcal{H}_{i,j} \neq 0$ if and only if $b^{fs}[i,j] > 0$, and:*

- *if buyer-s starts purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ gains one positive entry, α_{si}, while $\mathcal{H}_{i,j}$ gains a non-zero term for every $j \in \Phi_i$ with $\alpha_{sj} > 0$;*
- *if buyer-s stops purchasing from supplier-i, $\boldsymbol{\mathcal{A}}$ loses one positive entry, while $\mathcal{H}_{i,j}$ vanishes only for the suppliers $j \in \Phi_i$ whose only common buyer with i was s;*
- *relative to buyer-s's intermediate purchases, $\lambda_s(1-\alpha_s)$, the rivalry that s generates in $\boldsymbol{\mathcal{H}}$ is $|\sigma_s - 1|\left(1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2\right)$, where $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ is the Herfindahl index of s's purchases.*

Proof: by Theorem 2, for $i \neq j$, $\mathcal{H}_{i,j} = -\sum_{s\in\Phi}(\sigma_s - 1)\lambda_s(1-\alpha_s)\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$. A buyer with $\widehat{\alpha}_{si}\widehat{\alpha}_{sj} > 0$ purchases intermediate inputs, so that all non-zero terms share the same sign and cannot offset each other: $\mathcal{H}_{i,j} \neq 0$ if and only if $\alpha_{si}\alpha_{sj} > 0$ for some s, that is, for $b^{fs}[i,j] = \sum_{s\in\Phi}\alpha_{si}\alpha_{sj} > 0$. For the first claim, if α_{si} turns positive, the shares of s's other suppliers remain positive, and $\mathcal{H}_{i,j}$ gains the term $-(\sigma_s - 1)\lambda_s(1-\alpha_s)\widehat{\alpha}_{si}\widehat{\alpha}_{sj} \neq 0$ for every $j \in \Phi_i$ with $\alpha_{sj} > 0$. For the second claim, if α_{si} drops to zero, only the term of buyer-s disappears from $\mathcal{H}_{i,j}$, which therefore remains non-zero if and only if i and j share another buyer. For the third claim, the off-diagonal entries of $\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)'\widehat{\boldsymbol{a}}_s^{row}$ are $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$, and, since $\sum_{i\in\Phi}\widehat{\alpha}_{si} = 1$, their absolute values sum to $\sum_{i\in\Phi}\widehat{\alpha}_{si}\left(1-\widehat{\alpha}_{si}\right) = 1 - \|\widehat{\boldsymbol{a}}_s^{row}\|^2$; multiplying by $|\sigma_s - 1|\lambda_s(1-\alpha_s)$ and dividing by $\lambda_s\sum_{i\in\Phi}\alpha_{si} = \lambda_s(1-\alpha_s)$ gives the result. With equal shares, $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ is the inverse of the number of s's suppliers. ■

Horizontal linkages are built out of vertical ones, but not one for one. Each vertical relation of a buyer supports a horizontal relation with each of the buyer's other suppliers, and each horizontal relation is supported by every buyer that the two sectors share. Counted once for every buyer, horizontal relations grow with the square of buyers' portfolios, vertical relations only linearly. The first claim is the margin of this asymmetry. As the network expands, every new link adds one vertical relation and many horizontal ones, so that horizontal relations proliferate even as supply chains themselves expand. The second is its mirror image. As transactions are removed, vertical relations are lost one at a time, whereas a horizontal relation is lost only when the last buyer that supports it is gone, so that horizontal relations survive sparsification. The same counting applies to common suppliers: when s adds supplier-i, it becomes a neighbour, in $\boldsymbol{\mathcal{B}}^{fd}$, of every other buyer of i. Finally, the last claim shows that the asymmetry is not only a matter of counting: $\|\widehat{\boldsymbol{a}}_s^{row}\|^2$ is zero for a buyer with a single supplier, and rises towards $|\sigma_s - 1|$ as s spreads its purchases evenly over more suppliers. In other words, as a buyer spreads a given intermediate budget over an increasing number of downstream suppliers, the vertical intensity of its purchases does not change, whereas the rivalry it generates among them rises. A sectoral shock reaches the buyers of the shocked sector vertically, but all the other suppliers of these buyers horizontally: in larger and denser networks, shocks travel less as cascades along supply chains, and more as reallocations within clusters of horizontally related sectors.

3.3. SECTORAL EMPLOYMENT AND AGGREGATE OUTPUT

Theorem 2 describes how a shock reallocates sales shares across sectors. I now draw its implications for the two objects: the cross-section of sectoral employment, and aggregate output. A single operator links the two: the matrix $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ that moves Domar weights also moves workers across sectors, so that horizontal transmission becomes directly observable in employment. And since Domar weights are the slopes of aggregate output, the same matrix governs how aggregate output bends when many sectors are hit at once.

Sectoral employment.– Horizontal transmission reallocates a fixed resource across sectors. By property *A1.iii* of Assumption 1, a sector never substitutes between workers and its intermediate bundle as a whole, so that it spends a constant fraction α_i of its revenue on labour. Its employment can therefore move relative to the aggregate only if its Domar weight moves, and Theorem 2 states how Domar weights move.

PROPOSITION 3 (Employment reallocation) *To first order, sectoral employment responds as*

$$d\log \boldsymbol{n} = d\log N\,\mathbf{1} + \boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}^{v}\,d\log \boldsymbol{z} + \boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}^{h}\,d\log \boldsymbol{z}$$

where $\boldsymbol{\Lambda} = \mathrm{Diag}\,[\boldsymbol{\lambda}]$, $\boldsymbol{\mathcal{R}}^{v} = \boldsymbol{\mathcal{L}}'\big(\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}\big)\boldsymbol{\mathcal{L}}$ *and* $\boldsymbol{\mathcal{R}}^{h} = \boldsymbol{\mathcal{L}}'\big(\boldsymbol{\mathcal{H}} - \boldsymbol{I}\circ\boldsymbol{\mathcal{H}}\big)\boldsymbol{\mathcal{L}}$, *with* $\circ$ *the element-wise product, so that* $\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$ *retains the diagonal of* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}}^{v} + \boldsymbol{\mathcal{R}}^{h} = \boldsymbol{\mathcal{R}}$. *Moreover, if* $\sigma_s \geq 1$ *(*$\sigma_s \leq 1$*),* $\forall s \in \Phi$, *all entries of* $\boldsymbol{\mathcal{R}}^{v}$ *and the diagonal entries of* $\boldsymbol{\mathcal{R}}$ *are non-negative (non-positive), and all entries of* $\boldsymbol{\mathcal{R}}^{h}$ *are non-positive (non-negative).*

Proof: employment in sector-i is $n_i = \alpha_i \frac{p_i y_i}{w} = \alpha_i\,\lambda_i\,N$, with α_i constant, so that $d\log n_i = d\log N + \frac{d\lambda_i}{\lambda_i}$, and $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{R}}\,d\log \boldsymbol{z}$ by Theorem 2; splitting $\boldsymbol{\mathcal{H}}$ into its diagonal, $\boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$, and its off-diagonal part, $\boldsymbol{\mathcal{H}} - \boldsymbol{I}\circ\boldsymbol{\mathcal{H}}$, gives the decomposition. Since $\sum_{i\in\Phi}\alpha_i\lambda_i = \frac{1}{wN}\sum_{i\in\Phi} w n_i = 1$ at every equilibrium and $\boldsymbol{\alpha}$ is constant, then $\boldsymbol{\alpha}'d\boldsymbol{\lambda} = 0$ for every shock, that is $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$. If $\sigma_s \geq 1, \forall s \in \Phi$, the diagonal entries of $\boldsymbol{\mathcal{H}}$, $\mathcal{H}_{i,i} = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si}\big(1-\widehat{\alpha}_{si}\big)$, are non-negative, its off-diagonal entries are non-positive (Corollary 1), and $\boldsymbol{\mathcal{L}} \geq \mathbf{0}$. Hence $\mathcal{R}^{v}_{j,i} = \sum_{s\in\Phi}\ell_{sj}\,\mathcal{H}_{s,s}\,\ell_{si} \geq 0$ and $\mathcal{R}^{h}_{j,i} = \sum_{s\neq m}\ell_{sj}\,\mathcal{H}_{s,m}\,\ell_{mi} \leq 0$, while $\mathcal{R}_{i,i} = \big(\boldsymbol{\ell}_i^{col}\big)'\boldsymbol{\mathcal{H}}\,\boldsymbol{\ell}_i^{col} \geq 0$, since $\boldsymbol{\mathcal{H}}$ is positive semi-definite. The case $\sigma_s \leq 1$ reverses all signs. ■

The decomposition separates three forces. The common component, $d\log N$, shifts all sectors in the same proportion and carries no information on the network. The two network components are the employment counterpart of Theorem 2: pre-multiplying by $\boldsymbol{\Lambda}^{-1}$ turns a change in sales shares into a percentage change in employment, so that the same shift in spending matters more for a small sector than for a large one. Both components are switched on by substitution at buyers, and both vanish under Cobb-Douglas intermediates, but they create different relations between sectors. The vertical component transports the own response of each commodity along supply chains: when a commodity becomes cheaper, it gains share at its buyers if they treat their inputs as substitutes, and loses share if they treat them as complements, and every sector embodied in it moves with it. Hence $\mathcal{R}^{v}_{j,i} \neq 0$ only if some commodity embodies both i and j, that is, only if the two sectors lie along a common supply chain. The horizontal component is the rivalry between distinct commodities at their common buyers, transported to the sectors they embody: $\mathcal{R}^{h}_{j,i} \neq 0$ only if some buyer requires i and j through different inputs. By $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$, the two components offset each other in the aggregate: a shock reallocates workers between supply chains that compete for the same budgets, and the jobs gained along the chains that embody

the shocked sector equal those lost along the chains of its rivals.[7] Three predictions follow, for substitutes and complements alike. First, by Corollary 1, a shock that moves the employment of a sector moves that of its first-tier common-buyer neighbours in the opposite direction, unless they also lie along a common supply chain: shared budgets make reallocation zero-sum, and the elasticities only determine who gains. Second, sectors that are horizontally distant but vertically connected, for which $\boldsymbol{\mathcal{R}}^h$ is small relative to $\boldsymbol{\mathcal{R}}^v$, comove positively through the production network: they gain or lose together with the shocked sector, and the elasticities only determine whether they expand or contract. Third, sectors that are distant along both dimensions comove only through the common component. In general, comovement between two sectors is positive when their vertical connection outweighs their horizontal proximity, and opposite otherwise.

Aggregate output: what Hulten's theorem hides.– Proposition 3 shows that sectoral shocks move workers across sectors, and one might expect this reallocation to move aggregate output as well. To first order, it does not. In efficient economies Hulten's (1978) theorem

$$d\log Y = \boldsymbol{\lambda}' d\log \mathbf{z}$$

holds at every equilibrium. Aggregate output responds through Domar weights alone, whatever the arrangement of the Input-Output relationships behind them: economies with equal Domar weights but different horizontal geometries (as in Corollary 2) would respond identically to any vector of shocks. The reason is an envelope argument: the reallocation described by Theorem 2 is an optimal response of buyers to new relative prices where, moving labour and spending between intermediate uses that at the optimum have the same marginal value, leaves output unchanged to first order. At the aggregate level, horizontal geometry seems to vanish. This is not the case: it is hidden inside Hulten. While the Hulten's argument holds at every equilibrium, the weights it uses are themselves equilibrium objects, and they move with the shocks. Indeed, the theorem postulated by Hulten gives the slope of aggregate output with respect to sectoral productivities, $\frac{\partial \log Y}{\partial \log \mathbf{z}'} = \boldsymbol{\lambda}'$; Theorem 2 gives how this slope changes, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{R}}\, d\log \mathbf{z}$.

Horizontal geometry therefore leaves the first-order response unaffected but determines how that response changes with the shock itself, that is, the curvature of aggregate output – how the economy's production structure transforms dispersion in intermediate input prices into aggregate gains or losses. The logic is that of a single buyer. To first order, the cost of a buyer's intermediate bundle responds to input prices as their share-weighted average, so that how the bundle is composed does not matter. To second order, when intermediate input prices move unevenly, a buyer whose intermediate inputs are complements is held back by the inputs whose price fell least, and its cost falls by less than the average; a buyer whose inputs are substitutes shifts towards the input that became cheapest, and its cost falls by more. Aggregate output inherits these buyer-level curvatures, weighted by the size of each buyer's intermediate budget.

Consider a vector of independent and identically distributed sectoral shocks, $\boldsymbol{\varepsilon} = d\log \mathbf{z} \overset{\text{iid}}{\sim} \left(0, \nu^2\right)$, across $m = |\Phi|$ sectors; hold labour fixed, and let $\log Y^{det}$ be the deterministic level of real GDP (the Lucas's 1977 diversification: no role for idiosyncratic shocks). Positive and negative shocks offset each other in the first-order term, $\boldsymbol{\lambda}'\boldsymbol{\varepsilon}$; they do not offset each other in

[7] In the case of $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}} = \mathbf{0}$, the employment-weighted sum of relative responses is zero, so that $\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}}^v = -\boldsymbol{\alpha}'\boldsymbol{\mathcal{R}}^h$.

the dispersion of input prices within buyers' bundles, which is positive whenever shocks differ across sectors. Through the curvature, dispersion shifts the mean of aggregate output.

THEOREM 3 (On horizontal geometry and aggregate output) *The impact of the network's horizontal structure on aggregate GDP is as follows:*

- *to first order,* $d \log Y = \boldsymbol{\lambda}' \boldsymbol{\varepsilon}$*, so that* $\operatorname{Var}[d \log Y] = \nu^2 \|\boldsymbol{\lambda}\|^2$ *does not depend on* $\boldsymbol{\mathcal{H}}$ *and vanishes as* $m \to \infty$ *when* $\lambda_i \asymp \frac{1}{m}$*;*
- *to second order,* $d^2 \log Y = \boldsymbol{\varepsilon}' \boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon} = \sum_{i \in \Phi} (\sigma_i - 1) \lambda_i (1 - \alpha_i) \operatorname{Var}_{\widehat{\mathbf{a}}_i^{row}} [\boldsymbol{\mathcal{L}} \boldsymbol{\varepsilon}]$*;*
- *consequently, it holds* $\mathbb{E}[\log Y] - \log Y^{det} = \frac{1}{2} \nu^2 \operatorname{Trace} [\boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}] + o[\nu^2]$.

Proof: provided in Appendix A. ■

The three claims follow the argument above. The first establishes the implication of Hulten's (1978) theorem for aggregate volatility. Since the first-order response depends on the network only through Domar weights, their distribution determines whether idiosyncratic shocks average out. When balanced , $\lambda_i \asymp \frac{1}{m}$, the variance of aggregate output decays at rate $\frac{1}{m}$, as in Lucas (1977). When the network makes a few sectors large suppliers, directly or along supply chains, Domar weights are concentrated, $\|\boldsymbol{\lambda}\|$ decays more slowly, and diversification fails, as in Acemoglu, Carvalho, et al. (2012). In either case, holding Domar weights fixed, the downstream composition of intermediate input bundles is irrelevant: horizontal geometry plays no role and cannot be inferred from first-order aggregate volatility, nor does it alter its rate of decay.

The second part reveals what the Hulten's logic hides. The curvature of aggregate output is a size-weighted sum, across buyers, of the dispersion of their intermediate input prices, after shocks have travelled down the supply chains. Vertical structure, $\boldsymbol{\mathcal{L}}$, determines how shocks translate into intermediate input-price changes; horizontal geometry $\boldsymbol{\mathcal{H}}$ determines how these changes are distributed across the shared suppliers of each buyer. The resulting dispersion is then translated into cost changes according to common-buyers' elasticities: complementarity makes buyers' costs fall less than their average input-price change, while substitutability allows them to shift towards cheaper inputs and makes costs falling more. Thus, even with identical Domar weights and vertical structures, economies with different horizontal geometries can exhibit different aggregate output curvatures. In particular, if all intermediate inputs of a buyer become cheaper in the same proportion, that buyer contributes nothing to curvature, regardless of the magnitude of the shock. The overall reallocation term $\boldsymbol{\mathcal{L}}' \boldsymbol{\mathcal{H}} \boldsymbol{\mathcal{L}}$ therefore captures how the horizontal structure of the network transforms sectoral shocks in aggregate output curvature.

Finally, the third claim draws the implication for the level of GDP. Lucas's diversification eliminates aggregate volatility, $\operatorname{Var}[d \log Y] \to 0$, without eliminating micro volatility on *mean* output. Averaging shocks across many sectors does not necessarily reduce the resulting reallocation at each buyer, and horizontal transmission moves average output away from its deterministic benchmark: $\mathbb{E}[\log Y] - \log Y^{det} \neq 0$. As the number of sectors increases, positive and negative shocks can offset each other in the aggregate (Lucas 1977), but the resulting dispersion within each buyer's bundle need not to vanish. The effect on output remains bounded away from zero, whether Domar weights are balanced or concentrated, provided that buyers' upstream purchases are diversified and intermediate input elasticities are not unitary. As such, the

magnitude and sign of this effect depend jointly on horizontal geometry and elasticities: with complementarities, which empirical estimates support (*e.g.*, Atalay 2017; Boehm, Flaaen, et al. 2019), micro volatility lowers average output because of the network's horizontal structures.[8]

4. Data and Horizontal Anatomy

From the United States' Bureau of Economic Analysis (BEA), available data display information, for 3-digit U.S. 2017 North American Industrial Classification System (NAICS) sectors, on value-added, employment, net exports, and inter-sectoral Input-Output tables. The (balanced) panel data built covers 65 private sectors over the period 1998-2022. For additional details refer to Supplemental Appendix B.

Distance measures are derived from the *Total Requirements Table*, which records both direct and indirect Input-Output trade along each supply chain. The theory motivates this choice. By Theorem 2, the rivalry component of the reallocation operator, $-\boldsymbol{\mathcal{T}}'\widetilde{\boldsymbol{\mathcal{W}}}\boldsymbol{\mathcal{T}}$, is a common-buyer Gram matrix on total requirements, $\boldsymbol{\mathcal{T}} = \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{L}} - \boldsymbol{I}$: two commodities are rivals when the same buyers require both, directly or indirectly. On the supplier side, the covariance of sectoral responses is $\boldsymbol{\mathcal{L}}\boldsymbol{\mathcal{L}}'$, a common-supplier relation on the rows of the Leontief inverse. In both cases two sectors are close when they share partners both locally and throughout the depth of their supply chains. Accordingly, the distance between two sectors reflects the extent to which they share similar buyers/suppliers locally and throughout the depth of production linkages. Each cell $d^q[i,j] \in [0,1]$ of matrix $\boldsymbol{\mathcal{D}}^q$, for $q = \{fd, fs\}$ and $\forall i,j \in \Phi$, identifies the Euclidean distance (square root of the sum for squared differences of network intensities; see Section 2) between dyads of sectors in demand or supply relative to common sectors.

The empirical estimation of Section 5 classifies sectors by network economic distances at the *extensive margin*, $\boldsymbol{\mathcal{D}}^q_{ext}$, whose generic entry is $d^q_{ext}[i,j] = \{1,2,\ldots,d_{max}\}$. Differently from the theory, values closer to one correspond to shorter horizontal distances. As for their characterization, network distance-d among any two sectors $\{i,j\}$ is computed as a *shortest path problem* on the *intensive margin* distance matrices, $\boldsymbol{\mathcal{D}}^q$, implementing an algebraic version (*i.e.*, using matrix multiplication) of the traditional Dijkstra or Roy-Floyd-Warshall algorithms, since it is *(i)* suitable for discrete-value adjacency matrices – as the ones for intensive margin distances –, and *(ii)* fairly efficient for small to medium-sized graphs – as a sector-level production network. Supplemental Appendix B reports the technical version of the implemented algorithm which, in plain words, reads as follows.

(Shortest path algorithm) *Central idea is to find, for each single node, all the other nodes that can be reached in one step, then in two steps, and so on. Each time a node is reached for the first time, it records the number of steps it took to get there – this is the shortest path*

[8] Theorem 3 shares its starting point with Baqaee and Farhi (2019): Hulten's (1978) theorem is a first-order result, and departures from it are governed by how Domar weights respond to shocks. The results differ in how this object is read. The authors summarize it through unobservable reduced-form sufficient statistics and emphasize the degree of substitutability to which production factors can be reallocated. In my framework labour is fully mobile and the object factorizes into observed vertical transport and horizontal reallocation of buyers' budgets. The size of elasticities switches the allocative channel on and scale it, while horizontal geometry determines where reallocation takes place and how much of it occurs. Importantly, the last claim of the theorem adds that these effects on mean aggregate output survives the Lucas's (1977) diversification argument.

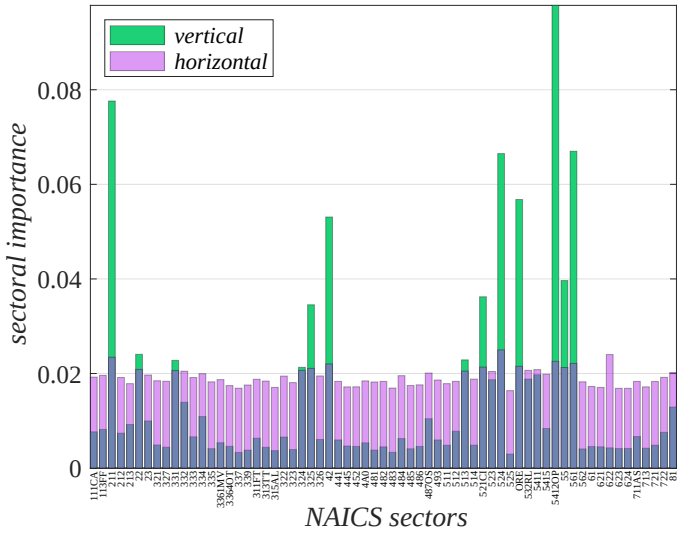


FIGURE 4: SECTORAL INFLUENCE IN SHOCK TRANSMISSION

Note: importance of each sector in reverberating idiosyncratic shocks through their vertical and horizontal network structures as dictated by the U.S. Input-Output propagation matrices in year 2007. *Source*: BEA and own calculations.

approach. In other words, find the shortest number of steps it takes to get from every node to every other nodes in a network. In particular:

(a) identify the direct (one-step) connections between any pair of nodes;

(b) for not-connected pairs, increase the path length by 1 in each round (two steps, then three steps, etc.). Multiply the graph by itself to discover which new pairs are now connected;

(c) if two nodes becomes connected, record the length as the shortest distance;

(d) repeat this process until any new reachable pairs of nodes is found. For any pair of nodes that are still uncategorised, set their distance to infinity.

4.1. PROPAGATING PATTERNS

I now analyse the mechanics of the horizontal transmission. Exploiting the observed propagation matrices, Figure 4 compares the importance of each sector in the vertical network, built on the Leontief inverse, with its importance in the horizontal network, built on network distances, for the 2007 table. In particular, it visualizes the sectoral propagation patterns using centrality measures from spectral norms, to be read in terms of how important each sector is for the transmission of shocks considering the dual network structures. The vertical importance from the Leontief transmission highlights few key sectors that act as propagation hubs: these retain a systemic role in providing inputs – *e.g.*, energy (211), risk intermediation (524), knowledge-intensive services (5412OP), and organizational support (561) –, underpinning a wide range of production activities. Asymmetric shocks originating in these sectors are recursively amplified along parallel supply chains, generating strong cascading effects while other sectors transmit shocks only weakly. In contrast, the horizontal propagation system spreads shocks almost uniformly. Most sectors share buyers or suppliers with many others, so that horizontal transmission is diffuse rather than hub-driven: each contributes to comovement with others, producing

widespread responses without recursive amplification. Overall, the image demonstrates the distinction between cascade-driven amplification along key nodes and diffuse propagation across economically similar sectors: vertical chains concentrate and amplify shocks from central nodes whereas horizontal complementarities induce diffuse comovement without feedback loops.

Table 1 in Appendix B, instead, documents the same contrast along a different dimension, since it investigates how vertical and horizontal geometries respond to varying levels of network sparsity. Progressively removing weaker transactions (increasing the threshold-τ for the volume of total sectoral shares), makes the network sparser and steadily lowers the spectral norm of the vertical operator, as supply chains are cut, thereby reflecting the diminishing role of cascade effects. Differently, the spectral norm of the horizontal propagation remains remarkably stable across thresholds, and it gradually becomes the predominant channel of shock transmission as τ increases, so that horizontal complementarities dominate in sparser networks. These results illustrate that, when weak linkages (*i.e.*, observed trade intensities) are trimmed, horizontal interdependencies exceed the vertical ones. They sustain substantial propagation as they rest on the overlap of many moderate links rather than on a few strong ones, and therefore survive sparsification. Table 1 thus lends empirical support to Corollary 2: as Input-Output economies become sparser, shocks increasingly travel along clusters of horizontally related sectors rather than along elongated vertical chains, confirming the principle of horizontal dominance.

4.2. Network exposure and aggregate shocks

The discussion so far has shown that the two geometries of the U.S. production network are distinct propagation objects, a precondition for the exercises that follow. The headline treatment of the network, however, is centred on the idea that horizontal geometries capture dimensions of economic exposures that are unconditionally distinct from vertical Input-Output linkages. If horizontal linkages capture an economically meaningful dimension of interdependence, they should also explain differences in sectoral responses to shocks "above and beyond" the information already embedded in Leontief inverse exposures, retaining a predictive power that conditionally complements them. Moreover, by Propositions 1-2, the relevant horizontal channel should depend on the nature of the shock: common-buyer distance transmits price movements, while common-supplier distance transmits demand impulses.

A Bartik (1991) shift-share design comes to the rescue. Consider an aggregate shock that affects all sectors simultaneously. Since the shock is widespread, identification cannot come from time-series variation alone. Instead, the relevant source of variation is cross-sectional: sectors may be differently exposed to the same aggregate disturbance because of their difference in Input-Output exposures. Formally, I first construct a measure of exposure by combining the weight of each network structure with a measure of predetermined sectoral sensitivity to the aggregate disturbance,

$$exposure_i^{(\mathcal{V},\mathcal{D}^q)} = \sum_{j \in \Phi_i} weight_{ij}^{(\mathcal{V},\mathcal{D}^q)} \, sensitivity_j \qquad for \quad q = \{fd, fs\}$$

This formulation encompasses all network structures, in that $weight_{ij}^{\mathcal{V}}$ are Leontief weights, and $weight_{ij}^{\mathcal{D}^q}$ are factor input demand and factor input supply distances. A sector is highly

exposed when it is strongly connected to sectors that are themselves highly exposed to the aggregate disturbance. Exposure therefore measures how embedded a sector is to the other sectors' sensitivities relative to the aggregate disturbance.

A natural concern is that horizontal and vertical exposures may be highly collinear and partly reflect similar objects since both belong to the same primitive Input-Output matrix (Section 2). Hence, the component of horizontal exposure that is orthogonal to vertical one is residualised from the auxiliary projection

$$exposure_i^{\mathcal{D}^q} = \beta_{exposure}^{cons} + \beta_{\mathcal{V}}\, exposure_i^{\mathcal{V}} + \gamma_i' \mathcal{Z}_i + r_i^{\mathcal{D}^q}$$

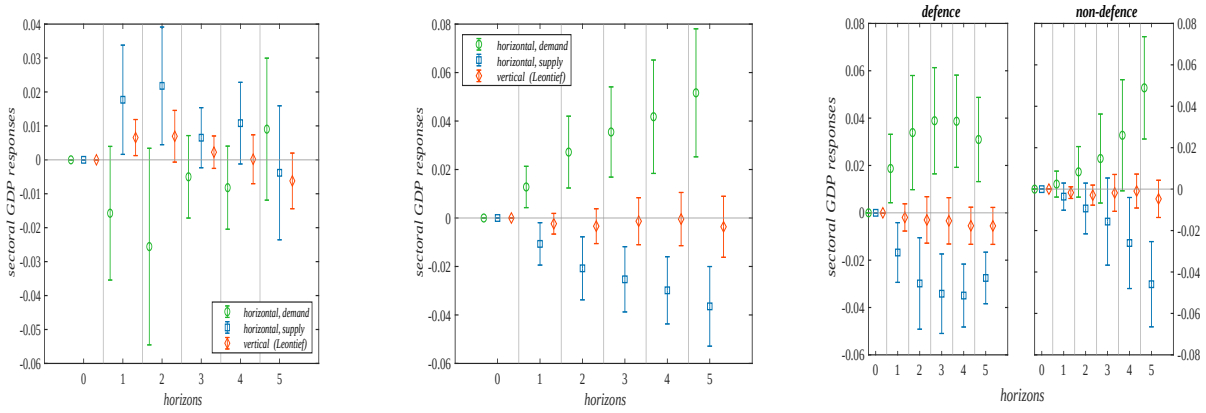
where $\mathcal{Z}_i$ are sectoral controls. Then, let G_t denote an aggregate shock. The Bartik exercise is the Local Projection (LP)

$$\begin{aligned}\Delta y_{i,t+h} \;&= \beta_h^{\mathcal{D}^{fd}} \left(G_t \times \widehat{r}_i^{\mathcal{D}^{fd}}\right) + \beta_h^{\mathcal{D}^{fs}} \left(G_t \times \widehat{r}_i^{\mathcal{D}^{fs}}\right) \\ &\quad + \beta_h^{\mathcal{V}} \left(G_t \times exposure_i^{\mathcal{V}}\right) + \phi_{t+h} + \phi_i + v_{t+h}^{y_i}\end{aligned}$$

over $h = \{1, \ldots, H\}$ time-horizons, where ϕ_i and ϕ_{t+h} denote sector and time fixed effects, respectively, and $v_{t+h}^{y_i}$ the horizon-specific error term. The interaction term captures the identified intensity with which the aggregate disturbance affects each sector through the network. Importantly, the aggregate shock itself is absorbed by time fixed effects, so that each coefficient β does not measure the dynamic response of the economy to the aggregate shock. Instead, each measures whether sectors with greater vertical/horizontal network exposure systematically react more ($\beta > 0$) or less ($\beta < 0$) strongly to the same shock than sectors with lower exposure. The sequence of coefficients traces the evolution of these differences over horizons. Consequently, the identifying variation is cross-sectional: a positive coefficient implies that highly exposed sectors react more strongly to the aggregate disturbance, whereas a negative coefficient indicates that network exposure dampens the transmission of the shock; the sequence of such coefficients across horizons traces the evolution of these cross-sectional differences over time. Therefore, the key empirical content of Figure 5 is not the dynamic unconditional effect of horizontal exposures, but rather the per-period conditional effect of horizontal distances after controlling for vertical Input-Output exposure.

Aggregate TFP shock.– I first consider an aggregate TFP disturbance, obtained as the unexpected component of aggregate total factor productivity after removing persistence and low-frequency trends, with sectoral sensitivity measuring how close each sector is to sectors that are historically sensitive to aggregate TFP innovations.[9] The estimated cross-sectional LP coefficients in Panel 5a reveal a sharp contrast across transmission channels. While the Leontief-based exposure has some predictive power at short horizons, its magnitude is smaller and its effects dissipate rapidly. By contrast, the supply-based distance measures generate larger

[9] Using the annual series from Fernald (2014), I extract the TFP innovation as the fitted residual from the AR(1) regression with time trend: $\log TFP_t = \beta_{tfp}^{cons} + \beta_{tfp} \log TFP_{t-1} + \beta_{time} \cdot t + \varepsilon_t^{tfp}$. This series is also exploited to construct sectoral exposures. First, I estimate sector-specific historical sensitivities to aggregate TFP innovations, capturing how strongly the sector tends to react when aggregate productivity unexpectedly changes. The sector-specific sensitivities are then propagated through the alternative network structures considered.



(A) TFP (B) GOVERNMENT SPENDING

FIGURE 5: "NETWORKING UP" AGGREGATE SHOCKS

Note: cross-sectional coefficients measure how differently sectors respond positively (amplification) or negatively (attenuation), due to vertical and horizontal network exposures, to the same aggregate shock, as in one standard deviation. Bars correspond to intervals at 90% of bootstrapped confidence level, clustered by sectors. *Source*: various and own calculations.

and more persistent differences in sectoral responses: sectors that are more closely connected, through common downstream buyers, to TFP-sensitive sectors react more strongly to aggregate productivity shocks over several horizons. Demand-based horizontal exposure does not generate statistically significant effects. Both patterns are those predicted by the theory. A positive aggregate TFP shock hits asymmetrically due to pre-determined exposures, increasing productivity more in some sectors than others. Through the horizontal supply channel, more productive sectors expand production and capture a larger share of downstream demand, leading to a reallocation of expenditure across competing suppliers. This competitive displacement implies that aggregate demand is satisfied by increasingly efficient producers, so that total economic activity rises despite unchanged downstream demand, as production is concentrated in higher-productivity sectors. By Proposition 2 and Theorem 2, the price change transmits to common-buyer neighbours, and the sign of their response reveals shared buyers' substitutability: a positive coefficient means that a sector gains when its rivals become more productive, that is, that shared buyers treat their inputs as complements at annual frequency, in line with the estimates of Atalay (2017) and Boehm, Flaaen, et al. (2019). By contrast, the horizontal demand channel does not play a significant role, since TFP shocks operate through productivity differences rather than own demand expansion. In fact, a productivity shock lowers the relative price of the most sensitive sectors without directly raising anyone's intermediate input demand, making upstream propagation through shared suppliers weak and non-systematic: by Proposition 1, common-supplier transmission requires a demand impulse at shared suppliers and is therefore second order. These findings therefore suggest that horizontal network structures constitute an independent and economically relevant dimension of production networks.

Government spending shock.– Panel 5b, instead, plots the coefficients for fiscal disturbances, constructed similarly to the TFP case.[10] The results reveal a markedly different pattern

[10] Pre-determined exposures are constructed from the 1997 U.S. Input-Output table as the government purchases in the sectoral supply of intermediates, capturing the extent to which a sector is connected to government demand and therefore likely to benefit from expansions in aggregate public expenditure.

across propagation channels. Government spending shocks diffuse not only through directly affected industries but also through common Input-Output relationships, and appear to diffuse across sectors sharing common suppliers, generating positive horizontal spillovers in consistency with the intrinsic demand-based nature of these shocks, while sectors sharing common buyers exhibit weaker responses, consistent with competitive reallocations of downstream demand. Reassuringly, the results are also robust to disentangling aggregate government expenditure into defence and non-defence components, whose pre-determined spending categories are concentrated in targeted industries: despite the different cross-sectional allocation of network exposures, the pattern of sectoral responses is preserved. Spending is a demand impulse that raises the scale of directly exposed sectors. By Proposition 1, the impulse reaches sectors that share suppliers with them, with the sign of $-\varphi_s$: the positive coefficient indicates that shared suppliers expand at decreasing marginal cost during fiscal expansions, as with slack capacity or increasing returns, so that co-customers are crowded in. By Proposition 2, the same impulse reaches common-buyer neighbours only through the price of the exposed sectors: if these face rising marginal costs, their prices rise together with their output, and complementarity in shared buyers – the same elasticity pattern inferred from TFP shocks – implies that their rivals contract, as found. Hence, these types of shocks operate as positive demand adjustments that propagate through the Input-Output network via two distinct horizontal channels. On the demand side, higher input requirements from directly affected sectors increases demand faced by shared upstream suppliers, which can adjust either through quantities or prices. If they respond by expanding production at given prices, the increase in input supply can benefit other sectors connected through the same supplier, or leave their production unaffected. If instead they raise prices, downstream sectors may either continue purchasing at higher cost or substitute toward alternative suppliers, reallocating production across the network. In both cases, the adjustment of common upstream suppliers to higher downstream demand generates positive spillovers as it propagates through Input-Output linkages or substitution effects, and leads to an overall expansion in economic activity. On the supply side, targeted spending induces sectors to expand their production to meet higher government demand. This allows these sectors to meet their supply more effectively, affecting how downstream expenditure is allocated across upstream suppliers. Given the unaltered demand of common buyers, higher absorption by directly exposed sectors crowds out others serving the same customers: horizontally exposed sectors experience negative spillovers through competitive pressures for downstream markets. In this case, there is no compensating expansion from shared downstream demand, so that effects are driven by within-buyer reallocation, resulting in overall dampened responses in horizontally exposed sectors.

Taken together, the two exercises reinforce the broader message of the paper: that the horizontal geometry is a distinct dimension of an Input-Output system, as Corollary 2 anticipates, whose relevant horizontal channel depends on the nature of the shock, as Propositions 1-2 predict. Aggregate productivity (supply) shocks primarily propagate through the supply-based horizontal structure since they move relative prices faced by shared downstream buyers, while aggregate fiscal (demand) shocks operate through a distinct network dimension, namely the demand-based horizontal geometry, since they move quantities and travel through shared upstream suppliers. The fact that the dominant propagation channel varies across shocks suggests

that horizontal geometries do not merely replicate classical Input-Output effects, but instead capture economically meaningful dimensions of sectoral interdependence that become relevant under different types of aggregate disturbances, and contain additional information on shock propagation that is not fully captured by standard vertical production linkages.

5. EMPIRICS

Besides Input-Output tables, sector-level data for the U.S. economy are exploited to test the predictions about sectoral comovement in Section 3. Estimation relies on the two-stage procedure outlined by Barattieri and Cacciatore (2023): first, I perform a series of sectoral panel Fixed-Effect (FE) regressions to isolate the exogenous dynamics in a given input of production not due to other (sector-specific, other sectors-specific and aggregate) factors; then, each estimated residual is exploited as an identified structural shock into a Local Projection (LP) analysis, in order to asses the network-based short-run cumulative variations in sectoral employment.

5.1. INTERMEDIATE INPUTS AND SECTORAL EMPLOYMENT

Sections 2 and 3 rest on the premise that adjustments in the stock of circulating intermediate inputs lead to synchronized sectoral employment responses. In fact, in Proposition 3, relative sectoral employment responds to the reallocation of buyers' budgets, which reaches a sector directly, through its horizontal neighbours, and indirectly, through the supply chains that embody it. Considering variations within a sector and across its economic neighbours, the first step isolates changes in input sourcing plausibly orthogonal to employment dynamics. The following panel fixed-effect regression over $i \in \Phi$ sectors is estimated:

$$\begin{aligned} x_{i,t} = \beta_x^{cons} + \beta_n \, \dot{n}_{i,t} + \sum_d \beta_{n(d)} \, \dot{n}_t\big(\Phi_i, d\big) + \sum_l \beta_{n(l)} \, \dot{n}_t\big(\Phi_i, l\big) \\ + \gamma_i' \mathcal{Z}_{i,t} + \Gamma_i' \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + \; u_{i,t}^x \end{aligned} \tag{4}$$

where $x_{i,t}$ identifies the set of intermediate inputs bought by sector-i (as a share of its total production), while $\dot{n}_{i,t}$, $\dot{n}_t\big(\Phi_i, d\big)$, and $\dot{n}_t\big(\Phi_i, l\big)$ represent its employment growth, that of its economically closer and further sectors, and that of its sectors located upstream or downstream, respectively. The latter two measures are defined as

$$\begin{aligned} \dot{n}_t\big(\Phi_i, d\big) &= \sum_{j \in \Phi_i} \Delta n_j\,(d) \qquad\qquad , \quad \forall d \in \boldsymbol{\mathcal{D}}_{ext}^q \quad \text{and} \quad q = \{fd, fs\} \\ \dot{n}_t\big(\Phi_i, l\big) &= \sum_{j \in \Phi_i} \ell_{i,j}\,(l)\; \Delta n_j\,(l) \quad , \quad \forall l = \{up, dw\} \end{aligned}$$

where $\Delta n\,(\cdot) = n_t\,(\cdot) - n_{t-1}\,(\cdot)$. As for vertical propagation, each change is weighted by the sales-normalised Leontief inverse relation, $\ell_{i,j}\,(l)$, for every sector-j located upstream or downstream of sector-i, thereby capturing direct and indirect network exposure. Changes in horizontally-related sectors, sorted by extensive margin distance, enter without Leontief weights: by Theorem 2, first-tier horizontal rivalry, $\mathcal{H}_{i,j}$, is not weighted by vertical exposure; by Corollary 2, horizontal relationships are not leveraged by the Leontief inverse weights.

A set of control variables characterizes the second line of eq. (4). $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$ comprise the labour force size of the sector, both its level and growth rate of value-added, and its net exports that control for changes in demand/supply not due to internal factors. The set of aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, is made of aggregate value-added and aggregate employment in size as well as in growth terms. In this and in the identification in the next subsection I control for variations in aggregates since comovement between aggregate and sectoral variables is well established (*e.g.*, Stock and Watson 1999; Huffman and Wynne 1999; Rebelo 2005; DiCecio 2009). Finally, while β's are coefficients to be estimated, ϕ_i imposes fixed-effects to control for unobserved heterogeneity across sectors. Standard errors are clustered at sector-level. For each sector, the estimated residual $\widehat{u}^x_{i,t}$ identifies the exogenous variation in the share of production of sector-i that is bought from Input-Output connections.

Impulse Response Functions (IRFs) of sectoral employment to changes in the sector-specific intermediate inputs rely on Local Projections. Given $h = \{1,\dots,H\}$ time-horizons, and for all sectors $i, j \in \Phi = \{1,\dots,m\}$, a battery of panel FE predictive regressions of the identified sector-specific structural shock, $\widehat{u}^x_{i,t}$, on the cumulative difference in sectoral levels is performed:

$$\dot{n}^{cum}_{i,t+h} = \beta_{u^x,h}\ \widehat{u}^x_{i,t} + \phi_{t+h} + \phi_i + \nu^x_{i,t+h} \tag{5}$$

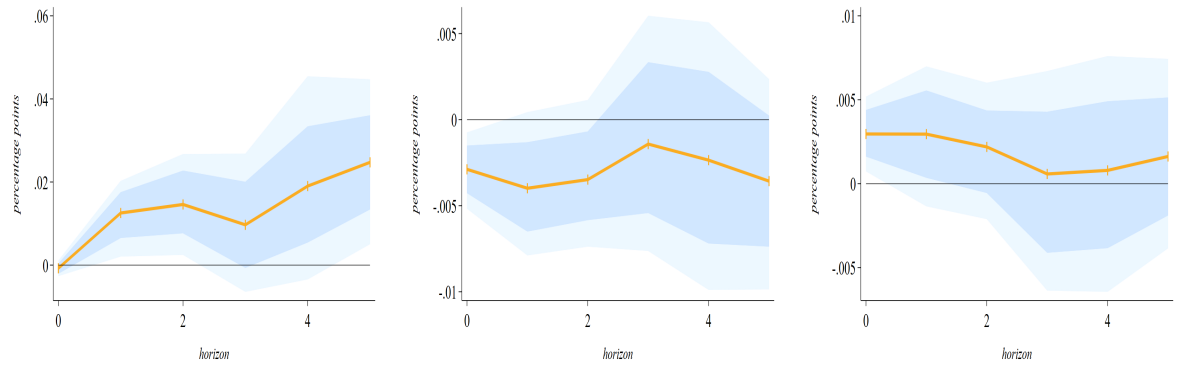
with $\dot{n}^{cum}_{i,t+h} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector-i employment at each horizon. Estimated coefficient $\beta_{u^x,h}$ is the response of the variation in the cumulative difference of sectoral employment at time $t+h$ given the identified shock hitting in period-t. Prediction error term $\nu^x_{i,t+h}$ is horizon-specific, while standard errors and bootstrapped Confidence Interval (CI) of $\beta_{u^x,h}$ are clustered by sectors. To avoid measurement errors due to unobserved heterogeneity in response, sectoral (ϕ_i) and horizon (ϕ_{t+h}) fixed-effects – that remove common trends in the comovement of sectoral employment –, are imposed. Differently from the first-stage (which controls common time trends using aggregate variables), ϕ_{t+h} removes the possibility of common trend in the *response* of sector-specific employment to idiosyncratic shocks.

Outcomes from eq. (5) estimate the cumulative responses of sectoral employment to changes in their set of intermediate inputs. Analogous estimation is performed to variations in nearby sectors, considering the dth-distance between sectors:

$$\dot{n}^{cum}_{i,t+h} = \sum_{j\in\Phi_i} \beta_{u^x_j,h}\,(d)\ \ \widehat{u}^x_{j,t}\,(d) + \phi_{t+h} + \phi_i + \nu^x_{i,t+h}\,(d) \tag{6}$$

Estimating eqs. (5) and (6), under *Leontief inverse* and *factor input supply* distances, delivers Figure 6. The first panel shows the cumulative (short-term) response of sector-specific employment to a 1% change in the output share of its intermediate input: averaging across sectors, the increase induces a positive statistically-significant change in sectoral employment, where each additional unit of intermediate inputs requires less than one worker so that factors of production are at least substitute, as the Cobb-Douglas specification instructs.

Changes in intermediate inputs from other sectors do not yield consistent outcomes. Results are almost always not statistically significant at 90% except when the shock hits, and the magnitude of the variation is negligible. Interesting is to note the different response between Panels 6b and 6c: for intermediate inputs changes in closer sectors there is an average negative effect, while changes in further sectors are positively related with variations in sectoral employment. In



(A) SECTOR-SPECIFIC (B) $d = 1$ (C) $d = 2$

FIGURE 6: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel 6a represents the sector-level employment response to changes in its intermediate inputs, while Panel 6b-6c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (5)-(6). *Source*: BEA and own calculations.

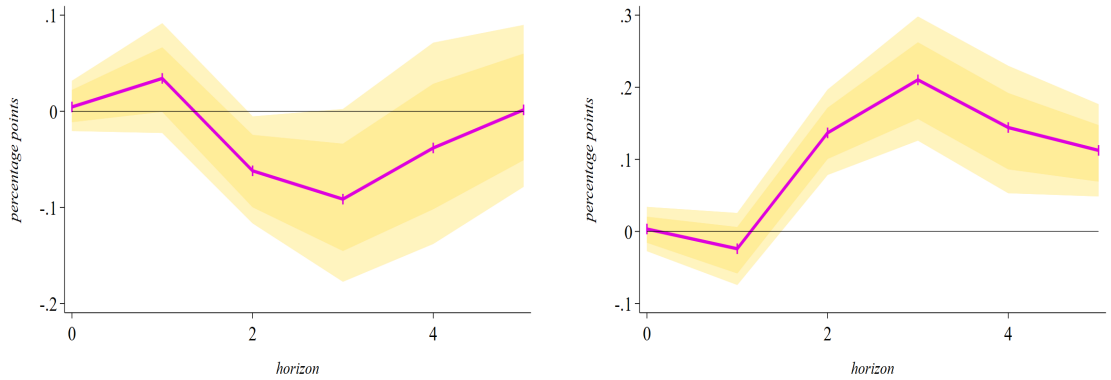
Supplemental Appendix C, under *Leontief inverse* and *factor input demand* network distances, I show how that the same average impact still holds, while results for distances' effects identify stronger but opposite results: increasing for closer sectors, declining for further ones. Such observations might suggest that certain types of intermediate inputs are more essential than others in production (*e.g.*, Carvalho and Voigtländer 2015), but it is actually the type of economic distance that determines their importance both in the production of sectoral output and in their network centrality of being larger buyers or suppliers.

5.2. SECTORAL COMOVEMENT OF EMPLOYMENT

I now analyse how sectoral employment adjusts in response to employment changes in other sectors, depending on their network economic distance. The previous identification and estimation scheme is developed, but now purging the dynamics of sector-specific employment with changes in its and other sectors' value-added, since employment is not exogenous to growth rates of sectoral output (*e.g.*, Hornstein and Praschnik 1997; Acemoglu, Akcigit, et al. 2015; Sandqvist 2017). Note that the decomposition in Proposition 3 maps into the empirical design. The common component is absorbed by time fixed effects. The transported components of $\boldsymbol{\mathcal{R}}$ are controlled for by Leontief-weighted exposures. What network economic distance captures, among sectors with no vertical connection, is direct rivalry at common buyers ($\boldsymbol{\mathcal{H}}$).

Identification.– To identify exogenous variations in sectoral employment, $n_{i,t}$, a panel fixed-effects regression over each sector $i \in \Phi$ is estimated:

$$\begin{aligned} n_{i,t} = \beta_n^{cons} + \beta_y\, \dot{y}_{i,t} + \sum_d \beta_{y(d)}\, \dot{y}_t\big(\Phi_i, d\big) + \sum_l \beta_{y(l)}\, \dot{y}_t\big(\Phi_i, l\big) \\ + \gamma_i' \mathcal{Z}_{i,t} + \Gamma_i' \dot{\mathcal{Z}}_{i,t} + \gamma' \mathcal{Z}_t + \Gamma' \dot{\mathcal{Z}}_t + \phi_i + u_{i,t}^n \end{aligned} \tag{7}$$



(A) $d = 1$ (B) $d = 2$

FIGURE 7: EMPLOYMENT COMOVEMENT UNDER DEMAND LINKAGES

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 7a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 7b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

where $\dot{y}_{i,t}$ is the value-added growth for sector-i, $\dot{y}_t\left(\Phi_i, d\right)$ identifies the value-added growth of its related sectors according to network distances, and $\dot{y}_t\left(\Phi_i, l\right)$ represents the value-added growth of its upstream and downstream sectors. Each measure is defined as that of eq. (4) but using y_i instead of employment. Moreover, for this specification, sector-level controls, $\mathcal{Z}_{i,t}$ and $\dot{\mathcal{Z}}_{i,t}$, comprise the labour force size of the sector, its value-added level and net exports, while aggregate controls, $\mathcal{Z}_t$ and $\dot{\mathcal{Z}}_t$, are analogous to eq. (4). Estimated coefficients are the β's, given sectoral fixed-effects, ϕ_i, and sectoral standard errors. Sector-specific estimated residual $\widehat{u}^n_{i,t}$ from eq. (7) identifies the exogenous variation in sectoral employment.

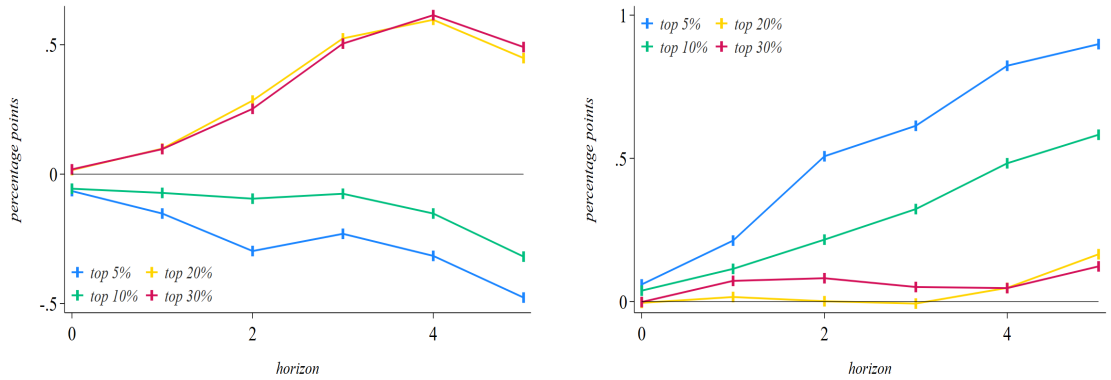
Local Projection analysis.– Sectoral comovement is defined as the response of employment in a given sector to changes in employment of other sectors:

$$\dot{n}^{cum}_{i,t+h} = \sum_{j \in \Phi_i} \beta_{u^n_j,h}\left(d\right)\ \widehat{u}^n_{j,t}\left(d\right) + \phi_{t+h} + \phi_i + \nu^n_{i,t+h}\left(d\right) \tag{8}$$

with $\dot{n}^{cum}_{i,t+h} = n_{i,t+h} - n_{i,t-1}$ denoting the cumulative change in sector-i employment at each horizon given the identified employment shock in closer or further sectors at the dth-distance. Coefficient $\beta_{u^n_j,h}\left(d\right)$ is the h-step-ahead response of sector-specific employment in cumulative difference due to the identified shock $\widehat{u}^n_{j,t}\left(d\right)$. Still, sector and horizon fixed effects, ϕ_i and ϕ_{t+h} respectively, are imposed to remove unobserved heterogeneity and common trends across sectors in the comovement of sectoral employment. Prediction error term $\nu^n_{i,t+h}\left(d\right)$ is horizon- and distance-specific in each predictive panel regression, with standard errors and bootstrapped Confidence Interval (CI) of $\beta_{u^n_j,h}\left(d\right)$ clustered by sectors.

5.2.1. Results for factor input demand distances

Considering sectors whose distance is defined in terms of buying from the same set of upstream suppliers, estimating eq. (8) yields Figure 7. By Proposition 2, demand-based complementari-



(A) $d = 1$ (B) $d = 2$

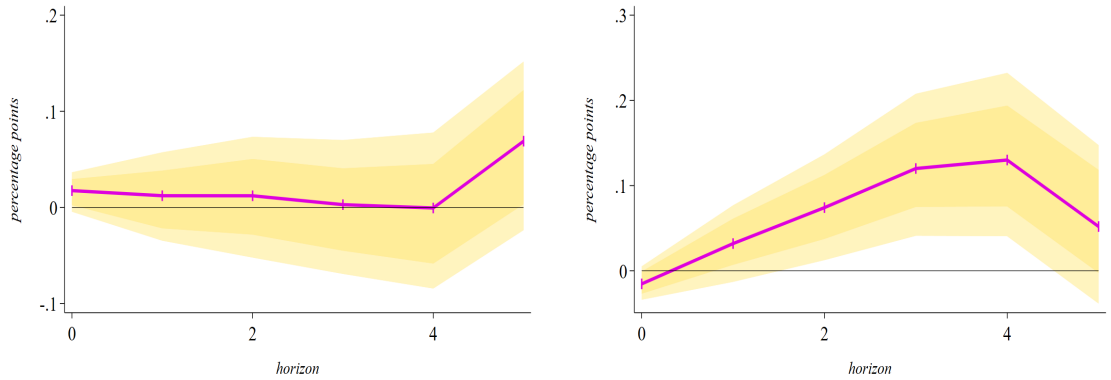
FIGURE 8: DEMAND LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 8a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 8b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (8). If a plot is not appearing, it means there is no response for the specified distance value. *Source*: BEA and own calculations.

ties are inherently ambiguous to interpret since they depend on the joint action of vertical propagation and horizontal spillovers: when buyers can easily substitute inputs, a positive shock in one sector strengthens the demand for the shared supplier and raises the upstream price, thereby crowding out the other sector; by contrast, when inputs are complements, the increase in the supplier's price erode the demand competitiveness of the other sector, dampening comovement. Where these forces conflict or remain weak, observed correlations are muted or ambiguous.

These mechanisms echo in observed responses. Panels 7a and 7b show that a 1% increase in employment in closely linked sectors ($d = 1$) produces an ambiguous response, whereas more distant sectors ($d = 2$) consistently display positive comovement. However, sectoral shocks in Panel 7a do not produce any statistically significant effects, as the estimated responses fail to reach significance at the 90% confidence levels. This lack may be driven by including all sectors indiscriminately, as sectors with fewer inter-sectoral linkages might introduce noise or bias to the estimation, masking and diluting the true magnitude of measured comovement in sectoral employment: restricting the sample to sectors with most Input-Output linkages could reveal more meaningful propagation effects where network effects are more pronounced, under which horizontal transmission scales with the pair's shares in shared partners.

Nevertheless, ambiguous patterns persist when partitioning sectors into finer subsets by the number of Input-Output linkages (Figure 8): comovement depends on the specific cluster considered. This confirm that demand-driven horizontal complementarities and vertical propagation operate jointly, often obscuring one another. Such mixed evidence ultimately reflects the inherent ambiguity of demand-based distances discussed in Subsection 3.1: because they capture substitution patterns across upstream intermediate inputs rather than genuine horizontal complementarities, horizontal propagation remains weak and easily masked by vertical effects. This is what Proposition 1 leads one to expect when the curvature of shared suppliers' costs



(A) $d = 1$ (B) $d = 2$

FIGURE 9: EMPLOYMENT COMOVEMENT UNDER SUPPLY LINKAGES

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 9a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 9b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

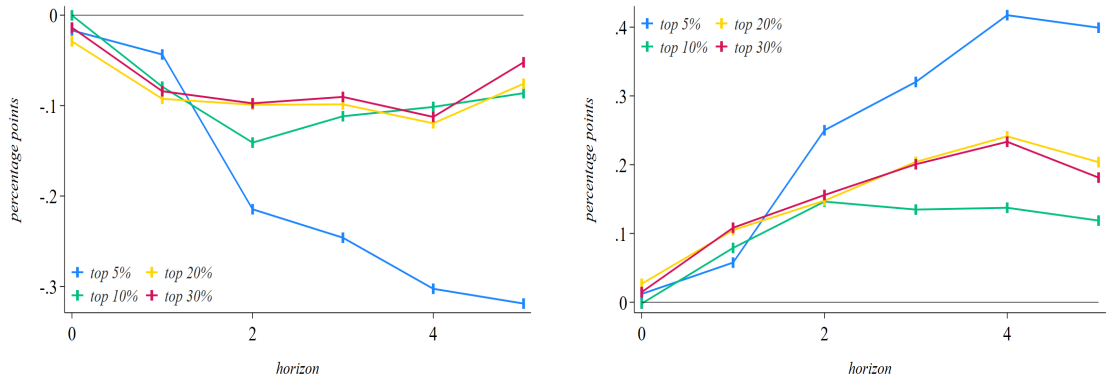
differs across clusters and common-supplier transmission overlaps with vertical propagation.

Positive comovement is most likely to emerge between sectors that are "economically" distant, for which the reallocation term of Proposition 3 is small and the common component dominates. Among closely connected sectors, the sign depends on the curvature of shared suppliers' costs and the balance between vertical and horizontal channels is more delicate, producing the mixed responses observed in the data. This suggests that observed employment comovement mirrors an interplay of vertical and horizontal propagation, thereby highlighting the inherent complexity in interpreting demand-based complementarities.

5.2.2. Results for factor input supply distances

Considering now sectors whose distance is defined in terms of selling to common downstream buyers, Figure 9 plots the responses from the estimation of eq. (8). Interpreting the effect of further ($d = 2$) sectors' variations in employment delivers the same results of the factor input demand case: from Panel 9b, a 1% increase in their employment increases sectoral employment as well, thus determining positive comovement and a positive transmission of the sector-idiosyncratic shock. Different is the case of closer ($d = 1$) sectors, where it appears a positive and then a negative cumulative effect on sector-specific employment with a small magnitude.

The lack of significance in Panel 9a might have two complementary interpretations: *(i)* changes in employment in nearby sectors have no measurable impact on others when the sectors are linked through a common set of downstream buyers; and *(ii)* the factor input supply network distance seems to dampen the vertical propagation of shocks across the production network, thereby weakening the expected comovement in production inputs across sectors. Specifically, when sectors share the same downstream buyers, a change in their relative price faced by the common buyer leads to opposite changes in their relative supply of intermediate inputs.



(A) $d = 1$ (B) $d = 2$

FIGURE 10: SUPPLY LINKAGES AND COMOVEMENT IN HIGHLY INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel 10a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel 10b plots the response to employment changes in further (distance equal to 2) ones. Each line corresponds to the average response of employment across sectors with different numbers of linkages, robust to 68% and 90% significance levels of bootstrapped Confidence Interval (CI) computed from eq. (8). *Source*: BEA and own calculations.

This generates an inverse comovement in sectoral production and, consequently, an opposite comovement in sectoral employment. In the context of Panel 9a, these opposing forces roughly cancel each other out when averaged across all sectors, resulting in the non-significant response. Accordingly, clearer effects might be correctly identified when considering highly interlinked sectors. Note that the non-significance of Panel 9a aligns with Theorem 2: horizontal rivalry is concentrated where shared buyers are large relative to the pair, with weights $\lambda_s (1 - \alpha_s) \widehat{\alpha}_{si} \widehat{\alpha}_{sj}$; averaged over all pairs at $d = 1$, many of which share only minor buyers, it is diluted.

Impulse responses for the clustered analysis are presented in Figure 10. The results for major network distances, shown in Panel 10b, largely mirror those found using the full sample, but with larger point estimates responding to the structural employment shock. More strikingly, at short supply distances (Panel 10a), an increase in employment in closely linked sectors reduces employment in the sector – as Proposition 3 and Corollary 1 predict for first-tier common-buyer neighbours –, resulting in an opposite comovement in sectoral employment. This inverse relationship grows stronger as the analysis narrows to smaller groups of sectors with increasingly dense linkages, which share larger buyers and carry larger weights in $\mathcal{H}$: the closer the sectors are in the production network when supplying downstream, the more pronounced these opposing employment responses become.

All these theoretical and empirical results are in line with the non-linear theories in Atalay (2017) and Baqaee and Farhi (2019): downstream complementarities in intermediate inputs are crucial to generate opposite responses from an idiosyncratic shock – *i.e.*, the downstream demand pass-through effect from the supply-driven network economic distance is accordingly generating opposite comovement (Proposition 1). Yet, such pass-through is found to be weaker when sectors have dissimilar Input-Output structure in supply.

5.3. ROBUSTNESS

Results on comovement hold along different "economically-based" robustness checks, sequentially presented in Supplemental Appendix C. Initially, I shift the focus towards peripheral sectors within the production network. Much of the existing literature holds on the role of central sectors, typically measured using Bonacich-Katz centrality (*e.g.*, Carvalho 2014). The underlying premise is that a positive shock in a central sector transmits outward, potentially inducing broader economic expansions – *i.e.*, microeconomic shocks can scale up to macroeconomic consequences through network centrality. From a comovement perspective, periods of economic expansion (contraction) are expected only when a positive (negative) shock in a central sector effectively propagates to more peripheral sectors: fluctuating aggregate activity is not merely a direct realization of shocks to central nodes, but rather it stems from the manner in which such asymmetric shocks ripple across the network to other, more peripheral sectors. Stated differently, it is not the initial shock per se that generates business cycle fluctuations, but the subsequent consequences it produces as it transmits across the network to less central nodes. If network distances were irrelevant, comovement would be expected universally. However, should demand- and supply-side linkages prove significant, the comovement patterns ought to remain consistent with those documented thus far.

I further account for changes in sectoral inventories, since they introduce a short-run margin that may buffer the network-based transmission: when inventories are available, sectors smooth shocks through previously accumulated stocks, temporarily decoupling production adjustments from contemporaneous changes in the purchased intermediate input bundle. As a result, the transmission of shocks through shared Input-Output structures should be attenuated, breaking the simultaneity of horizontal complementarities and weakening the role of both demand- and supply-based network economic distances in shaping short-run comovement. By contrast, in the absence of inventories, changes in economically nearby sectors are expected to translate directly into the usual horizontal propagation effects.

Finally, I test the robustness of the presented findings by employing alternative base years for the U.S. production network. My main analysis is anchored at the 2007 Input-Output structure, which conveniently divides the sample period into two equal sub-periods. Theoretically, the results should remain valid provided that the production network structure exhibits limited evolution over time.[11] Empirically, I replicate the full set of analyses using network data from other benchmark years.

Centrality scores.– Figures C.6 and C.7 present the impulse responses of employment to factor input demand and supply distances, respectively, considering only the peripheral sectors of the U.S. production network. Results are broadly consistent with those discussed in earlier sections: opposite comovement under supply distances, and positive comovement for shocks originated in further sectors. The only noteworthy exception concerns the response of employment to changes in closer sectors under demand-based distances: rather than exhibiting the

[11] A condition broadly supported in the empirical literature. Carvalho (2014) and Acemoglu, Ozdaglar, et al. (2016) document relative stability of U.S. sectoral production network and its role on shock propagation. Classical Input-Output analysis (*e.g.*, Leontief 1986; Miller and Blair 2022) also emphasises the persistence of production linkages in the short to medium run.

ambiguity observed for all sectors (overlapping of vertical and horizontal dimensions), inspecting horizontal complementarities in demand for more peripheral sectors generate effects that resemble those observed under supply-based distances, with opposite comovement occurring across sectors sharing the same network structure in terms of buying from common suppliers. This refines the observed pattern under direct demand-based measures of Figure 7, suggesting that opposite employment comovement among sectors purchasing from similar upstream suppliers largely occurs in sectors not playing a key role in the propagation of shocks. These outcomes create a direct bridge with Acemoglu, Carvalho, et al. (2012): positive shocks in central sectors transmit unevenly to peripheral ones according to different network economic distance values, thereby inducing different types of aggregate behaviours.

Inventories.– The outlined hypothesis is strongly supported by the supply-based patterns in Figure C.8. Horizontal complementarities vanish for sectors holding inventories, and the transmission of a positive shock reverts to its vertical logic.[12] By contrast, in contexts without inventories, the baseline results are preserved, with positive shocks generating opposite comovement across sectors with nearly similar Input-Output structures. This sharp contrast indicates that horizontal effects are operative when production adjustments are directly tied to contemporaneous changes in circulating intermediate inputs, rather than when accumulated in storage. Inventories act as a short-run buffer that temporarily deactivates the horizontal propagation channel without altering the underlying network structure. These findings provide direct evidence that the documented horizontal complementarities reflect structural economic mechanisms, arising from contemporaneous adjustments through shared linkages, and not mechanical features of the production network.

Different base years.– All the documented results, derived using the 2007 U.S. tables, remain robust if replicated for alternative benchmark years (2002 and 2012), as depicted in Figures C.9, C.10, and C.11, supporting the inherent ambiguity of demand-based distances and the clearer, more systematic comovement patterns under supply-based distances. This temporal consistency suggests that the structural features of the U.S. production network have remained remarkably stable over time, documenting the enduring nature of Input-Output linkages and the slow-moving evolution of production networks, thus lending further credibility to the analysis.

6. Concluding Remarks

Initial inquiries into production networks chiefly centred on how disaggregation induced by inter-sectoral trade generates comovement of macroeconomic variables. Reviving this foundational perspective, this paper moves beyond traditional vertical linkages by theoretically introducing and empirically validating the horizontal geometry of Input-Output economies, measured through network "economic" distances reflecting similarities between sectors in terms of common upstream suppliers or downstream buyers. Constitutive premise is that sectors are densely interconnected not only by their observed trade flows but also through shared inter-

[12] Over the sample period, inventories are observed in NAICS agriculture (11), extraction (21), and manufacturing (31-33) sectors, as well as in transportation (481-484), publishing (511), and recording (512) industries.

sectoral relationships, providing a richer understanding of the complex architecture underlying any modern economy. Demand-based distances promote an overlap of horizontal and vertical propagation, as shocks transmitted through shared upstream suppliers interact with parallel supply chain effects. By contrast, supply-based distances directly threaten the design of vertical transmission, as revenue reallocation among sectors selling to the same downstream buyers counteracts the standard propagation of shocks. These mechanisms operate independently of trade intensities, offering a powerful refinement to prevailing network analysis. As for aggregate output, horizontal propagation reallocates rather than cascades. It is hidden in Hulten's (1978) theorem, which leaves first-order aggregate fluctuations to Domar weights alone, but it governs how these weights move with shocks: it shifts workers across sectors and bends aggregate output, so that micro volatility moves mean output by an amount that diversification across many sectors does not eliminate. By conceptualising these horizontal complementarities, the paper enriches the understanding of how independent and idiosyncratic shocks permeate the production system. Ultimately, it reveals that the demand/supply architecture of an Input-Output structure and its horizontal geometry critically shape the transmission of shocks, offering novel insights into the networked nature of economic activity.

Horizontal geometry may help to assess policy effectiveness by accounting for sectoral composition and interdependencies between targeted and untargeted sectors. Fiscal policy provides a natural application, as sectors sharing common upstream suppliers or downstream buyers with those directly exposed to government spending may experience additional effects that conventional Input-Output linkages (*e.g.*, Barattieri, Cacciatore, and Traum 2023; Bouakez et al. 2025) fail to capture. The same logic applies to tariff policy, as changes in relative prices redistribute revenues across sectors (*e.g.*, Barattieri and Cacciatore 2023; Antonova et al. 2025; Clausing and Obstfeld 2025) and may generate horizontal effects among sectors serving common downstream buyers. Likewise, industrial policies targeting strategic sectors may affect economically close but untargeted sectors through shared production relationships, extending the analysis of policy-induced distortions and downstream spillovers (*e.g.*, Liu 2019; Lane 2025). Finally, horizontal geometry may enrich the analysis of monetary policy by identifying how common production relationships shape the expectations on sectoral price behaviours and the propagation of policy-induced shocks, complementing existing production network approaches to monetary transmission (*e.g.*, Pasten et al. 2020; Kalemli-Özcan et al. 2025).

Beyond these policy applications, horizontal linkages offer a theoretical framework for investigating global production networks (*e.g.*, Caliendo et al. 2022; Huo et al. 2025) through sector-country economic distances, the complexity of firm-to-firm linkages (*e.g.*, di Giovanni et al. 2018; Boehm, Flaaen, et al. 2019) by linking otherwise unconnected firms, and the endogenous formation of production networks (*e.g.*, Carvalho and Voigtländer 2015; Oberfield 2018) under node-specific comparison across a spectrum of competing suppliers. More broadly, the principle of identifying economic proximity through shared interdependencies extends beyond Input-Output economies to investment (*e.g.*, vom Lehn and Winberry 2022), credit (*e.g.*, Delli Gatti et al. 2010), and financial (*e.g.*, Huremović, Jiménez, et al. 2026) networks, suggesting that horizontal geometry provides a general analytical framework for understanding how economic shocks and policy interventions propagate through interconnected systems.

APPENDIX

A. PROOF OF THEOREM 3

Let me derive the three claims one-by-one: in what follows, part *(i)* derives the slope of aggregate output with respect to sectoral productivities; part *(ii)* how this slope changes with productivities; and part *(iii)* what this change implies for the mean of aggregate output. Throughout, $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$, and, for any vector $\boldsymbol{u}$, $\mathrm{Var}_{\widehat{\boldsymbol{a}}_j^{row}}[\boldsymbol{u}] = \sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i^2 - \left(\sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i\right)^2$ is the variance of the entries of $\boldsymbol{u}$ across buyer-j's inputs, weighted by their shares in j's bundle.

(i) Step i.1: real GDP and the price index. With labour fixed at N and the wage w as numeraire, nominal GDP, wN, is constant, and Cobb-Douglas preferences give $\log P = \sum_{i\in\Phi}\omega_i \log p_i + \text{const}$, where P is the aggregate final-consumption price index and p_i is the price of commodity-i. Hence, $\log Y = \log[wN] - \log P$ implies $d\log Y = -\boldsymbol{\omega}' d\log \boldsymbol{p}$, where Y denotes real GDP and $\boldsymbol{p}$ is the vector of sectoral prices. *Step i.2: prices.* Unit costs give $d\log p_i = -d\log z_i + \sum_{j\in\Phi}\alpha_{ij}\, d\log p_j$, that is, $d\log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z}$. *Step i.3: Hulten's theorem.* Substituting, and since Domar weights are collected into $\boldsymbol{\lambda}' = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}$, then

$$d\log Y = \boldsymbol{\omega}'\boldsymbol{\mathcal{L}}\, d\log \boldsymbol{z} = \boldsymbol{\lambda}' d\log \boldsymbol{z}$$

holds at every equilibrium, with $\boldsymbol{\lambda}$ evaluated at current shares. *Step i.4: variance.* Since $\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{0}$ and $\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \nu^2\boldsymbol{I}$, where ν^2 is the common variance of sectoral shocks. then $\mathrm{Var}[\boldsymbol{\lambda}'\boldsymbol{\varepsilon}] = \boldsymbol{\lambda}'\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\boldsymbol{\lambda} = \nu^2\|\boldsymbol{\lambda}\|^2$, in which horizontal operator $\boldsymbol{\mathcal{H}}$ does not appear. *Step i.5: rate.* If $\lambda_i \asymp \frac{1}{m}$ uniformly across sectors, then $\|\boldsymbol{\lambda}\|^2 = \sum_{i\in\Phi}\lambda_i^2 \asymp m\cdot\frac{1}{m^2} = \frac{1}{m} \to 0$.

(ii) Step ii.1: the slope. By part *(i)*, the gradient of $\log Y$ at any vector of productivities is $\boldsymbol{\lambda}'[\boldsymbol{z}]$. *Step ii.2: the Hessian.* Differentiating once more, Theorem 2 gives

$$\frac{\partial^2 \log Y}{\partial \log \boldsymbol{z}\, \partial \log \boldsymbol{z}'} = \frac{\partial \boldsymbol{\lambda}}{\partial \log \boldsymbol{z}'} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$$

which is symmetric (as a canonical Hessian matrix). *Step ii.3: transport, then dispersion.* Along the shock direction, with $d\log \boldsymbol{z} = \boldsymbol{\varepsilon}$ evaluated at the deterministic benchmark, $d^2\log Y = \boldsymbol{\varepsilon}'\boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon} = (\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon})'\,\boldsymbol{\mathcal{H}}\,(\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon})$, where $\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon} = -d\log \boldsymbol{p}$ is the price fall of every commodity. *Step ii.4: covariance form.* For any $\boldsymbol{u}$, then $\boldsymbol{u}'\left[\,\mathrm{Diag}\left[\widehat{\boldsymbol{a}}_j^{row}\right] - \left(\widehat{\boldsymbol{a}}_j^{row}\right)'\widehat{\boldsymbol{a}}_j^{row}\right]\boldsymbol{u} = \sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i^2 - \left(\sum_{i\in\Phi}\widehat{\alpha}_{ji}u_i\right)^2 = \mathrm{Var}_{\widehat{\boldsymbol{a}}_j^{row}}[\boldsymbol{u}]$. Applying it with $\boldsymbol{u} = \boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}$ to each term of $\boldsymbol{\mathcal{H}}$:

$$d^2\log Y = \sum_{j\in\Phi}(\sigma_j - 1)\,\lambda_j\,(1-\alpha_j)\,\mathrm{Var}_{\widehat{\boldsymbol{a}}_j^{row}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right]$$

Each term is minus the curvature of buyer-j's cost index, $(1-\sigma_j)\,\mathrm{Var}_{\widehat{\boldsymbol{a}}_j^{row}}[d\log \boldsymbol{p}]$, weighted by j's intermediate spending relative to GDP.

(iii) Step iii.1: second-order expansion. Let Y^{det} denote real GDP at the deterministic benchmark, before sectoral shocks. Around this deterministic equilibrium, $\log Y = \log Y^{det} + \boldsymbol{\lambda}'\boldsymbol{\varepsilon} + \frac{1}{2}\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon} + R[\boldsymbol{\varepsilon}]$, with the remainder equal to $R[\boldsymbol{\varepsilon}] = o\left[\|\boldsymbol{\varepsilon}\|^2\right]$. As implied by the little-o $[\cdot]$ notation, the remainder is asymptotically negligible relative to $\|\boldsymbol{\varepsilon}\|^2$, meaning that $\lim_{\boldsymbol{\varepsilon}\to\boldsymbol{0}}\frac{R[\boldsymbol{\varepsilon}]}{\|\boldsymbol{\varepsilon}\|^2} = 0$. *Step iii.2: expectations.* The first-order term has mean zero, $\boldsymbol{\lambda}'\mathbb{E}[\boldsymbol{\varepsilon}] = 0$. The second-order term does not: by the cyclic property of the trace identity $\mathbb{E}[\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon}] = \mathrm{Trace}\left[\boldsymbol{\mathcal{R}}\,\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}']\right]$, where

Trace $[\cdot]$ denotes the sum of the diagonal entries of a square matrix, then

$$\mathbb{E}\left[\boldsymbol{\varepsilon}'\boldsymbol{\mathcal{R}}\boldsymbol{\varepsilon}\right] = \text{Trace}\left[\boldsymbol{\mathcal{R}}\,\mathbb{E}\left[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\right]\right] = \nu^2\,\text{Trace}\left[\boldsymbol{\mathcal{R}}\right]$$

Step iii.3: remainder. With $\boldsymbol{\varepsilon} = \nu\boldsymbol{\eta}$, where $\boldsymbol{\eta}$ is a normalized shock vector with $\mathbb{E}\left[\|\boldsymbol{\eta}\|^3\right] < \infty$, and bounded third derivatives of $\log Y$ in a neighbourhood of the deterministic equilibrium, Taylor expansion implies $\left|\mathbb{E}\left[R\right]\right| \leq \Xi\nu^3\,\mathbb{E}\left[\|\boldsymbol{\eta}\|^3\right] = o\left[\nu^2\right]$ for a fixed number of sectors, as $\nu^2 \to 0$, where Ξ is a constant independent of variance ν^2. Taking expectations in Step *iii.1* gives the third statement. Since $\nu^2\,\text{Trace}\left[\boldsymbol{\mathcal{R}}\right] = \sum_{j\in\Phi}\left(\sigma_j - 1\right)\lambda_j\left(1-\alpha_j\right)\mathbb{E}\left[\,\text{Var}_{\widehat{\boldsymbol{a}}_j^{row}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right]\right]$ by part *(ii)* and Step *iii.2*, the mean effect is the expected dispersion of input prices within buyers' bundles, which, unlike the first-order term, cannot average out.

B. EMPIRICS

TABLE 1: DIMENSIONAL DOMINANCE

	network sparsity				
	15% $\tau = 0.5\%$	30% $\tau = 1\%$	40% $\tau = 1.5\%$	50% $\tau = 2\%$	95% $\tau = 5\%$
vertical propagation	.2133	.1995	.1803	.1497	.0762
horizontal propagation	.1371	.1559	.1718	.1731	.1746

Spectral norms for a 3-digit U.S. Input-Output network in year 2007. Due to tresholding-τ, transactions are removed and sparsity is approximated. Source: BEA and own calculations.

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SUPPLEMENTAL APPENDIX

(Outline) *In this appendix I report all the material complementary to the main text; it is made of additional tables and figures, and of discussions on further analytical results. Section A deepens further the theoretical findings. Section B consolidates and further discusses the data presented in Section 4. Finally, Section C complements and enriches the exposition of the empirical findings in Section 5. All the proofs are presented to be intuitive and informative, expanding that in the main text.*

A. THE HOUSEHOLD AS AN ADDITIONAL "SECTOR"

The theory of Sections 2-3 gives the household Cobb-Douglas preferences, so that its expenditure shares across sectors, $\{\omega_i\}_{i\in\Phi}$, are constant and all horizontal transmission runs through the production network, isolating the supply side from the final demand. Instead, this appendix asks what changes when the household has CES preferences with elasticity $\sigma_0 \neq 1$. The answer is organized around a single observation: final demand becomes one more buyer of size one, with no labour share, and with a budget that does not respond to prices. Recall that, for notational reason, the representative household is referred to with index-0. Every result of the paper then holds with one more term, weighted by $\sigma_0 - 1$, while the terms that describe the supply side keep exactly the form derived and discussed in the main text. The benchmark is the special case $\sigma_0 = 1$, in which the household term is Cobb-Douglas and vanishes. When $\sigma_0 - 1$ has the same sign as the elasticities of the production network, the household reinforces the supply-side results; when it has the opposite sign, the two terms may offset each other, and the results of the paper describe the supply-side component of the response. Subsection A.1 sets up the household as a buyer. Subsection A.2 extends the local results of Subsection 3.1, while A.3 the factorization of Subsection 3.2. The results on employment, aggregate output and volatility of Subsection 3.3 are generalised in Subsection A.4.

A.1. The household as a buyer

The representative household inelastically supplies N units of labour at wage w, and consumes

$$C = \left(\sum_{i\in\Phi} \gamma_{0i}^{1/\sigma_0} c_i^{(\sigma_0-1)/\sigma_0} \right)^{\sigma_0/(\sigma_0-1)} \qquad \text{and} \qquad P = \left(\sum_{i\in\Phi} \gamma_{0i}\, p_i^{1-\sigma_0} \right)^{1/(1-\sigma_0)}$$

where P is the consumer price index and real GDP is $Y = \frac{wN}{P}$. Expenditure shares, $\omega_i = \frac{p_i c_i}{wN} = \gamma_{0i} \frac{p_i^{1-\sigma_0}}{P^{1-\sigma_0}}$, are no longer constant, and respond to prices as

$$d\log \omega_i = (1-\sigma_0) \left(d\log p_i - \sum_{j\in\Phi} \omega_j\, d\log p_j \right) \tag{A.1}$$

when a good becomes cheaper relative to the others, its share of final expenditure rises if goods are substitutes in consumption ($\sigma_0 > 1$) and falls if they are complements ($\sigma_0 < 1$).

***Final demand as buyer*-0.–** Let $\Phi_0 = \{0\} \cup \Phi$, and give final demand the attributes of a buyer: size $\lambda_0 = 1$, since its budget is GDP; labour share $\alpha_0 = 0$, since it spends all of its

budget on final goods; portfolio $\widehat{\boldsymbol{a}}_0^{row} = \boldsymbol{\omega}'$, that is, $\widehat{\alpha}_{0i} = \omega_i$; and demand elasticity $\eta_0 = 1$, since its nominal budget, wN, does not respond to prices. With this convention, market clearing in value includes final demand as one of the buyers of every good,

$$\lambda_i = \sum_{s \in \Phi_0} \lambda_s \left(1 - \alpha_s\right) \widehat{\alpha}_{si} \qquad \text{and, summing over } i, \qquad \sum_{s \in \Phi_0} \lambda_s \left(1 - \alpha_s\right) = \sum_{i \in \Phi} \lambda_i$$

Total downstream buyers' budget, household included, equals total sales from upstream. Let $\boldsymbol{\mathcal{C}}_s = \text{Diag}\left[\widehat{\boldsymbol{a}}_s^{row}\right] - \left(\widehat{\boldsymbol{a}}_s^{row}\right)' \widehat{\boldsymbol{a}}_s^{row}$ be the $m \times m$ covariance matrix of buyer-s's portfolio, so that $\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{u}\right] = \boldsymbol{u}' \boldsymbol{\mathcal{C}}_s \boldsymbol{u}$ and $\text{Cov}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{u}, \boldsymbol{v}\right] = \boldsymbol{u}' \boldsymbol{\mathcal{C}}_s \boldsymbol{v}$, and let $\boldsymbol{\ell}_i^{col}$ be the i-th column of the Leontief inverse $\boldsymbol{\mathcal{L}}$. For the household, $\boldsymbol{\mathcal{C}}_0 = \text{Diag}\left[\boldsymbol{\omega}\right] - \boldsymbol{\omega}\boldsymbol{\omega}'$.

Three properties carry over unchanged, because they depend on technology and on the household's budget, not on its preferences. First, prices: unit costs depend only on technology and on the wage, so that $d \log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\, d \log \boldsymbol{z}$. Second, the Hulten's (1978) theorem: by Shephard's lemma applied to P, $d \log P = \sum_{i \in \Phi} \omega_i \, d \log p_i$, so that $d \log Y = -\boldsymbol{\omega}' d \log \boldsymbol{p} + d \log N = \boldsymbol{\lambda}' d \log \boldsymbol{z} + d \log N$, with $\boldsymbol{\lambda}' = \boldsymbol{\omega}' \boldsymbol{\mathcal{L}}$, at every equilibrium. Third, employment: labour shares are constant, so that $n_i = \alpha_i \lambda_i N$ and $\sum_{i \in \Phi} \alpha_i \lambda_i = 1$. The household changes how Domar weights move, not how prices move: this is why every result below differs from the benchmark only through the response of Domar weights.

LEMMA 1 (Leontief columns and buyers' dispersion) *Considering the final demand as an additional common downstream buyer:*

- *for every i and every $j \neq i$, then $\ell_{ji} \leq \left(1 - \alpha_j\right) \ell_{ii}$;*
- *for every buyer $s \in \Phi_0$ and good $i \in \Phi$, $\text{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col}\right] \geq \underline{\alpha}^2 \widehat{\alpha}_{si}\left(1 - \widehat{\alpha}_{si}\right)$, where $\underline{\alpha} = \min_i \alpha_i > 0$; for the household, moreover, $\sum_{i \in \Phi} \text{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] \geq 1 - \|\boldsymbol{\lambda}\|^2$.*

Proof: for the first claim, column-i of $\boldsymbol{\mathcal{L}} = \boldsymbol{I} + \boldsymbol{\mathcal{A}}\boldsymbol{\mathcal{L}}$ gives, for $j \neq i$, $\ell_{ji} = \sum_{s \in \Phi} \alpha_{js} \ell_{si} \leq \left(1 - \alpha_j\right) \max_s \ell_{si} < \max_s \ell_{si}$, since $\max_s \ell_{si} \geq \ell_{ii} \geq 1$. The maximum of column-$i$ is therefore attained only at ℓ_{ii}, and $\ell_{ji} \leq \left(1 - \alpha_j\right) \ell_{ii}$. For the second claim, draw one of buyer-s's inputs, final good-j with probability $\widehat{\alpha}_{sj}$, and record ℓ_{ji}. Splitting the draws into final good-i, with probability $\widehat{\alpha}_{si}$, and the other final goods, the law of total variance bounds the variance from below by the dispersion between the two group means, $\widehat{\alpha}_{si}\left(1 - \widehat{\alpha}_{si}\right)\left(\ell_{ii} - \sum_{j \neq i} \widehat{\alpha}_{sj} \ell_{ji} / \left(1 - \widehat{\alpha}_{si}\right)\right)^2$; by the first claim, the second mean is at most $\left(1 - \underline{\alpha}\right) \ell_{ii}$, so that the gap is at least $\underline{\alpha}\, \ell_{ii} \geq \underline{\alpha}$. For the household, $\sum_j \omega_j \ell_{ji} = \lambda_i$, so that $\text{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] = \sum_j \omega_j \ell_{ji}^2 - \lambda_i^2 \geq \omega_i \ell_{ii}^2 - \lambda_i^2 \geq \omega_i - \lambda_i^2$; summing over i gives the result. ■

A.2. Local transmission at final demand

At the shared node, the household enters Propositions 1 and 2 in two ways: as a common buyer of the final goods it consumes, and as one of the buyers of a congested supplier.

PROPOSITION 4 (Final demand as a shared node) *At first round, the household's spending on good-i responds to the price of good-j as*

$$\delta_{\to 0}\left[i, j\right] = \frac{\partial \log\left[p_i \, c_i\right]}{\partial \log p_j} = \left(\sigma_0 - 1\right) \omega_j$$

and the response of i's total sales in Proposition 2 gains the term $\left(\sigma_0 - 1\right) \omega_i \frac{\omega_j}{\lambda_i}$, that is, the sum extends to all common buyers $s \in \Phi_0 \setminus \{i, j\}$. (ii) Proposition 1 holds with the pass-through

$$\widetilde{\varphi}_s = \frac{\varphi_s}{1 + \varphi_s\left(\mu_{s0}\,\chi_{0s} + \sum_{i\in\Phi}\mu_{si}\,\chi_{is}\right)} \qquad \textit{with} \quad \chi_{0s} = \sigma_0\,(1-\omega_s) + \omega_s$$

and the household's consumption of the shared supplier-s responds to the expansion of buyer-j as $\frac{\partial \log c_s}{\partial \tilde{\xi}_j} = -\,\chi_{0s}\,\widetilde{\varphi}_s\,\mu_{sj}$.

Proof: the household's budget, wN, is given, so that $d\log[p_i c_i] = d\log\omega_i$; by eq. (A.1), with only p_j moving, $d\log\omega_i = (1-\sigma_0)(0-\omega_j\,d\log p_j) = (\sigma_0-1)\omega_j\,d\log p_j$. Final demand accounts for a share $\frac{\omega_i}{\lambda_i}$ of i's sales, which gives the term in $\frac{\partial \log p_i y_i}{\partial \log p_j}$; it coincides with the summand of Proposition 2 at $s=0$, since $\lambda_0 = 1$, $\alpha_0 = 0$, $\eta_0 = 1$ and $(1-\alpha_0)\,\widehat{\alpha}_{0i} = \omega_i$. This proves the first part. Then, by eq. (A.1), with only p_s moving, $d\log c_s = d\log\omega_s - d\log p_s = \left[(1-\sigma_0)(1-\omega_s) - 1\right] d\log p_s = -\chi_{0s}\,d\log p_s$, which replaces the object $-d\log p_s$ in Step 4 of the proof of Proposition 1: market clearing now gives $d\log y_s = -\mu_{s0}\chi_{0s}\,d\log p_s + \sum_i \mu_{si}\,d\log x_{is}$, and the remaining steps are unchanged. ■

The household is the cleanest common buyer of the economy. Its budget is fixed, so that the vertical term of Proposition 2, $(1-\alpha_s)(1-\eta_s)$, is identically zero, and its transmission is purely horizontal: when good-j becomes cheaper, the household reallocates a given budget, and the spending on every other good moves by $(\sigma_0-1)\,\omega_j\,d\log p_j$, with a sign set by σ_0 alone. This is the horizontal channel of standard multi-sector models without intermediate inputs, as in the "half-spider" of Figure 2. At a common supplier, the household adds one more margin of substitution away from a congested good, χ_{0s}, which equals one in the benchmark: it changes the size of the price pass-through, not the vertical nature of the channel.

A.3. General equilibrium

The equilibrium reallocation across the whole network (Theorem 2) changes accordingly.

THEOREM 4 (Factorization with a CES household) *To first order, Domar weights respond to sectoral shocks as*

$$d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0\right)\boldsymbol{\mathcal{L}}\,d\log\mathbf{z} = \left(\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right)d\log\mathbf{z}$$

with $\boldsymbol{\mathcal{H}}_0 = (\sigma_0-1)\left[\mathrm{Diag}\left[\boldsymbol{\omega}\right] - \boldsymbol{\omega}\boldsymbol{\omega}'\right]$, *where* $\boldsymbol{\mathcal{H}}$ *and* $\boldsymbol{\mathcal{R}} = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}\boldsymbol{\mathcal{L}}$ *are the operators of Theorem 2, and* $\boldsymbol{\mathcal{R}}_0 = \boldsymbol{\mathcal{L}}'\boldsymbol{\mathcal{H}}_0\boldsymbol{\mathcal{L}}$. *Equivalently,*

$$\frac{\partial\lambda_i}{\partial\log z_j} = \sum_{s\in\Phi}(\sigma_s-1)\,\lambda_s\,(1-\alpha_s)\,\mathrm{Cov}_{\widehat{\mathbf{a}}_s^{row}}\left[\boldsymbol{\ell}_i^{col},\boldsymbol{\ell}_j^{col}\right] + (\sigma_0-1)\,\mathrm{Cov}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col},\boldsymbol{\ell}_j^{col}\right]$$

where the last term is the household-based covariance.

Proof: let me derive each step separately. *Step 1*. Differentiating $\boldsymbol{\lambda}' = \boldsymbol{\omega}' + \boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$, $d\boldsymbol{\lambda}' = d\boldsymbol{\omega}' + \boldsymbol{\lambda}'d\boldsymbol{\mathcal{A}} + d\boldsymbol{\lambda}'\boldsymbol{\mathcal{A}}$: Domar weights move because the household reallocates its budget, because buyers reallocate their intermediate budgets, and because the size of each buyer changes. *Step 2*. By eq. (A.1), $d\omega_i = (1-\sigma_0)\,\omega_i\left(d\log p_i - \sum_j \omega_j\,d\log p_j\right)$, that is, $d\boldsymbol{\omega} = (1-\sigma_0)\,\boldsymbol{\mathcal{C}}_0\,d\log\boldsymbol{p}$. *Step 3*. Under CES minimization, $d\alpha_{si} = (1-\alpha_s)\,d\widehat{\alpha}_{si} = (1-\alpha_s)(1-\sigma_s)\,\widehat{\alpha}_{si}\left(d\log p_i - \sum_j \widehat{\alpha}_{sj}\,d\log p_j\right)$, so that the i-th entry of $\boldsymbol{\lambda}'d\boldsymbol{\mathcal{A}}$ is $\sum_{s\in\Phi}(1-\sigma_s)\,\lambda_s\,(1-\alpha_s)\left[\boldsymbol{\mathcal{C}}_s\,d\log\boldsymbol{p}\right]_i$. Since $\lambda_0(1-\alpha_0) = 1$, Steps 2 and 3 combine into a single sum over $s\in\Phi_0$. *Step 4*. Solving Step 1 for $d\boldsymbol{\lambda}$, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\sum_{s\in\Phi_0}(1-\sigma_s)\,\lambda_s\,(1-\alpha_s)\,\boldsymbol{\mathcal{C}}_s\,d\log\boldsymbol{p}$. *Step 5*. Substituting $d\log\boldsymbol{p} = -\boldsymbol{\mathcal{L}}\,d\log\mathbf{z}$ gives $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{L}}'\left(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0\right)\boldsymbol{\mathcal{L}}\,d\log\mathbf{z}$, where the terms $s\in\Phi$ form $\boldsymbol{\mathcal{H}}$ and the term $s=0$ is $\boldsymbol{\mathcal{H}}_0$. The covariance form follows from $\boldsymbol{\mathcal{L}}\boldsymbol{e}_i = \boldsymbol{\ell}_i^{col}$ and $\boldsymbol{u}'\boldsymbol{\mathcal{C}}_s\boldsymbol{v} = \mathrm{Cov}_{\widehat{\mathbf{a}}_s^{row}}[\boldsymbol{u},\boldsymbol{v}]$. ■

The theorem is the sense in which the household extends the results without changing them. The supply-side operator, $\boldsymbol{\mathcal{R}}$, has the form derived in the text, and the household adds one more

buyer to the sum that defines it: the largest buyer of the economy, of size one, whose portfolio is final demand. The household term inherits the logic of the factorization. Read from right to left, $\boldsymbol{\mathcal{L}}$ transports a shock to the price of every good, $\boldsymbol{\mathcal{H}}_0$ reallocates final expenditure according to how unevenly these prices have moved, and $\boldsymbol{\mathcal{L}}'$ transports the change in final demand for each good up to its suppliers. Two features distinguish it from the supply-side term. First, its rivalry component is rank one: since final spending embodies sector-i by $\sum_j \omega_j \ell_{ji} = \lambda_i$, then

$$\boldsymbol{\mathcal{R}}_0 = (\sigma_0 - 1)\left(\boldsymbol{\mathcal{L}}' \operatorname{Diag}[\boldsymbol{\omega}]\, \boldsymbol{\mathcal{L}} - \boldsymbol{\lambda}\boldsymbol{\lambda}'\right)$$

so that the household makes every pair of sectors rivals in proportion to the product of their Domar weights, including sectors that never sell to final demand and compete only through the consumer goods that embody them. The supply-side rivalry, $-\boldsymbol{\mathcal{T}}'\widetilde{\boldsymbol{\mathcal{W}}}\boldsymbol{\mathcal{T}}$, is instead shaped by which buyers require which goods, and it is this term that carries the horizontal geometry of the production network. Correspondingly, the unweighted counterpart of $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0$ is $\boldsymbol{\mathcal{B}}^{fs} + \boldsymbol{\omega}\boldsymbol{\omega}'$. Second, the household term is active without any production network: with $\boldsymbol{\mathcal{A}} = \mathbf{0}$, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{H}}_0\, d\log \boldsymbol{z}$, the reallocation of standard multi-sector models; and with Cobb-Douglas intermediates, $d\boldsymbol{\lambda} = \boldsymbol{\mathcal{R}}_0\, d\log \boldsymbol{z}$, so that upstream sectors inherit, through $\boldsymbol{\mathcal{L}}'$, the reallocation of the household among the consumer goods they supply.

COROLLARY 4 (Signs and proliferation with a CES household) *Considering the final demand as an additional common downstream buyer:*

- *under Cobb-Douglas intermediates and preferences ($\sigma_s = 1$, $\forall s \in \Phi_0$), $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0 = \mathbf{0}$; with only CES final demand, hence under Cobb-Douglas intermediates alone, $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0 = \boldsymbol{\mathcal{H}}_0$, which is non-zero whenever $\sigma_0 \neq 1$ and the household buys at least two final goods;*
- *if $\sigma_s \geq 1$ ($\sigma_s \leq 1$), $\forall s \in \Phi_0$, then $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0$ and $\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0$ are positive (negative) semi-definite, and all off-diagonal entries of $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0$ are non-positive (non-negative);*
- *if, in addition, $\sigma_s \neq 1$ and $\lambda_s > 0$ for all $s \in \Phi_0$, then, for $i \neq j$, $\mathcal{H}_{i,j} + [\boldsymbol{\mathcal{H}}_0]_{ij} \neq 0$ if and only if $b^{fs}[i,j] + \omega_i \omega_j > 0$, and the rivalry generated by the household, relative to its purchases, is $|\sigma_0 - 1|\left(1 - \|\boldsymbol{\omega}\|^2\right)$.*

Proof: for the first claim, note that every term of $\sum_{s\in\Phi_0}$ carries $\sigma_s - 1$; if only $\sigma_0 \neq 1$, the sum reduces to $\boldsymbol{\mathcal{H}}_0$, whose off-diagonal entries, $-(\sigma_0 - 1)\,\omega_i\omega_j$, are non-zero for any two goods with positive final demand. For the second claim, each $\boldsymbol{\mathcal{C}}_s$ is a covariance matrix, hence positive semi-definite, and it enters with a weight of the sign of $\sigma_s - 1$; the quadratic form of $\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0$ at $\boldsymbol{x}$ is that of $\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0$ at $\boldsymbol{\mathcal{L}}\boldsymbol{x}$, and has the same sign; the off-diagonal entries of $\boldsymbol{\mathcal{C}}_s$ are $-\widehat{\alpha}_{si}\widehat{\alpha}_{sj} \leq 0$. In the third claim, for $i \neq j$, then $\mathcal{H}_{i,j} + [\boldsymbol{\mathcal{H}}_0]_{ij} = -\sum_{s\in\Phi_0}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\,\widehat{\alpha}_{si}\widehat{\alpha}_{sj}$: a sum of terms of the same sign, non-zero if and only if some buyer (household included) purchases both goods, that is, if and only if $\sum_{s\in\Phi}\alpha_{si}\alpha_{sj} + \omega_i\omega_j > 0$. The off-diagonal entries of $\boldsymbol{\mathcal{C}}_0$ sum in absolute value to $\sum_i \omega_i\,(1-\omega_i) = 1 - \|\boldsymbol{\omega}\|^2$, and the household's purchases are $\lambda_0\,(1-\alpha_0) = 1$. ■

The sign results of Corollary 1 extend once the household's elasticity is included in the sign conditions. Without that condition, the supply-side statements continue to describe $\boldsymbol{\mathcal{R}}$, and $\boldsymbol{\mathcal{R}}_0$ is the household's contribution: with complementary intermediates and substitutable consumption goods, for instance, the network makes a sector's rivals at common buyers gain from its gain, while the household makes them lose. The last claim adds the household to Corollary 3. The household is a buyer with an exceptionally large portfolio: it supports a horizontal relation between every pair of goods with positive final demand, and its rivalry, $|\sigma_0 - 1|\left(1 - \|\boldsymbol{\omega}\|^2\right)$,

is close to its maximum when final expenditure is spread over many goods. A new consumer good becomes, at once, a rival of all the others.

A.4. Employment, aggregate output and volatility

Changes in final demand also reallocate workers upstream.

PROPOSITION 5 (Employment reallocation with a CES household) *To first order, sectoral employment responds to shocks as*

$$d\log \boldsymbol{n} = \underbrace{d\log N\,\mathbf{1}}_{\text{common}} + \underbrace{\boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}\,d\log \boldsymbol{z}}_{\text{production network}} + \underbrace{\boldsymbol{\Lambda}^{-1}\boldsymbol{\mathcal{R}}_0\,d\log \boldsymbol{z}}_{\text{household}}$$

with $\boldsymbol{\Lambda} = \operatorname{Diag}[\boldsymbol{\lambda}]$ *and* $\boldsymbol{\alpha}'(\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0) = \mathbf{0}$.

Proof: by the third property of Subsection A.1, $d\log n_i = d\log N + \frac{d\lambda_i}{\lambda_i}$, and $d\boldsymbol{\lambda}$ is given by Theorem 4. Since $\boldsymbol{\alpha}'\boldsymbol{\lambda} = 1$ at every equilibrium and $\boldsymbol{\alpha}$ is constant, $\boldsymbol{\alpha}' d\boldsymbol{\lambda} = 0$ for every shock. ■

Reallocation remains zero-sum, and the household adds a component that does not require intermediate inputs. In the half-spider, where $\boldsymbol{\mathcal{A}} = \mathbf{0}$ and $\alpha_i = 1$, $\boldsymbol{\lambda} = \boldsymbol{\omega}$ and the proposition reduces to $d\log n_i - d\log N = (\sigma_0 - 1)\left(d\log z_i - \sum_j \omega_j\, d\log z_j\right)$: workers move towards sectors whose productivity rises more than average if final goods are substitutes in consumption, and away from them if they are complements.

THEOREM 5 (Aggregate output with a CES household) *Let* $\boldsymbol{\varepsilon} = d\log \boldsymbol{z}$*, with independent shocks of mean zero and variance* ν^2*, and hold labour fixed. Then:*

- *to first order,* $d\log Y = \boldsymbol{\lambda}'\boldsymbol{\varepsilon}$*, so that* $\operatorname{Var}[d\log Y] = \nu^2\|\boldsymbol{\lambda}\|^2$*, as in Theorem 3;*
- *to second order,* $d^2\log Y = \boldsymbol{\varepsilon}'(\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0)\boldsymbol{\varepsilon} = \sum_{s\in\Phi}(\sigma_s - 1)\,\lambda_s\,(1-\alpha_s)\operatorname{Var}_{\widehat{\boldsymbol{a}}_s^{row}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right] + (\sigma_0 - 1)\operatorname{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}\right]$*;*
- *consequently, it holds* $\mathbb{E}[\log Y] - \log Y^{det} = \frac{1}{2}\nu^2 \operatorname{Trace}\left[\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right] + o\left[\nu^2\right]$*, where the household term has the sign of* $\sigma_0 - 1$ *and is bounded away from zero by*

$$\left|\operatorname{Trace}[\boldsymbol{\mathcal{R}}_0]\right| = |\sigma_0 - 1| \sum_{i\in\Phi} \operatorname{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right] \geq |\sigma_0 - 1| \max\left\{1 - \|\boldsymbol{\lambda}\|^2,\ \underline{\alpha}^2\left(1 - \|\boldsymbol{\omega}\|^2\right)\right\}$$

Proof: the first claim follows from the second property of Subsection A.1. Since the shocks are independent, have mean zero, and variance ν^2, $\operatorname{Var}[d\log Y] = \nu^2\|\boldsymbol{\lambda}\|^2$. For the second claim, the same property holds at every equilibrium, so the Hessian of $\log Y$ with respect to log productivities is $\frac{\partial\boldsymbol{\lambda}}{\partial \log \boldsymbol{z}'} = \boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0 = \boldsymbol{\mathcal{L}}'(\boldsymbol{\mathcal{H}} + \boldsymbol{\mathcal{H}}_0)\boldsymbol{\mathcal{L}}$, by Theorem 4. Here, $\boldsymbol{\mathcal{R}}$ captures firms' production curvature, while $\boldsymbol{\mathcal{R}}_0$ captures the additional curvature induced by the CES household. Expanding the quadratic form buyer by buyer gives the stated expression in terms of weighted variances, where $\widehat{\boldsymbol{a}}_s^{row}$ denotes sector-s's normalized input shares and $\boldsymbol{\omega}$ the household's expenditure shares. The corresponding elasticities of substitution are σ_s for sectoral buyers and σ_0 for the household. The household term is the curvature of the CES consumer price index. Differentiating Shephard's lemma once more gives $d^2\log P = \sum_{i\in\Phi}\omega_i d^2\log p_i + (1-\sigma_0)\operatorname{Var}_{\boldsymbol{\omega}}\left[d\log \boldsymbol{p}\right]$, where P is the consumer price index and p_i is the price of sector-i. Since $d\log \boldsymbol{p} = -\boldsymbol{\mathcal{L}}\boldsymbol{\varepsilon}$, this yields the household variance term in the theorem. For the third claim, the Taylor expansion in the proof of Theorem 3 applies with $\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0$, giving $\mathbb{E}\left[\log Y\right] - \log Y^{det} = \frac{1}{2}\nu^2 \operatorname{Trace}\left[\boldsymbol{\mathcal{R}} + \boldsymbol{\mathcal{R}}_0\right] + o\left[\nu^2\right]$. Moreover, $\operatorname{Trace}\left[\boldsymbol{\mathcal{R}}_0\right] = \sum_{i\in\Phi}\left(\boldsymbol{\ell}_i^{col}\right)'\boldsymbol{\mathcal{H}}_0\boldsymbol{\ell}_i^{col} = (\sigma_0 - 1)\sum_{i\in\Phi}\operatorname{Var}_{\boldsymbol{\omega}}\left[\boldsymbol{\ell}_i^{col}\right]$, where $\boldsymbol{\ell}_i^{col}$ is the i-th column of $\boldsymbol{\mathcal{L}}$. Its sign is therefore determined by $\sigma_0 - 1$, and the lower bound follows from Lemma 1, summed over $i \in \Phi$ and using $\sum_{i\in\Phi}\omega_i = 1$. ■

The first-order results mirror those of the main text (Theorem 3): the household changes neither Hulten's (1978) theorem nor the variance of aggregate output, so that Corollary 2 and

the diversification argument for aggregate volatility are unaffected. The household enters the curvature as one more buyer whose input prices move unevenly, with weight one. Two differences from the supply-side bound are worth noting. First, the factor $1 - \overline{\alpha}$ disappears because the household's budget is the whole of GDP, so that the bound holds even without intermediate inputs. Second, the household term is bounded below by a statistic of Domar weights alone, $1 - \|\boldsymbol{\lambda}\|^2$, which tends to one whenever Domar weights are balanced. When $\sigma_0 - 1$ and the intermediate elasticities have opposite signs, the two components of the mean effect offset each other, and its sign is an empirical matter.

B. Data, figures and tables

B.1. Data details

In order to build the main dataset with a panel of 65 private 3-digit U.S. 2017 North American Industry Classification System (NAICS) sectors used in Sections 4-5, I rely on different data sources from the Bureau of Economic Analysis (BEA), otherwise stated.[I] For years from 1998 to 2022, these information are:

(a) employment (Persons Engaged in Production by Industry, in thousands. Last revised on: September 29, 2023);

(b) value-added (Value-Added by Industry, in billions of U.S. dollars. Last revised on: December 19, 2024);

(c) aggregate Total Factor Productivity (from Fernald 2014) and Government Consumption Expenditures and Gross Investment (*e.g.*, Ramey and Zubairy 2018);[II]

(d) inter-sectoral trade and Input-Output linkages, in particular:

- sectoral share of own production, and thus the sectoral share of intermediate inputs from other sectors (Industry-by-Commodity Market Share Matrix, After Redefinitions – Summary, in producers' prices);
- total inputs required (directly and indirectly) in order to deliver one dollar of output to final users (Commodity-by-Commodity Total Requirements, After Redefinitions – Summary, in producers' prices);
- sector-specific imports and exports of goods and services, government (total, defence, non-defence) expenditure shares, as well as changes in private inventories (The Use of Commodities by Industries, Before Redefinitions – Summary, in producers' prices and in millions of U.S. dollars).

Note that "Summary" defines the level of sectoral disaggregation considered in the analysis (3-digit approximately). The BEA also provides information on Input-Output matrices to a

[I] Note that I refer to 3-digit but, according to the BEA classification, four sectors are at 2-digit ("construction", "management of companies and enterprises", "educational services", and "other services, except government"), while five sectors related to finance and insurance are at 4-digit level.

[II] Here is the `website` for the TFP series. The U.S. government expenditure series is downloaded from the Federal Reserve Economic Data (FRED) database, with reference name *GCEA*; such series, net of federal defence expenditure (named *FDEFX*), represents the non-defence expenditure. Last revised on: June 15, 2026.

more detailed level (6-digit), but these are on a five-yearly basis (thus it would impossible to run the analysis in Subsection 5.1), and data on other variables are not collected. Moreover, the sample begins in 1998 so to have all sectors classified given the NAICS system, and ends in 2022, the last year available when I started this project.[III] Last time I accessed these public production network data was in March 2025.

Turn to the manipulation of Input-Output matrices. To recover the production share of intermediate inputs of a given sector coming from the production of other sectors, each cell of the yearly squared matrix contains the intermediate output share from a given sector to another and, on the main diagonal, the share of intermediate output of a given sector directly produced by that sector. Henceforth, labelling by $\boldsymbol{X} = \left[x_{ij}\right]$, with $x_{ij} \geq 0$, this Input-Output matrix I compute, for each sector, $x_i = 1 - x_{ii}$. In other words, to extract the total share of intermediate inputs that sector-i buys from other sectors, I compute $\boldsymbol{x} = \mathbf{1} - \left[x_{ii}\right]_{i \in \Phi}$, where $\mathbf{1}$ is an $m \times 1$ vector of ones; the resulting $m \times 1$ vector $\boldsymbol{x}$ reports, for every considered year, the total production of a sectoral good which is due to intermediate inputs bought through Input-Output relationships. For what concerns the production network structure, I rely on the main Input-Output matrix (*Commodity-by-Commodity Total Requirements, After Redefinitions*) which, consistently with the main text, I label as $\boldsymbol{\mathcal{L}} = \left[\ell_{ij} \geq 0\right]$.[IV] Such Leontief inverse matrix investigates its deep characteristics and, following the literature (*e.g.*, Acemoglu et al. 2012; Carvalho 2014) I disregard small transactions: this allows me to account for a narrower network amplification due to the removal of bias from meaningless inter-sectoral trade intensity. Disregarding small transactions implies that $\ell_{ij} = \frac{\ell_{ij}}{\ell_i}$ is set to 0 if $\ell_{ij} \leq 0.01$, where $\ell_i = \sum_{j \in row} \ell_{ij}$ is sector-i's total input purchases (*i.e.*, *row sum* of matrix $\boldsymbol{\mathcal{L}}$), or $\ell_i = \sum_{j \in column} \ell_{ij}$ is sector-i's total input sales (*i.e.*, *column sum* of matrix $\boldsymbol{\mathcal{L}}$).[V] The resulting matrix would therefore display each cell to be greater than 0.01 – *i.e.*, larger than 1%. Leontief inverse matrix, upstream and downstream sectors, and the weight associated to each of them (as in Subsections 5.1-5.2) are identified through it.

To compute network distances between sectors, I rely on the promptly-downloadable BEA Input-Output table (*i.e.*, the *Commodity-by-Commodity Total Requirements – After Redefinitions*), which reports the total output of each commodity required to satisfy one unit of final demand for any given commodity. By incorporating both direct and indirect production requirements, this table captures Input-Output linkages extending throughout the production network and along the entire supply chain. This representation is particularly suited to my purpose, as it allows me to compare sectors according to common Input-Output structures not only at the local level, through their direct relationships, but also across the full length of their respective supply chains, through the indirect linkages connecting them to further upstream and downstream stages of production. Similarities between sectors reflect commonalities in their production net-

[III] The 3-digit classification is the most granular if one wants to keep track of the evolution in the set of intermediate inputs, and 1998 is the second year in which the NAICS system has been adopted. Before 1997, classification follows the U.S. Standard Industrial Classification (SIC) system.

[IV] Row sectors identify (upstream) suppliers, while column sectors the (downstream) buyers: row sector is the origin and column sector is the destination of the circulating intermediate input.

[V] The "sales-based" Leontief inverse (in the main text) is a normalization of the baseline Input-Output matrix by sectoral total sales, so that each entry represents the share of a sector's inputs relative to its output.

work exposure at different depths of the supply chain, rather than being driven solely by their immediate bilateral connections. Accordingly, the Euclidean distances (eqs. B.1-B.2) between sectors are computed from differences in their total-requirements profiles, capturing the extent to which sectors share similar buyers/suppliers across the production network. From these distances, I compute the *intensive margin* values for both network "economic" distance metrics.

(Shortest path algorithm, technical but suitable for coding) *Given element* $\boldsymbol{\mathcal{G}}$ *being an* $m \times m$ *adjacency matrix with non-negative discrete values, the idea is to compute a distance matrix* $\boldsymbol{\mathcal{D}}$ *where each cell* $d[i,j]$ *is the length of the shortest path between node-i and node-j. The numerical algorithm is as follows:*

(a) from the Input-Output matrix $\boldsymbol{\mathcal{A}}$*, build an associated identity matrix,* $\boldsymbol{I}$*;*

(b) set the initial length of the path, $k = 1$*, and build a length matrix* $\boldsymbol{\mathcal{G}}_i = \boldsymbol{\mathcal{G}}$*;*

(c) build a Boolean matrix, $\boldsymbol{B}$*, whose elements are* $b_{ij} = 1$ *if* $\boldsymbol{\mathcal{G}}_i \neq 0$*;*

(d) as long as matrix $\boldsymbol{B}$ *contains a value different from zero (i.e., there are connections among nodes not considered):*

- *update identity matrix* $\boldsymbol{I}$ *by adding* k *whenever the condition on* $\boldsymbol{B}$ *is true, that is,* $I_{ij} = I_{ij} + k$ *if* $b_{ij} = 1$*;*
- *update lengths,* $k = k + 1$*;*
- *algebraic matrix calculation:* $\boldsymbol{\mathcal{G}}_i = \boldsymbol{\mathcal{G}}_i' \boldsymbol{\mathcal{G}}$*;*
- *update* $\boldsymbol{B}$ *with new paths: set* $b_{ij} = 1$ *if* $\mathcal{G}_{ij,k} \neq 0$ *and* $d[i,j] = 0$*;*

(e) once this loop concludes, set to infinite all the disconnected nodes, that is $d_{ij} = \infty$ *if* $d[i,j] = 0$*, to check consistency;*

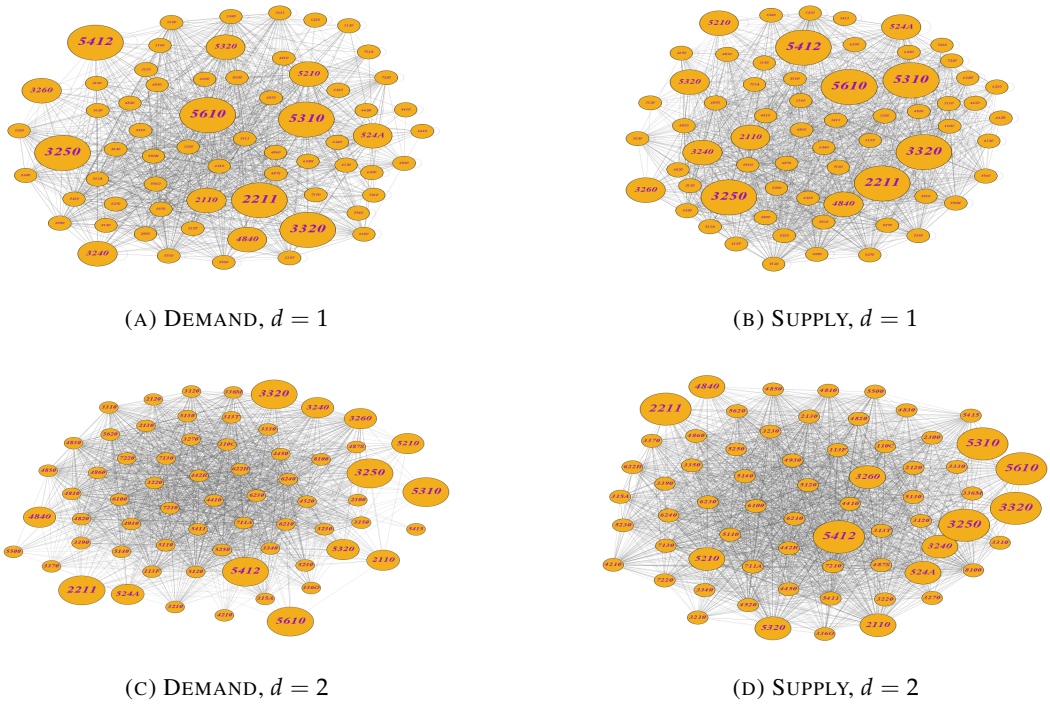
(g) set to the minimum distance the elements on the main diagonal, $\boldsymbol{\mathcal{D}} = \boldsymbol{\mathcal{D}} - \boldsymbol{I}$*.*

Outside this routine set to zero all the disconnected nodes (those at ∞*). Matrix* $\boldsymbol{\mathcal{D}}_{ext}$*, whose values are at the extensive margin and identify network distances among any pair* $\{i,j\}$*, is built. The generic element identifying the existence of a link between node-i and node-j in the distance matrix* $\boldsymbol{\mathcal{D}}$ *is then* $d_{ext}[i,j] = \{1, 2, \ldots, d_{max}\}$*.*

B.2. Characteristics of the distance networks

To illustrate the constructed distance matrices Figure B.1 displays, for each extensive margin value, the horizontal linkages among sectors from both common demand and supply network ties. Importantly, each connection does not necessarily represent inter-sectoral trade between two sectors, but rather it reflects an horizontal linkage based on demand- or supply-driven network distance relationships.

A few concatenated points are then worth noting. First, the distribution of horizontal complementarities is uneven across distance levels. At very short range ($d = 1$), the demand-based network appears slightly denser than the supply-based one, suggesting that sectors are more frequently linked through common upstream suppliers than through shared downstream buyers. However, this asymmetry vanishes as the distance increases ($d = 2$), where both types



(A) DEMAND, $d = 1$

(B) SUPPLY, $d = 1$

(C) DEMAND, $d = 2$

(D) SUPPLY, $d = 2$

FIGURE B.1: VISUALIZATION OF PRODUCTION NETWORK DISTANCES

Note: each panel of the figure represents the distance-based inter-sectoral production network corresponding to the (*commodity-by-commodity total direct requirements*) U.S. Input-Output matrix in the year 2007. *Factor input demand* and *factor input supply* network distances are computed implementing the *shortest path algorithm* on distance matrices whose values are at the intensive margin, as well derived from the *directed* configuration of the production network. Panels B.1a and B.1c correspond to the demand-based network distances, with $d = 1$ for low distance (closest sectors) and $d = 2$ for major distance (further sectors), among any pair of sectors; analogously it is for the supply-based network distances in Panels B.1b and B.1d. There are 65 3-digit U.S. 2017 NAICS private sectors, and each orange node corresponds to one of them. Two sectors connected under $d = 1$ cannot be linked under $d = 2$, and otherwise; constructed distance matrices are full (Theorem 1). Biggest nodes correspond to top 10% of mostly connected sectors (*i.e.*, major number of inter-sectoral connections), while intermediate ones the additional top 20%. Figure drawn with the software package *Gephi*, version 0.10, exploiting the *ForceAtlas2* layout algorithm. *Source*: BEA and own calculations.

of horizontal ties spread more diffusely across the network. Second, sectors with the highest number of Input-Output linkages tend to concentrate at the very core of the production system when considering only close connections ($d = 1$). As shown in Panels B.1a and B.1b, both the largest hubs (top 10% by connectivity) and the mid-sized hubs (top 20%) are similarly central, occupying short distances from the network's nucleus. Third, these same highly connected sectors tend to become relatively more peripheral once longer distances ($d = 2$) are considered. Panels B.1c and B.1d show that, at this range, their positions are more evenly distributed across the network, reflecting the fact that their trade structures – both upstream and downstream – are highly similar to those of other sectors. In other words, most connected sectors are less unique at longer distances, as commonalities in demand and supply linkages become more widespread.

Altogether, these observations highlight that the structure of horizontal complementarities depend crucially on the notion of distance. When short, horizontal linkages cluster around the key sectors whereas, at longer distances, such complementarities are more evenly distributed. This pattern confirms that sectors typically share both upstream suppliers and downstream buyers at smaller but not at a greater distance, and the same vertical production system generates distinct horizontal geometries.

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES

denomination	***codes***	
	NAICS 2017	*BEA*
Farms	111, 112	111CA
Forestry, Fishing and Related Activities	113, 114, 115	113FF
Oil and Gas Extraction	211	211
Mining, except Oil and Gas	212	212
Support Activities for Mining	213	213
Utilities	22	22
Construction	23	23
Wood Products	321	321
Nonmetallic Mineral Products	327	327
Primary Metals	331	331
Fabricated Metal Products	332	332
Machinery	333	333
Computer and Electronic Products	334	334
Electrical Equipment, Appliances, and Components	335	335
Motor Vehicle, Bodies and Trailers, and Parts	3361-3	3361MV
Other Transportation Equipment	3364-9	3364OT
Furniture and Related Products	337	337
Miscellaneous Manufacturing	339	339
Food and Beverage and Tobacco Products	311, 312	311FT
Textile Mills and Textile Product Mills	313, 314	313TT
Apparel and Leather and Allied Products	315, 316	315AL
Paper Products	322	322
Printing and Related Support Activities	323	323
Petroleum and Coal Products	324	324
Chemical Products	325	325
Plastics and Rubber Products	326	326
Wholesale Trade	42	42
Motor Vehicle and Parts Dealers	441	441
Food and Beverage Stores	445	445
General Merchandise Stores	452	452
Other Retail	442-4, 446-8, 45 ex. 452	4AO
Air Transportation	481	481
Rail Transportation	482	482
Water Transportation	483	483
Truck Transportation	484	484
Transit and Ground Passenger Transportation	485	485

...

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

TABLE B.1: LIST OF CONSIDERED SECTORS AND THEIR CODES (*continued*)

denomination	***codes***	
	NAICS 2017	*BEA*
...		
Pipeline Transportation	486	486
Other Transportation and Support Activities	487, 488, 492	487OS
Warehousing and Storage	493	493
Publishing Industries, except Internet (includes Software)	511	511
Motion Picture and Sound Recording industries	512	512
Broadcasting and Telecommunications	515, 517	513
Data Processing, Internet Publishing, and Other Information Service	518, 519	514
Federal Reserve Banks, Credit Intermediation, and Related Activities	521, 5221, 5222, 5223	521CI
Securities, Commodity Contracts, and Investments	523	523
Insurance Carriers and Related Activities	5241, 5242	524
Funds, Trusts, and Other Financial Vehicles	525	525
Housing (not considered in the analysis)	·	HS
Other Real Estate	531	ORE
Rental and Leasing Services and Lessors of Intangible Assets	532, 533	532RL
Legal Services	5411	5411
Computer Systems Design and Related Services	5415	5415
Miscellaneous Professional, Scientific, and Technical Services	541 ex. 5411, 5415	5412OP
Management of Companies and Enterprises	55	55
Administrative and Support Services	561	561
Waste Management and Remediation Services	562	562
Educational Services	61	61
Ambulatory Health Care Services	621	621
Hospitals	622	622
Nursing and Residential Care Facilities	623	623
Social Assistance	624	624
Performing Arts, Spectator Sports, Museums, and Related Activities	711, 712	711AS
Amusement, Gambling, and Recreation industries	713	713
Accommodation	721	721
Food Services and Drinking Places	722	722
Other Services, except Government	81	81

Sectors are defined according to the nomenclature used by the U.S. Bureau of Economic Analysis (BEA).

B.3. Network distances at the intensive margin

Section 4 of the main text characterizes distance matrices at their "extensive margin", $\boldsymbol{\mathcal{D}}^{fd}_{ext}$ and $\boldsymbol{\mathcal{D}}^{fs}_{ext}$, whose values are $d^{q}_{ext}[i,j] = \{1,2,\ldots,d_{max}\}$ for $q = \{fd, fs\}$, exploiting the *shortest path algorithm* on the "intensive margin" distances from the Leontief inverse production network, $\boldsymbol{\mathcal{L}}$, that accounts for direct and indirect sectoral connections.

To characterize the "intensive margin" matrices $\boldsymbol{\mathcal{D}}^{fd}$ and $\boldsymbol{\mathcal{D}}^{fs}$ I exploit the system of *Cartesian coordinates* typical of any production network (*e.g.*, Conley and Dupor 2003). Given $\alpha^{fd}_{ij} = \frac{\alpha_{ij}}{\sum_{j\in\Phi}\alpha_{ij}}$, the generic element of *factor input demand* distance matrix is computed as

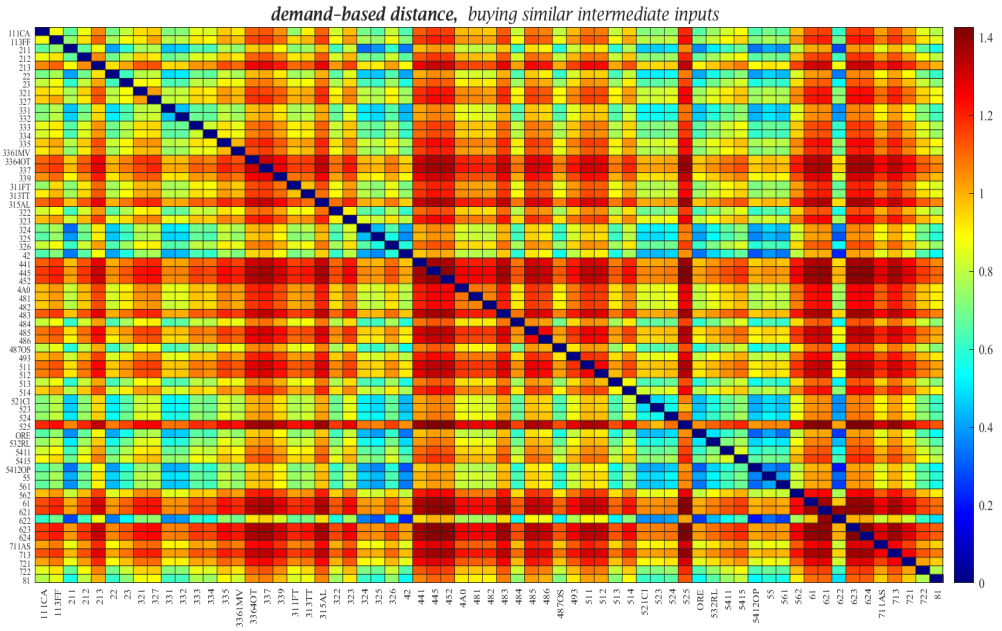
$$d^{fd}[i,j] = \left\{ \sum_{s\in\Phi} \left[\alpha^{fd}_{is} - \alpha^{fd}_{js} \right]^2 \right\}^{\frac{1}{2}} \tag{B.1}$$

which defines the length of the line segment (*i.e.*, the Euclidean distance) connecting sectors i and j when both are buying from the same other sector-s, $\forall s \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{fd} = \left[d^{fd}[i,j] \right]$ with generic element $d^{fd}[i,j] > 0$. Demand distance is the difference between columns constructed by dividing each cell by its row sum. Why? A row sum represents the total sales of the sector in question, so dividing each entry by its associated row sum yields the share of that (row) sector's output purchased by each of the column sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (column) sectors in terms of their purchases from common upstream suppliers, thereby pointing to the strength of the upstream pass-through (share of total sales of the common sector).

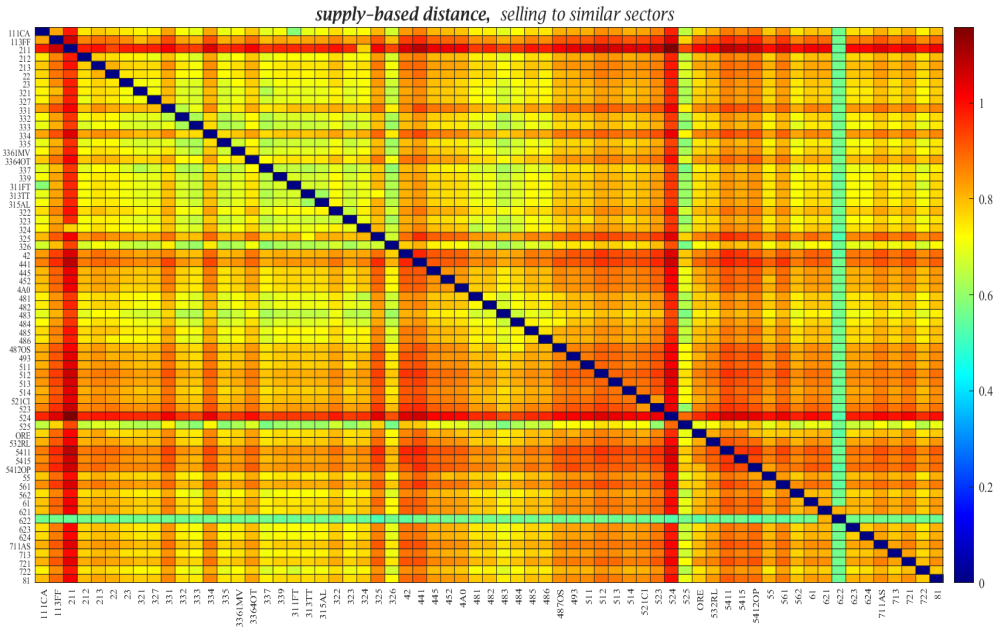
In an analogous way, given $\alpha^{fs}_{ij} = \frac{\alpha_{ij}}{\sum_{i\in\Phi}\alpha_{ij}}$, the generic element of *factor input supply* distance matrix is shaped by

$$d^{fs}[i,j] = \left\{ \sum_{s\in\Phi} \left[\alpha^{fs}_{si} - \alpha^{fs}_{sj} \right]^2 \right\}^{\frac{1}{2}} \tag{B.2}$$

tracing a Euclidean distance between sectors $\{i,j\}$ when both are selling to the same sector-s, $\forall s \in \Phi$, so that $\boldsymbol{\mathcal{D}}^{fs} = \left[d^{fs}[i,j] \right]$ with $d^{fs}[i,j] > 0$. Supply distance is the difference between rows constructed by dividing each cell by its column sum. Why? A column sum represents the total purchases of the sector in question, so dividing each entry by its associated column sum yields the share of that (column) sector's input purchased from each of the row sectors. This normalization allows for a meaningful comparison across dyads of sectors: by examining these shares, one can assess the relative distance between two (row) sectors in terms of their traded quantities to common downstream buyers, thereby pointing to the strength of the downstream pass-through (share of total purchases of the common sector).



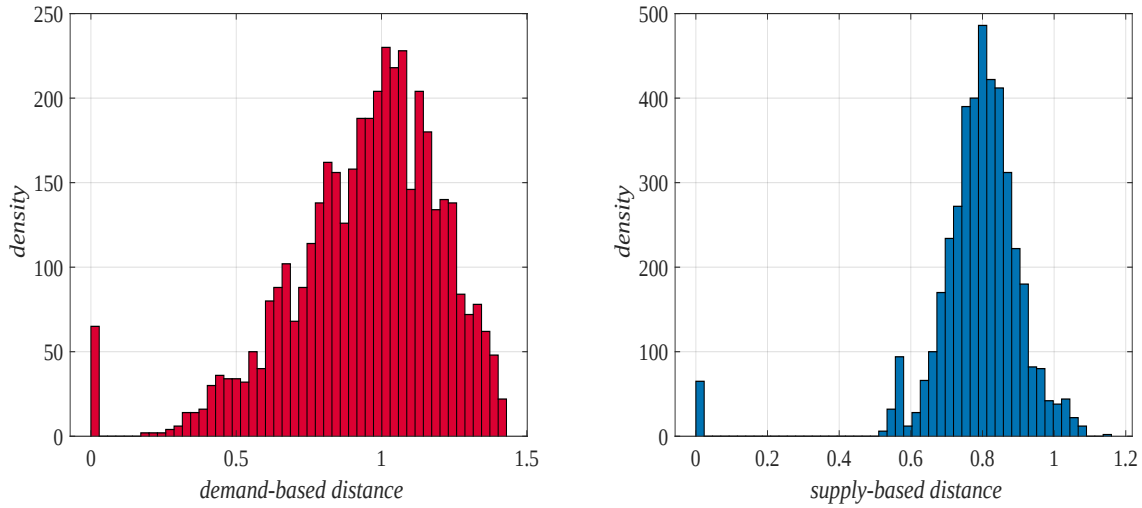
(A) FACTOR INPUT DEMAND



(B) FACTOR INPUT SUPPLY

FIGURE B.2: INTENSIVE MARGIN MATRICES FOR NETWORK ECONOMIC DISTANCE

Note: heatmaps of sectoral distances characterizing the U.S. production network in year 2007 for 3-digit U.S. 2017 NAICS sectors. Entry (i, j) represents the intensive marginal value, $d^q[i, j] \geq 0$ for $q = \{fd, fs\}$, of the horizontal relationship between sector-i and sector-j under *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, Panel B.2a) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel B.2b). Diagonal sparsity (*i.e.*, lots of 0s), is a direct consequence of the fact that the difference of inter-sectoral trade intensities between a sector and itself is zero. Distance matrices are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network. *Source*: BEA and own calculations.



(A) FACTOR INPUT DEMAND (B) FACTOR INPUT SUPPLY

FIGURE B.3: DISTRIBUTION OF NETWORK ECONOMIC DISTANCES

Note: these figures depict the distributions for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, Panel B.3a) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, Panel B.3b). Distance values on the x-axis are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

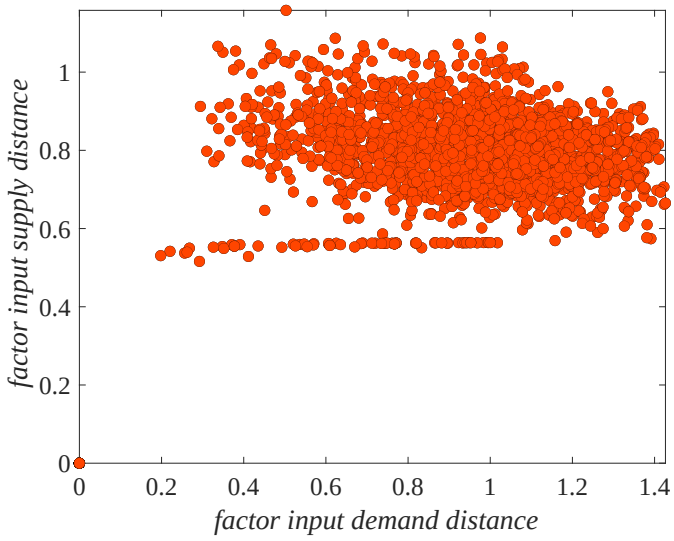
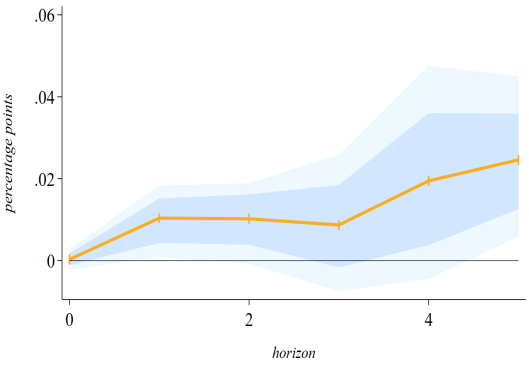


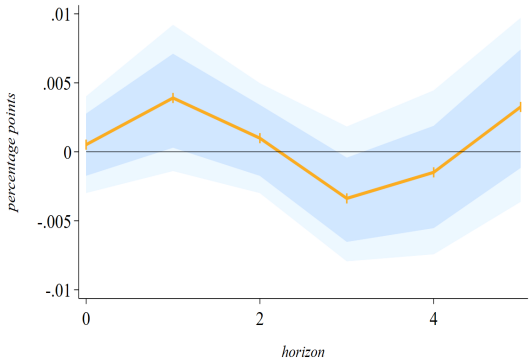
FIGURE B.4: CORRELATION OF NETWORK ECONOMIC DISTANCES

Note: the figure shows the correlation between the measures for *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers, x-axis) and *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers, y-axis). Distances are at the *intensive margin*, as in eqs. (B.1)-(B.2), computed from the *directed* sectoral production network *Source*: BEA and own calculations.

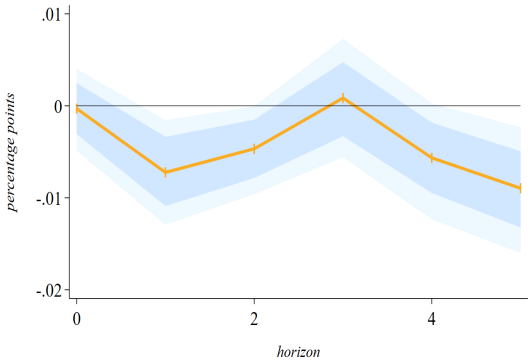
C. COMPLEMENTARY RESULTS



(A) SECTOR-SPECIFIC



(B) $d = 1$



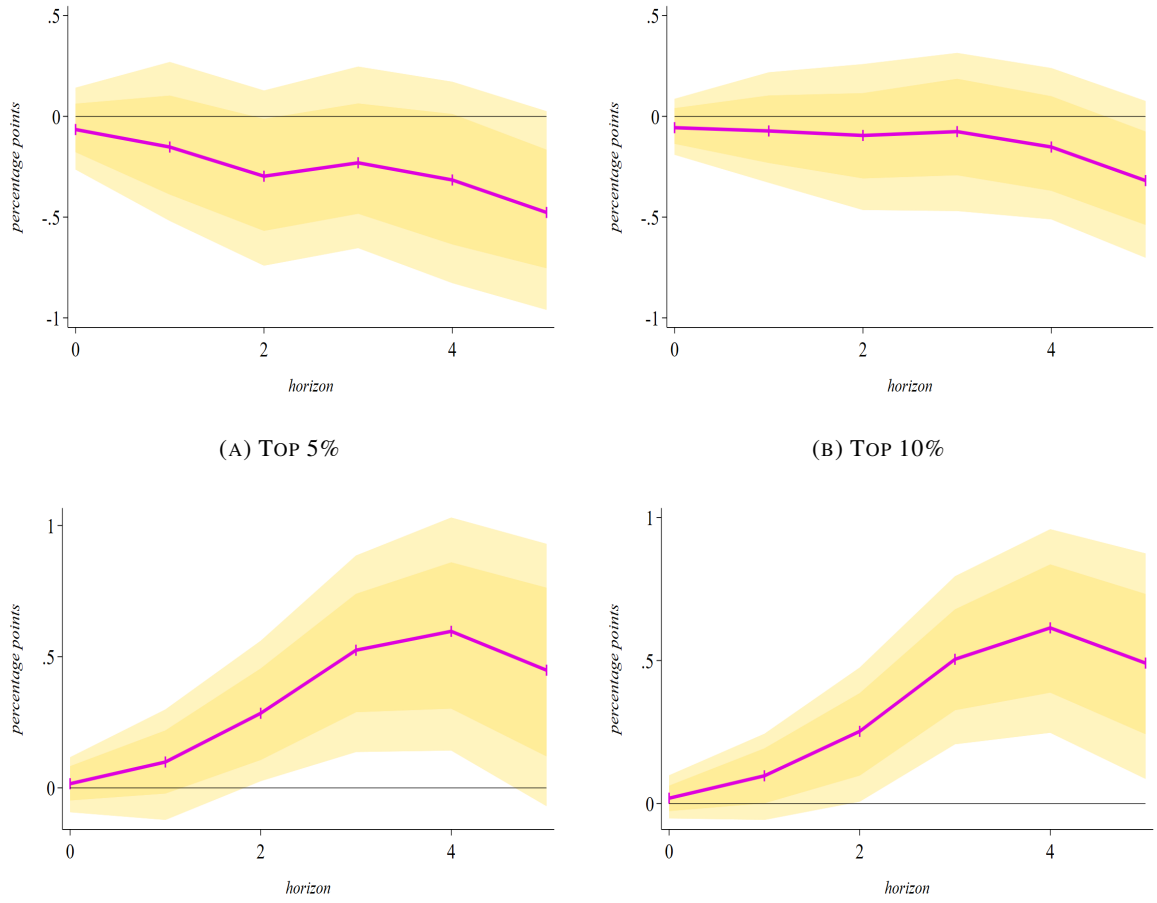
(C) $d = 2$

FIGURE C.1: SECTORAL EMPLOYMENT RESPONSE TO INTERMEDIATE INPUTS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment to a 1% increase in the sector-specific set of intermediate inputs as a share of its value-added for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. In particular, Panel C.1a represents the sector-level employment response to changes in its intermediate inputs, while Panel C.1b-C.1c to changes in the set of intermediates in closer (distance equal to 1) and further (distance equal to 2) sectors. The solid-orange line corresponds to the average response of employment across sectors, while shadow-blue and shadow-light blue areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eqs. (5)-(6). *Source*: BEA and own calculations.

C.1. Plotting separated results for highly interlinked sectors

Unpacking all the combined plots, below I report the estimated Local Projection (LP) dynamics for top 5%, top 10%, top 20%, and top 30% of more connected sectors in the production network (*i.e.*, major number of Input-Output linkages) separately.[VI] The purpose is to show the significance of the responses appearing in Panels 10a-10b of Figure 10 in Subsection 5.2 of the main text, and in Panels 8a-8b of Figure 8 in Appendix C.



(A) TOP 5%

(B) TOP 10%

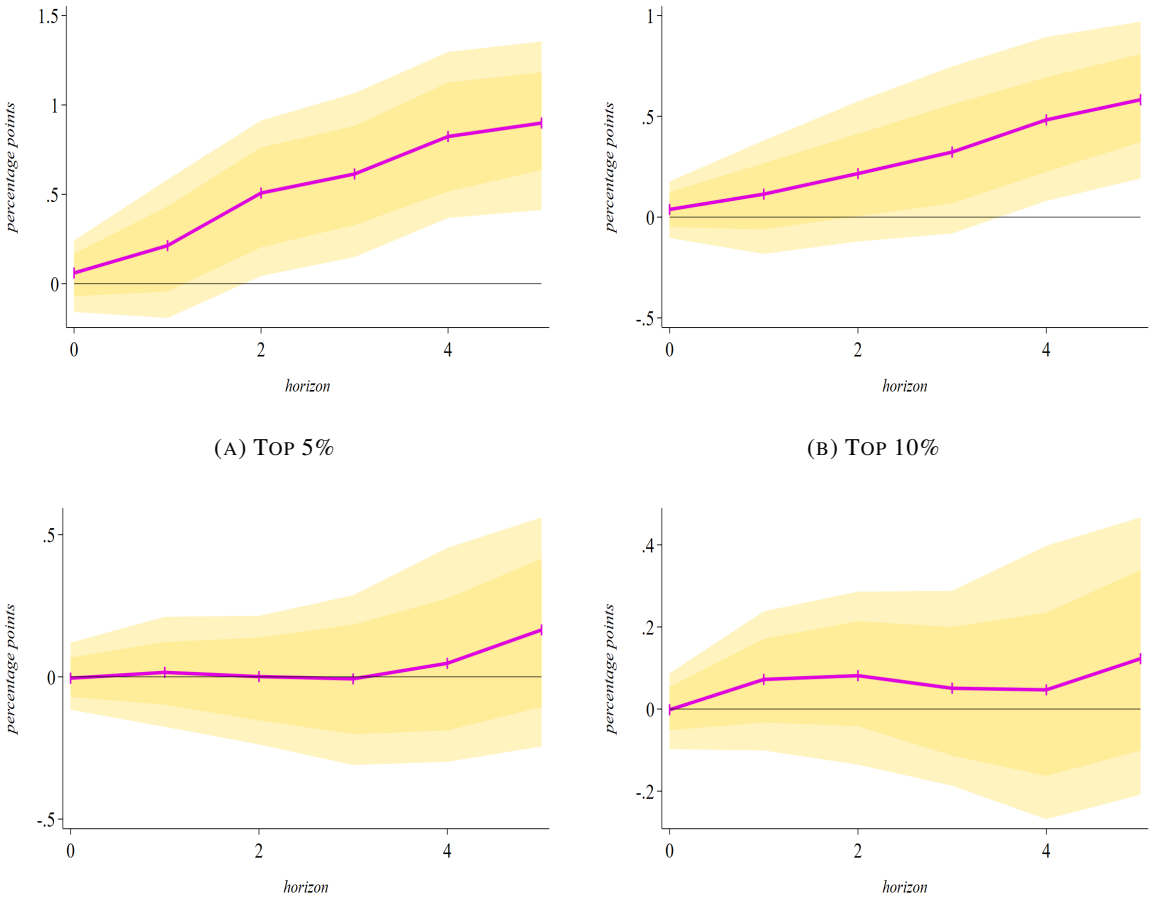
(C) TOP 20%

(D) TOP 30%

FIGURE C.2: DEMAND ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 8a of Figure 8, each panel of this figure shows the employment response of top 5% (Panel C.2a), 10% (Panel C.2b), 20% (Panel C.2c), and 30% (Panel C.2d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

[VI] Besides my network-based effects outlined in the main text, further theories beyond employment comovement are in Rogerson (1987) and Boldrin et al. (2001).



(A) TOP 5%

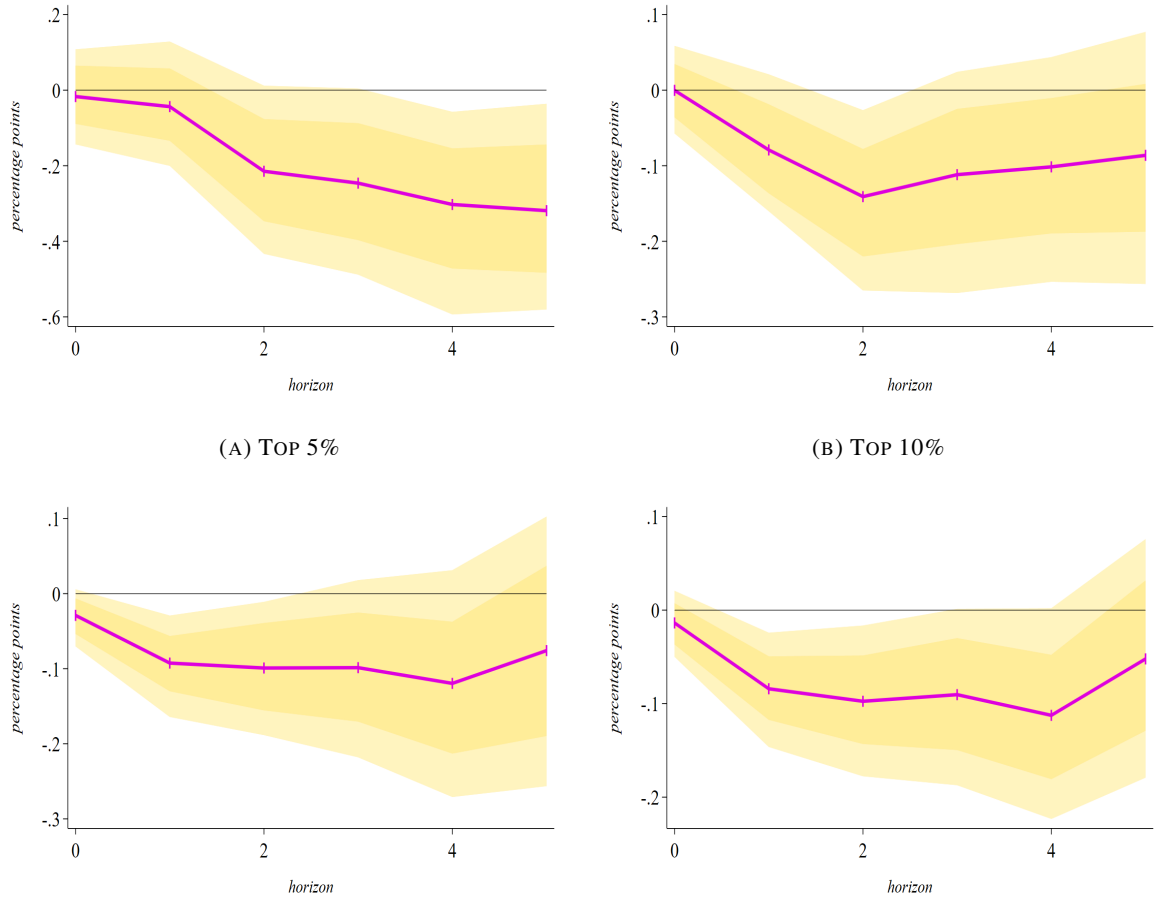
(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE C.3: DEMAND ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 8b of Figure 8, each panel of this figure shows the employment response of top 20% (Panel C.3c) and 30% (Panel C.3d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



(A) TOP 5%

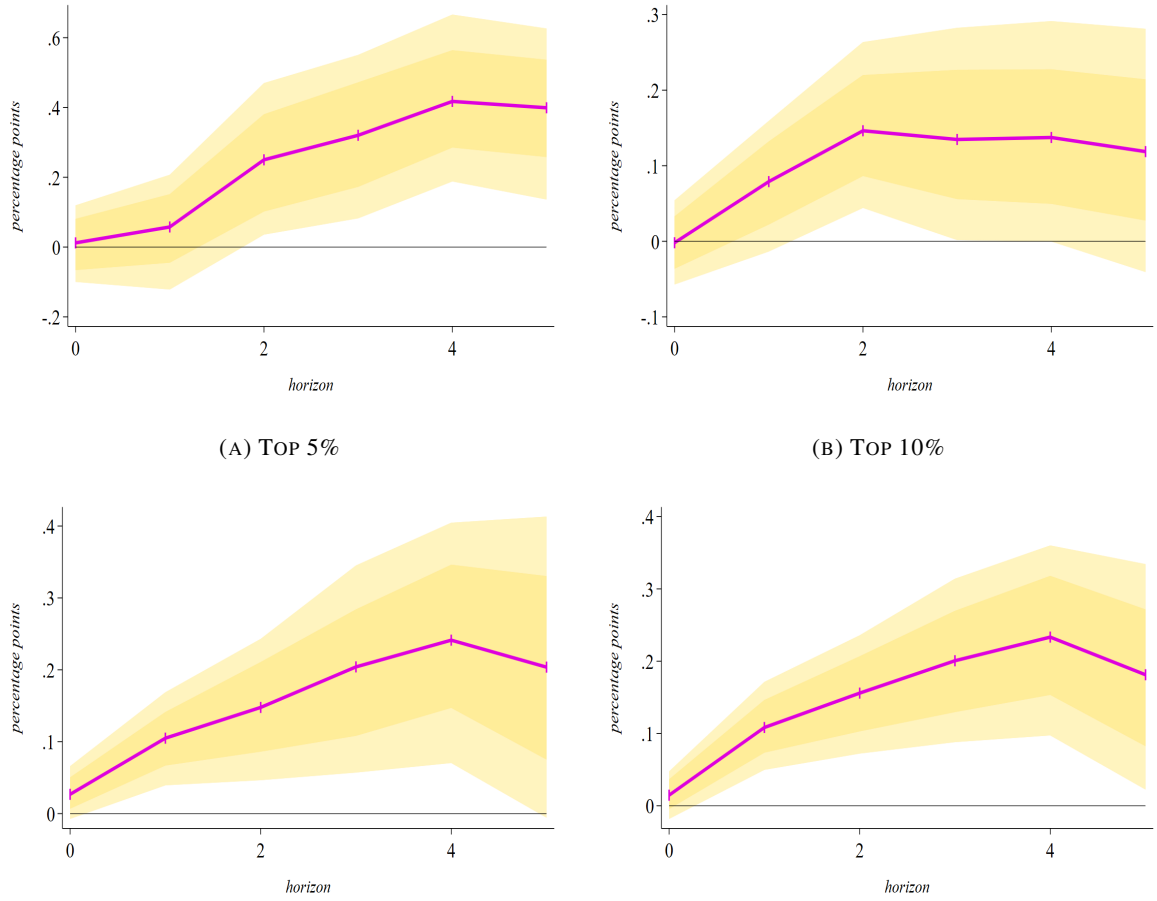
(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE C.4: SUPPLY ($d = 1$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 10a of Figure 10, each panel of this figure shows the employment response of top 5% (Panel C.4a), 10% (Panel C.4b), 20% (Panel C.4c), and 30% (Panel C.4d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



(A) TOP 5%

(B) TOP 10%

(C) TOP 20%

(D) TOP 30%

FIGURE C.5: SUPPLY ($d = 2$) EFFECTS FOR INTERLINKED SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Unpacking Panel 10b of Figure 10, each panel of this figure shows the employment response of top 5% (Panel C.5a), 10% (Panel C.5b), 20% (Panel C.5c), and 30% (Panel C.5d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their further (distance equal to 2) sectors. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

C.2. Robustness with centrality measures

To identify the centrality degree of each sector, I follow the approach outlined in Carvalho (2014) to measure the sector-level Bonacich-Katz centrality.[VII] Using the Leontief-inverse matrix $\boldsymbol{\mathcal{L}}$ for the U.S. production network as of 2007, the Bonacich-Katz centrality $m \times 1$ vector, $\boldsymbol{bk}$, is defined as:

$$\boldsymbol{bk} = \sum_{k \geq 0} \zeta^k \boldsymbol{\mathcal{L}}^k \mathbf{1} = \left(\boldsymbol{I} - \zeta \boldsymbol{\mathcal{L}} \right)^{-1} \mathbf{1} \tag{C.1}$$

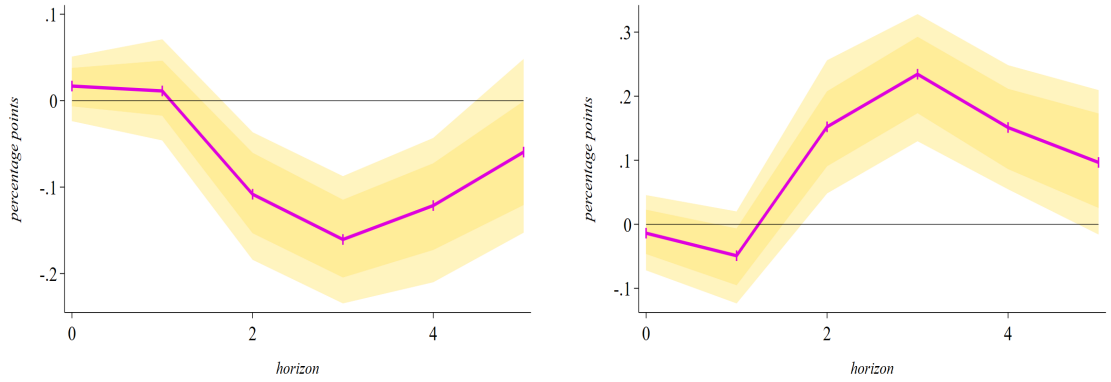
where index-k refers to the length of the paths in the production network – that is, the number of steps (or network "hops") connecting one sector to another – and, given a finite set of sectors $i, j \in \Phi = \{1, \ldots, m\}$:

- $\mathbf{1}$ is an $m \times 1$ vector of ones, reflecting the base influence;
- $\zeta \in \left(0, \frac{1}{\nu_{max}}\right)$ is a dampening scalar parameter, with ν_{max} being the largest eigenvalue of matrix $\boldsymbol{\mathcal{L}}$;
- $\boldsymbol{I}$ is the $m \times m$ identity matrix.

The value of ζ determines the weight assigned to indirect linkages: smaller values place more emphasis on direct connections, while larger values account for deeper propagation within the network. Following Carvalho (2014), a value of $\zeta = 0.5$ is typically used, which lies safely below the inverse of the largest eigenvalue of most empirically observed Input-Output matrices, ensuring convergence of the series. This measure reflects the idea that a sector is central not only if it is directly connected to many others, but also if it is connected to sectors that are themselves highly central.

This formulation ensures that the centrality score of each sector captures recursive amplification along both its direct connections and its indirect influence through other sectors due to the use of matrix $\boldsymbol{\mathcal{L}} = (\boldsymbol{I} - \boldsymbol{\mathcal{A}})^{-1} = \left[\ell_{ij} \geq 0\right]$, where $\boldsymbol{\mathcal{A}} = \left[\alpha_{ij} \geq 0\right]$ is the directed production network matrix defining the intensity of good produced by sector-j in the total intermediate inputs used by sector-i, with $\alpha_{ij} = 0$ indicating that sector-i does not make use of the good produced by sector-j in producing its own intermediate good.

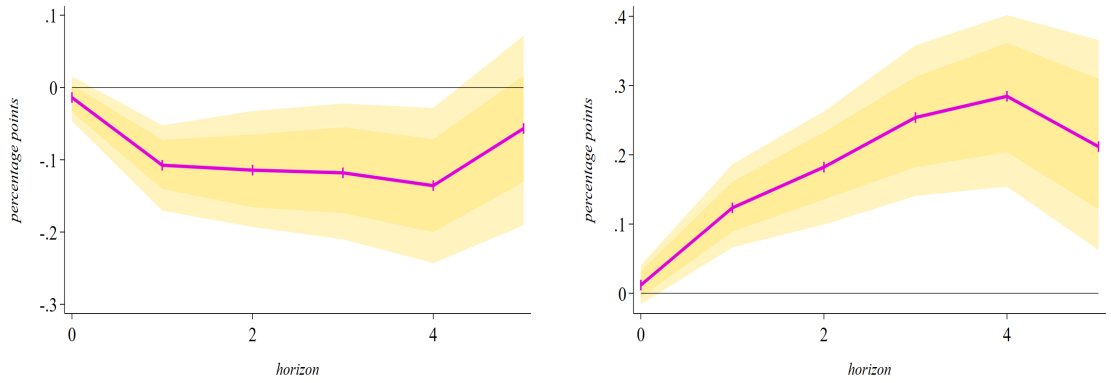
[VII] Outlined by Bonacich (1987), it is a measure of a sector's overall importance within an Input-Output system, capturing not only its direct connections to other sectors but also the centrality of those it is connected to. In essence, a sector is considered central if it is linked to other central sectors, creating a recursive structure of influence. This measure goes beyond simple counts of linkages by incorporating indirect connections – weighted by a dampening factor – to reflect the diminishing influence of more distant sectors in the network. In the context of production network, a sector with high Bonacich-Katz centrality is one that plays a key role in the propagation of shocks, as its position allows it to influence (or be influenced by) large portions of the economy through both direct and indirect channels. Compared to other centrality measures, Bonacich-Katz is particularly well-suited for economic applications, as it accounts for the intensity of inter-sectoral relationships and the structure of the network in a realistic and nuanced way.



(A) $d = 1$ (B) $d = 2$

FIGURE C.6: DEMAND LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.6a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.6b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.



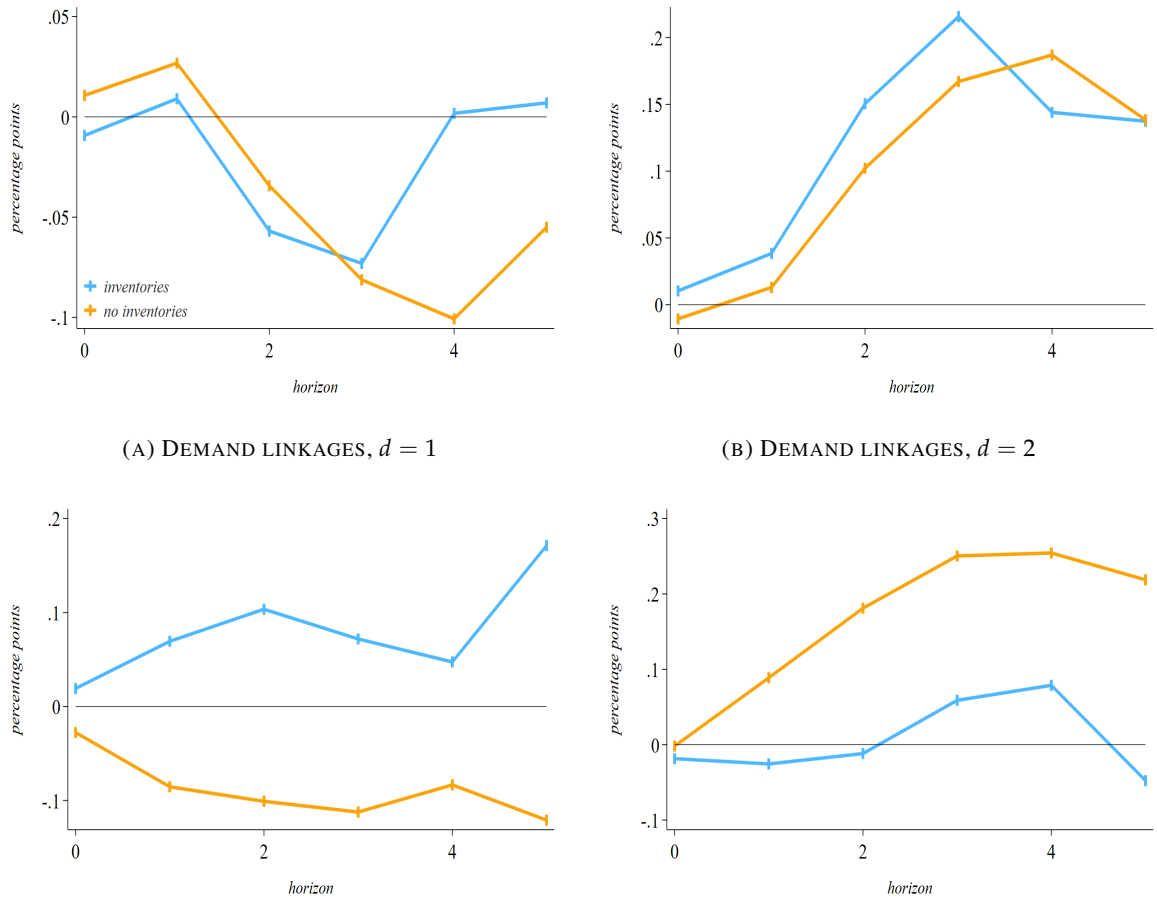
(A) $d = 1$ (B) $d = 2$

FIGURE C.7: SUPPLY LINKAGES AND COMOVEMENT IN PERIPHERAL SECTORS

Note: given the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of most central sectors (with *centrality* defined according to sector's relevance in final consumption) to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Panel C.7a represents the sector-level employment response to employment changes in closer (distance equal to 1) sectors, while Panel C.7b plots the response to employment changes in further (distance equal to 2) ones. The solid-purple line corresponds to the average response of employment across sectors, while shadow-gold and shadow-light gold areas correspond to 68% and 90% significance levels of bootstrapped Confidence Interval (CI), respectively, computed from eq. (8). *Source*: BEA and own calculations.

C.3. Robustness with private inventories

Responses for comovement in sectoral employment levels when controlling for changes in sectoral inventories, distinguishing further between sectors with inventories and those without. The results are shown in Figure C.8. Note that the plots refer dynamic changes in inventories and the Input-Output matrix held at the baseline year 2007.



(A) DEMAND LINKAGES, $d = 1$

(B) DEMAND LINKAGES, $d = 2$

(C) SUPPLY LINKAGES, $d = 1$

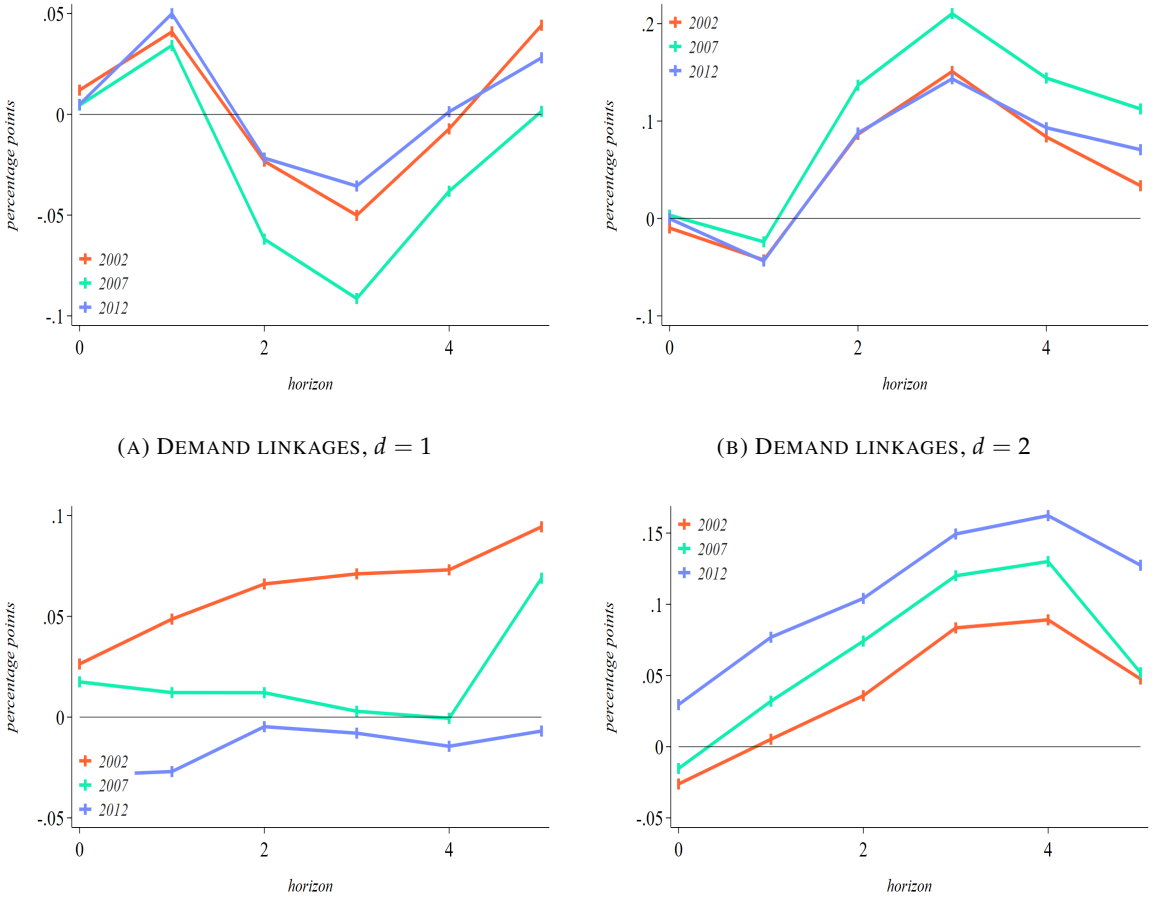
(D) SUPPLY LINKAGES, $d = 2$

FIGURE C.8: COMOVEMENT UNDER CHANGES IN SECTORAL INVENTORIES

Note: given both the *factor input demand* (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *factor input supply* (*i.e.*, supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network in year 2007, the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors when controlling for the changes in private inventories, for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Solid lines corresponds to the average response of employment across sectors, given 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8). Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. Since not all sectors have inventories, the panels are disentangled by subgroups that make use or not of private inventories. *Source*: BEA and own calculations.

C.4. Robustness with production network at different base years

Responses for comovement in sectoral employment levels in the main analysis of Section 5 are centred on the implementation of a sectoral production network represented by an Input-Output matrix in the baseline year 2007. Differently, Figures C.9, C.10 and C.11 perform again all the main analysis by referring, instead, to a production network structure five years before (2002) and five years after (2012) the baseline year.



(A) DEMAND LINKAGES, $d = 1$

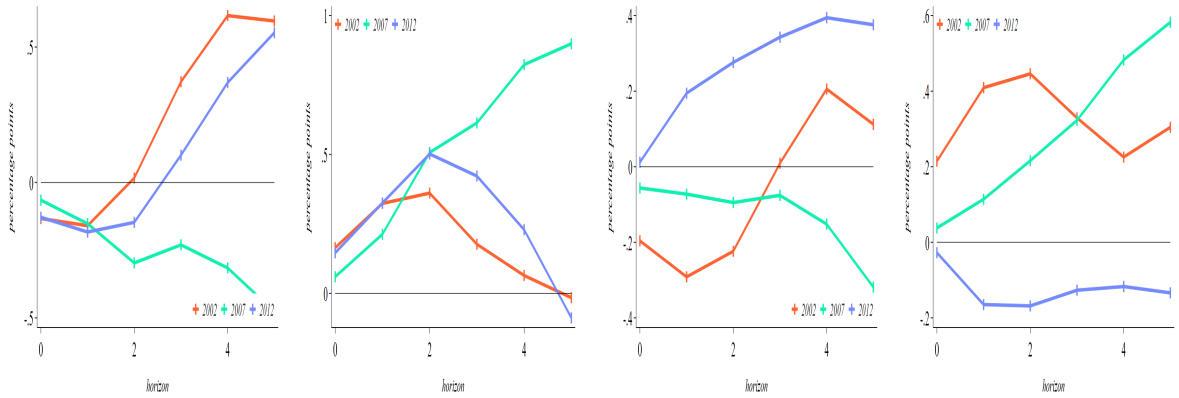
(B) DEMAND LINKAGES, $d = 2$

(C) SUPPLY LINKAGES, $d = 1$

(D) SUPPLY LINKAGES, $d = 2$

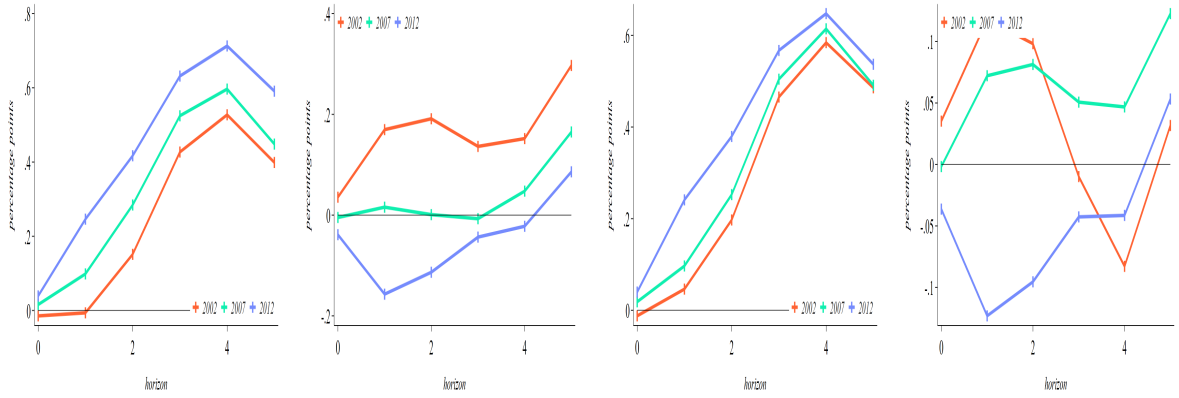
FIGURE C.9: COMOVEMENT OVER DIFFERENT BASELINE YEARS

Note: given both the *factor input demand* (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *factor input supply* (*i.e.*, supply linkages across sectors given their common downstream buyers) network distances, and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the results in the main text: Panels C.9a-C.9b expand their relatives of Figure 7, while Panels C.9c-C.9d do so for their relatives in Figure 9. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figures in the main text. Responses are referred to employment changes in closer (distance equal to 1) and further (distance equal to 2) sectors. *Source*: BEA and own calculations.



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(A) TOP 5% (B) TOP 10%

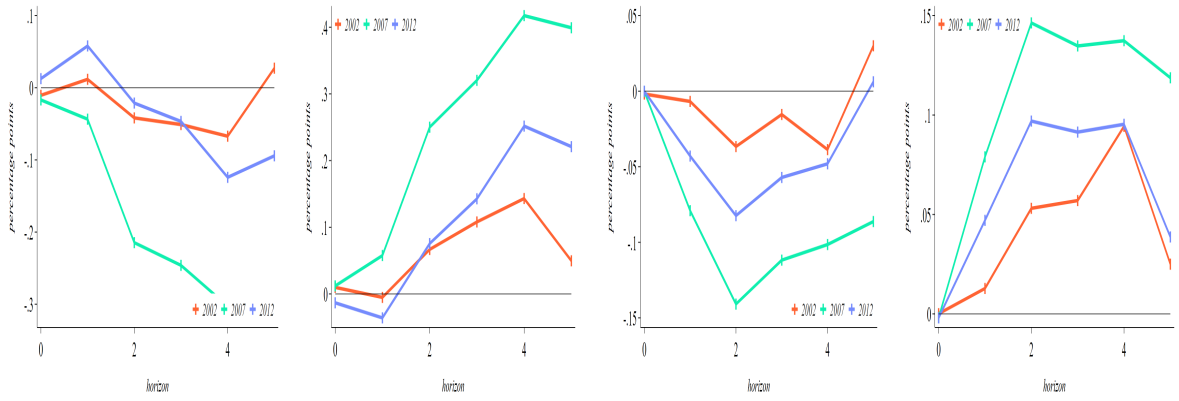


$d = 1$ $d = 2$ $d = 1$ $d = 2$

(C) TOP 20% (D) TOP 30%

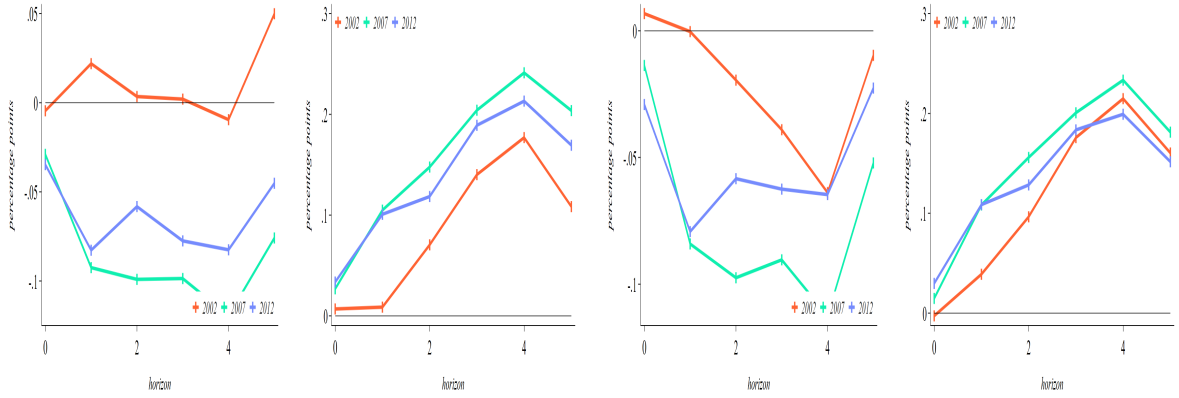
FIGURE C.10: DEMAND LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input demand* network distance (*i.e.*, demand linkages across sectors given their common upstream sellers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 8a-8b of Figure 8, each panel of this figure shows the employment response of top 5% (Panel C.10a), 10% (Panel C.10b), 20% (Panel C.10c), and 30% (Panel C.10d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figure. *Source*: BEA and own calculations.



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(A) TOP 5% (B) TOP 10%



$d = 1$ $d = 2$ $d = 1$ $d = 2$

(C) TOP 20% (D) TOP 30%

FIGURE C.11: SUPPLY LINKAGES AND INTERLINKED SECTORS OVER THE YEARS

Note: given both the *factor input supply* network distance (*i.e.*, supply linkages across sectors given their common downstream buyers) and the *Leontief inverse* transmission (*i.e.*, sectoral direct and indirect network exposure) characterizing the North American production network over different years (2002, 2007, and 2012), the figure shows the response of sectoral employment of sectors with most Input-Output linkages to a 1% increase in the employment level of other sectors for 3-digit U.S. 2017 NAICS sectors over the period 1998-2022. Expanding to different production network structures the responses of Panels 10a-10b of Figure 10, each panel of this figure shows the employment response of top 5% (Panel C.11a), 10% (Panel C.11b), 20% (Panel C.11c), and 30% (Panel C.11d) more connected sectors in the production network (*i.e.*, major number of I-O linkages) – from top-left to bottom-right, respectively – to employment changes in their closer (distance equal to 1) or further (distance equal to 2) sectors. Solid lines corresponds to the average response of employment across sectors, while 68% and 90% significance levels of bootstrapped Confidence Interval (CI), computed from eq. (8), roughly correspond to the associated figure. *Source*: BEA and own calculations.

D. Illustrative episodes of horizontal transmission

In conclusion, a set of recent episodes that exemplify the formalized mechanisms of horizontal shock transmission are presented. Firms/sectors not directly linked through bilateral Input-Output transactions nonetheless exhibit simultaneous outcomes because they share a common downstream buyer (supply-based distance) or a common upstream supplier (demand-based distance). These episodes illustrate how network-based horizontal economic distance, rather than observed vertical exposure, can generate comovement.

Technological shock in Samsung and supply-based spillovers to competing memory suppliers (2026).– On February 2, 2026, Samsung Electronics announced significant technological progress in the development and production of next-generation High Bandwidth Memory 4 (HBM4) chips. HBM is a vertically stacked dynamic memory technology optimized for high-speed data processing and is a critical input for Artificial Intelligence (AI) accelerators produced by firms such as NVIDIA. The announcement constituted a positive, firm-specific productivity shock for Samsung in the high-performance memory segment. Because Samsung and other memory producers – most notably Micron and SK Hynix – compete to supply advanced memory chips to the same downstream buyers (including NVIDIA), market expectations regarding future procurement patterns were immediately revised. Financial press coverage reported that investors anticipated a potential reallocation of orders toward Samsung, reflecting its improved technological position. Importantly, Samsung and its rival suppliers do not maintain vertical trade relationships with one another. Nevertheless, the positive shock in Samsung generated adverse valuation effects for competing firms, reflecting expected substitution by the common downstream buyer.[VIII]

Supply-based Apple's reallocation of AirPods assembly orders (2022).– In 2022, Apple reduced and partially suspended assembly orders for AirPods 2 at Goertek, a major Chinese contract manufacturer. The shock was idiosyncratic to Goertek, stemming from operational issues that impaired its ability to meet Apple's standards and delivery schedules. Rather than reducing overall AirPods production proportionally, Apple responded by shifting demand from the negatively shocked supplier to its competitor, reallocating a substantial share of the affected orders to Luxshare Precision Industry, another contract manufacturer producing similar components for Apple. As a result, Luxshare experienced an increase in expected output and revenues, while Goertek faced declining orders and downward revisions in market expectations. Goertek and Luxshare are not vertically linked to each other; however, they are horizontally proximate because they depend on the same downstream buyer (Apple). The negative shock in one supplier therefore translated into a positive demand shift toward its competitor: Luxshare experienced an increase in expected orders and revenues despite having no direct vertical linkage with Goertek.[IX]

[VIII] Reported by financial outlets such as `Barron's`, `investing.com`, and `MarketWatch` in February 2026.

[IX] Covered by `Yicai Global`, `Euronew` and `MacRumors` in November 2022.

Supply-based Apple's reallocation for OLED panels (2026).– In January 2026, Apple shifted a significant portion of its iPhone OLED (Organic Light-Emitting Diode) panel orders from BOE Technology Group in China to the american company Samsung Display after persistent production and yield issues affected BOE's ability to meet quality and volume requirements. OLED panels are critical display components used in iPhones, and Apple relies on multiple suppliers to ensure supply stability and bargaining power. The operational difficulties at BOE constituted a negative firm-specific shock. In response, Apple reallocated millions of display panel orders to Samsung Display, which had available capacity and superior production reliability. The reallocation of downstream demand increased expected revenues for Samsung Display while reducing BOE's projected sales and market share. As in the AirPods episode, the two suppliers are not connected through vertical production relationships; rather, they are horizontally linked because they sell to the same downstream buyer. The transmission of the shock occurred through Apple's procurement decision, not through direct trade between BOE and Samsung Display.[X]

AI-driven surge in memory chips demand (2025).– In 2025, the rapid expansion of Artificial Intelligence (AI) applications led to an unprecedented increase in global demand for advanced memory chips, particularly DRAM (Dynamic Random Access Memory) and HBM (High Bandwidth Memory). DRAM constitutes the primary volatile memory used in computing systems, while HBM is a high-performance, vertically stacked memory technology optimized for AI accelerators and data-intensive workloads. The surge in AI-related computing, especially for large language models and generative AI systems, generated a sharp increase in procurement by major cloud service providers and AI hardware producers. Because global production capacity for advanced memory is concentrated among a small number of firms (*e.g.*, Samsung, SK Hynix, Micron), the downstream demand shock tightened upstream supply conditions and increased prices. Downstream sectors relying on these same memory inputs – including consumer electronics manufacturers, enterprise hardware producers, and specialized computing firms – faced rising input costs and constrained availability, even though they were not directly linked to each other through trade.[XI]

Cloud-based services expansion (2023-2024).– Between 2023 and 2024, large-scale investment in cloud computing infrastructure – driven by AI adoption and data center expansion – significantly increased downstream demand for advanced semiconductors, including graphics processing units (GPUs) and specialized AI accelerators. This expansion placed substantial pressure on upstream semiconductor manufacturing capacity. Because leading-edge chip fabrication is highly concentrated – particularly in firms such as Taiwan Semiconductor Manufacturing Company (TSMC) – the increase in cloud-sector demand reduced effective upstream supply available to other industries relying on the same fabrication processes. Automotive manufac-

[X] Reported by `MacRumors`, `TechNode`, and `DisplayDaily` in January 2026.

[XI] Reported in 2025 by `investing.com`, `Business Standard`, and `Reuters`.

turers, consumer electronics firms, and industrial equipment producers faced delays and supply constraints despite having no direct trade relationship with cloud service providers. The resulting synchronized production adjustments across otherwise unrelated sectors reflected shared exposure to upstream fabrication capacity rather than vertical trade interdependence.[XII]

Reallocation of TSMC supply toward AI customers (2026).– In 2023, the rapid expansion of Artificial Intelligence (AI) applications triggered a sharp increase in downstream demand for high-performance computing chips, particularly graphics processing units (GPUs). These chips rely on advanced fabrication nodes supplied by TSMC, the world's leading contract chip manufacturer, operating as a pure-play foundry, producing chips designed by numerous fabless semiconductor firms across industries, and whose cutting-edge production capacity is both technologically complex and capacity-constrained. Given the long planning horizon of semiconductor manufacturing – with wafer capacity typically allocated several quarters, and often years, in advance – the AI-driven surge in orders during 2023 induced TSMC to reprioritize production toward high-margin AI customers: in 2024, TMSC reallocated portions of its advanced fabrication capacity toward AI chip designers such as NVIDIA, responding to exceptionally strong demand for AI accelerators. Industry reports indicate that such reallocations affected fabrication schedules extending into 2025–2026, as a growing share of advanced-node capacity was reserved for AI-related chips. Given finite leading-edge fabrication capacity, TMSC's prioritization of AI-related chip production effectively reduced available output for other clients, including smartphone and consumer electronics chip designers. As a consequence, other downstream clients competing for the same foundry capacity faced relatively tighter access or lower priority in allocation. These experienced longer lead times and constrained access to advanced nodes, despite not interacting directly with the AI hardware sector. The shock therefore originated in heightened demand from one downstream segment (AI hardware) but propagated through TSMC as a shared upstream production node, generating horizontal effects on multiple unrelated industries.[XIII]

Across all episodes, the defining feature is that idiosyncratic variations are mediated by the horizontal geometry of the production network rather than by bilateral trade, where shared downstream buyers (*supply-based* distance) or shared upstream suppliers (*demand-based* distance) determine the strength and the direction of micro-level shock transmission. These cases provide qualitative empirical grounding for the paper's central theoretical insight: comovement can arise from horizontal network proximity even in the absence of vertical Input-Output linkages.

[XII] Documented by the CNBC, Reuters, and Financial Times during 2023-2024.

[XIII] Reported in January-February 2026 by MacRumors, Business Insider, and AppleInsider.

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